RAG Grounding Test Cases

Categories 1 to 5

RAG Answer Grounding / Hallucination Detection Evaluation

Prepared from the provided Streamlit pipeline screenshots

Reference Context: Renewable Energy - Solar and Wind Power

1. Purpose and Scope

This document consolidates the RAG answer grounding test-case evidence from Category 1 to Category 5. It is structured as an official validation document where each test case includes the question, expected output, actual output, result, grounded ratio, RAGAS score, and screenshot evidence inside the same table.

2. Test Coverage Overview

Category	Type	Description	Test Cases
Category 1	Fully Grounded	Questions answerable directly from the uploaded context.	15
Category 2	Partially Grounded	Multi-part questions with both supported and unsupported parts.	16
Category 3	Out of Context	Questions unrelated to the document or asking facts not provided.	14
Category 4	Mixed Multi-Part Stress Test	Questions combining valid and invalid parts to test hallucination removal.	5
Category 5	Edge Cases	Trick, ambiguous, and related-sounding prompts.	9
Total	**All Categories**	**Complete screenshot evidence set.**	**59**

3. Result Field Definitions

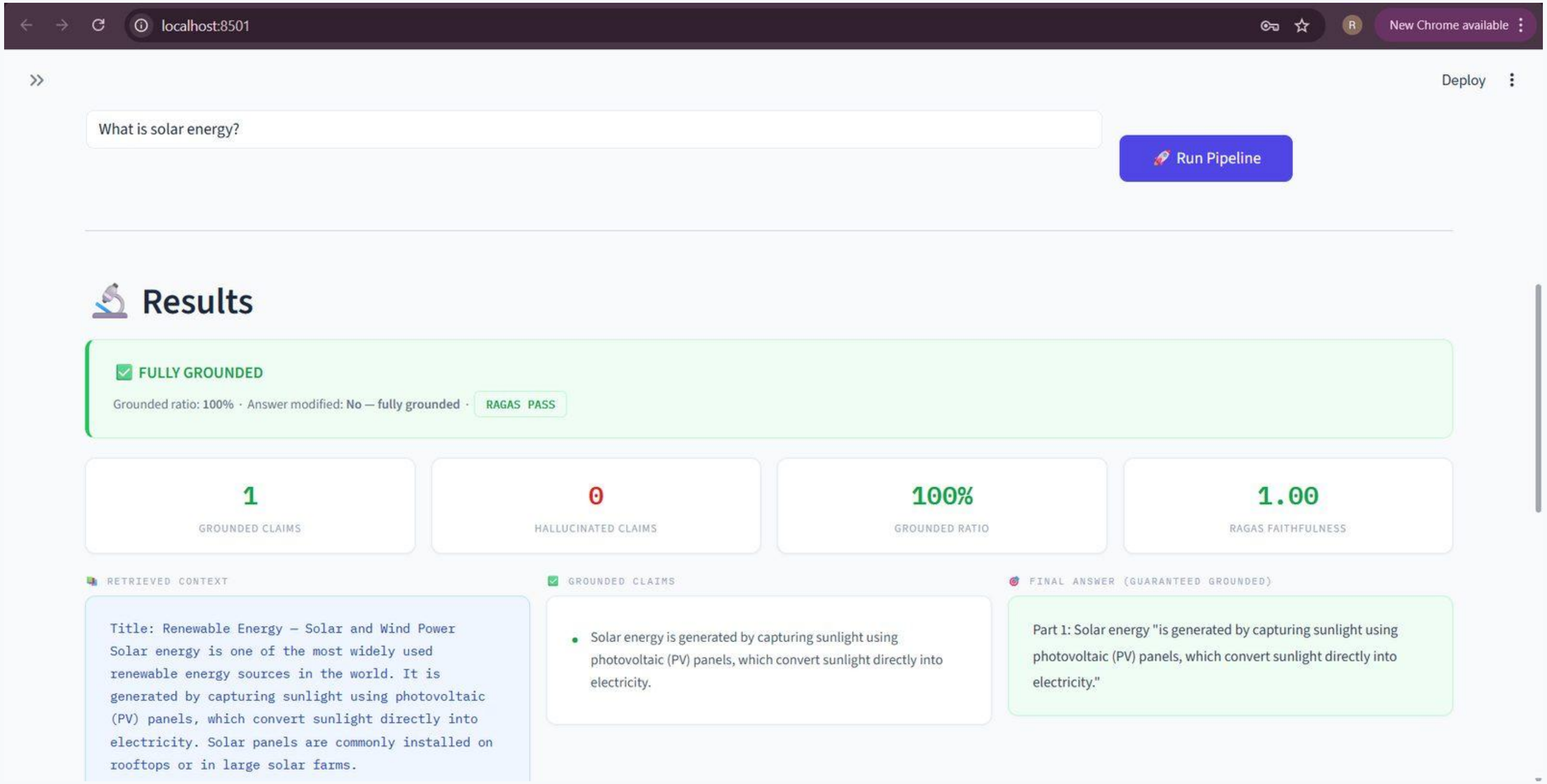
Field	Meaning
Expected Output	The correct behavior expected from the pipeline for the given question.
Actual Output	The observed behavior shown in the Streamlit test screenshot.
Grounded Claims	Claims supported by the retrieved document context.
Hallucinated Claims	Claims not supported by the document and therefore removed or blocked.
Grounded Ratio	Percentage of grounded claims in the generated answer.
RAGAS Faithfulness	Additional faithfulness score displayed in the local Streamlit interface.

Category 1 - Fully Grounded Test Cases

Single-part questions supported by the uploaded renewable-energy context.

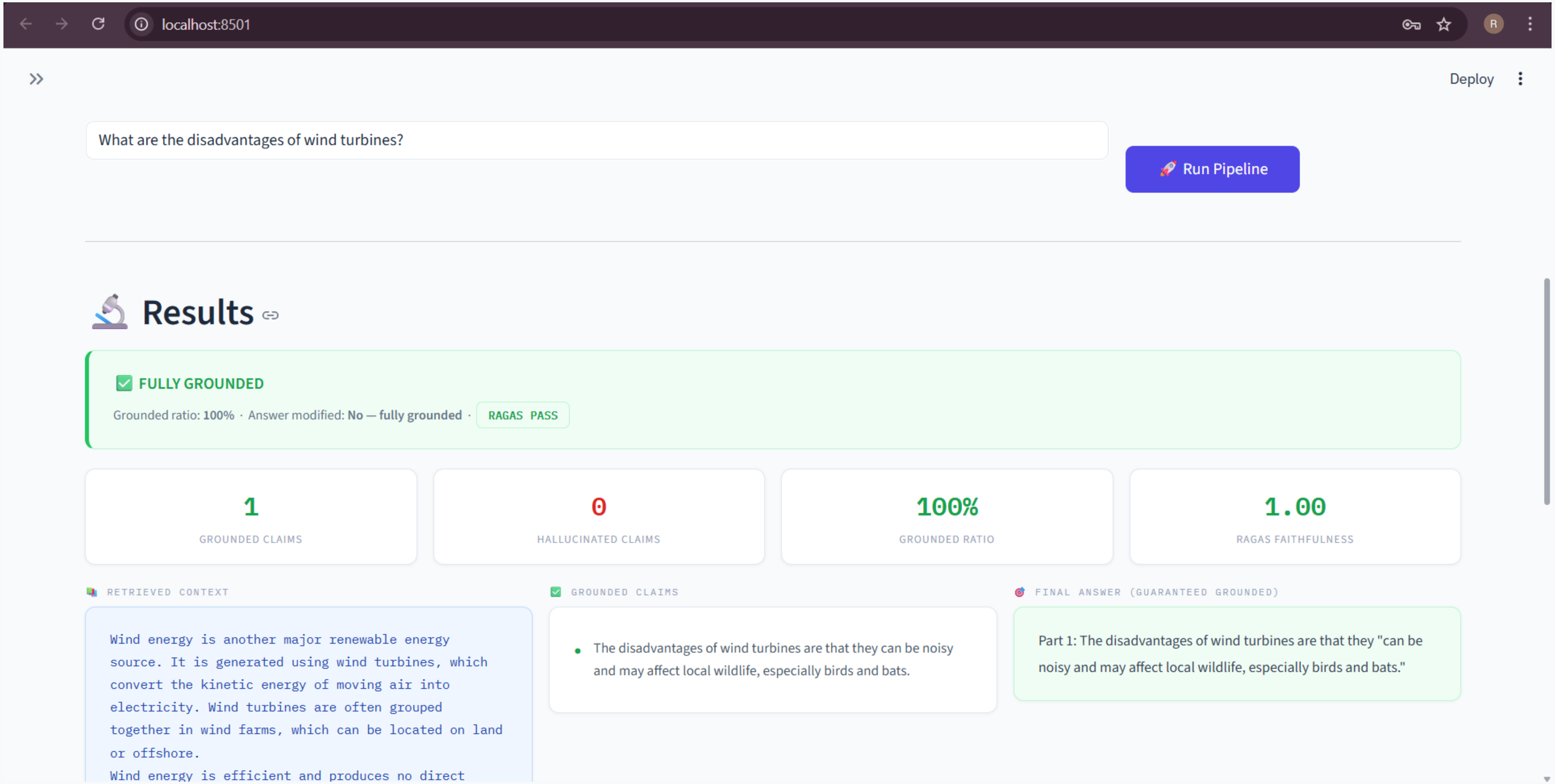
ID	Question	Expected Output	Actual Output	Result / Metrics
TC-FG-01	What is solar energy?	Answer should state that solar energy is generated by capturing sunlight using photovoltaic (PV) panels that convert sunlight directly into electricity.	The system returned the same PV-panel based explanation and marked it fully grounded.	FULLY GROUNDED Grounded Ratio: 100% RAGAS: 1.00

Screenshot Evidence - TC-FG-01



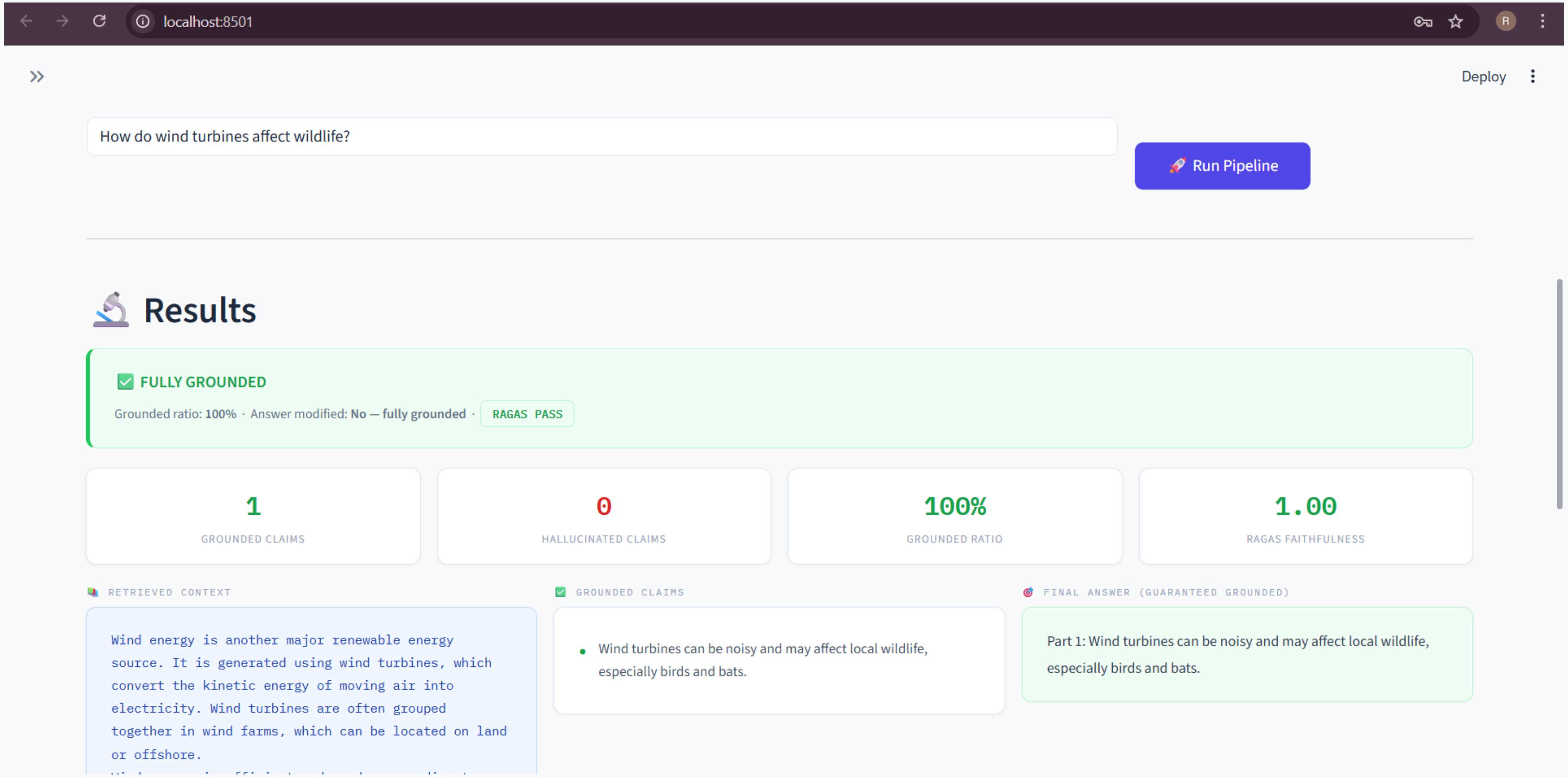
ID	Question	Expected Output	Actual Output	Result / Metrics
TC-FG-02	What are the disadvantages of wind turbines?	Answer should mention only supported disadvantages: wind turbines can be noisy and may affect local wildlife, especially birds and bats.	The system returned the supported disadvantage statement only.	FULLY GROUNDED Grounded Ratio: 100% RAGAS: 1.00

Screenshot Evidence - TC-FG-02



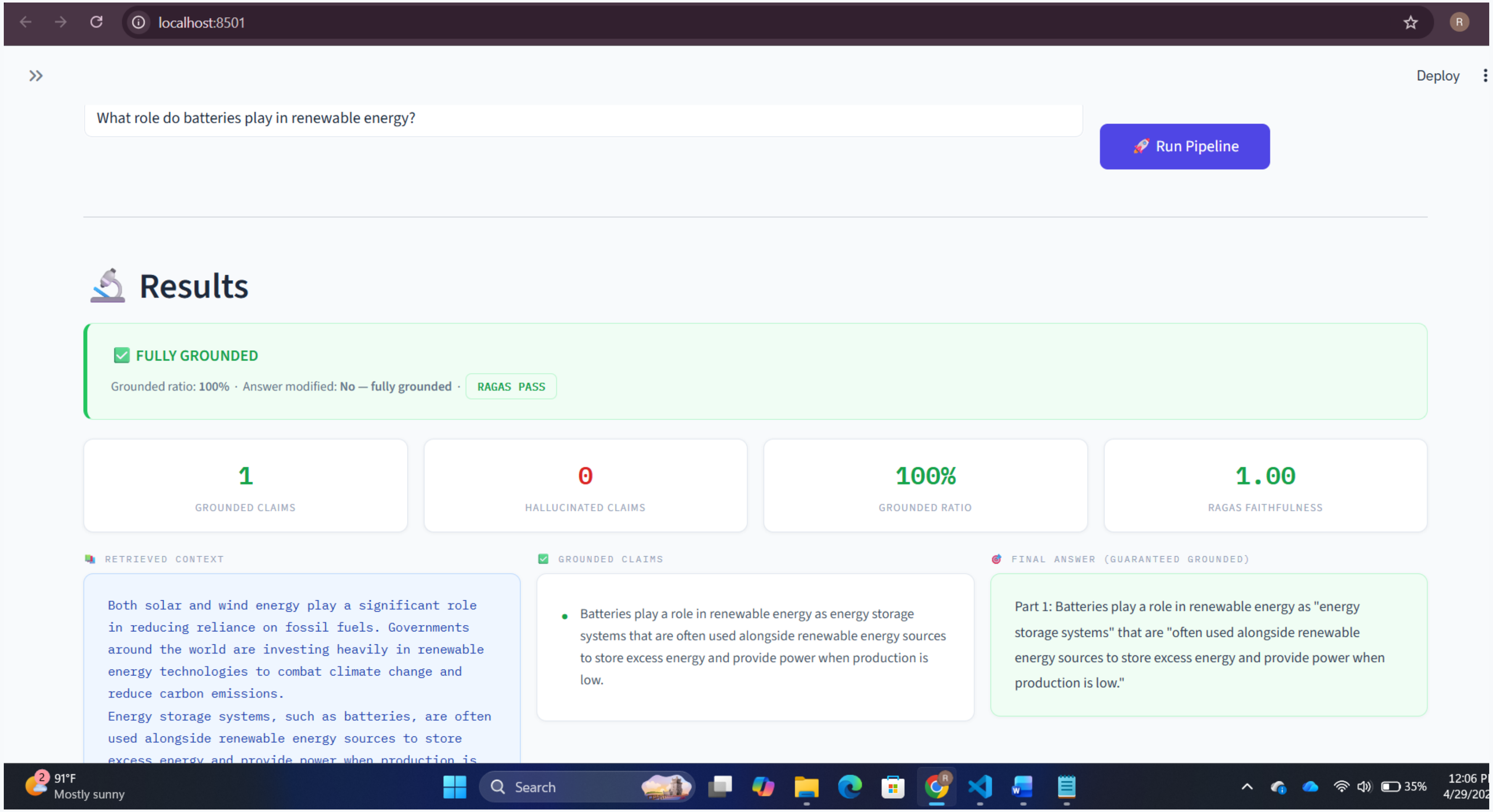
ID	Question	Expected Output	Actual Output	Result / Metrics
TC-FG-03	How do wind turbines affect wildlife?	Answer should state that wind turbines may affect local wildlife, especially birds and bats.	The system returned only the wildlife impact statement from the context.	FULLY GROUNDED Grounded Ratio: 100% RAGAS: 1.00

Screenshot Evidence - TC-FG-03



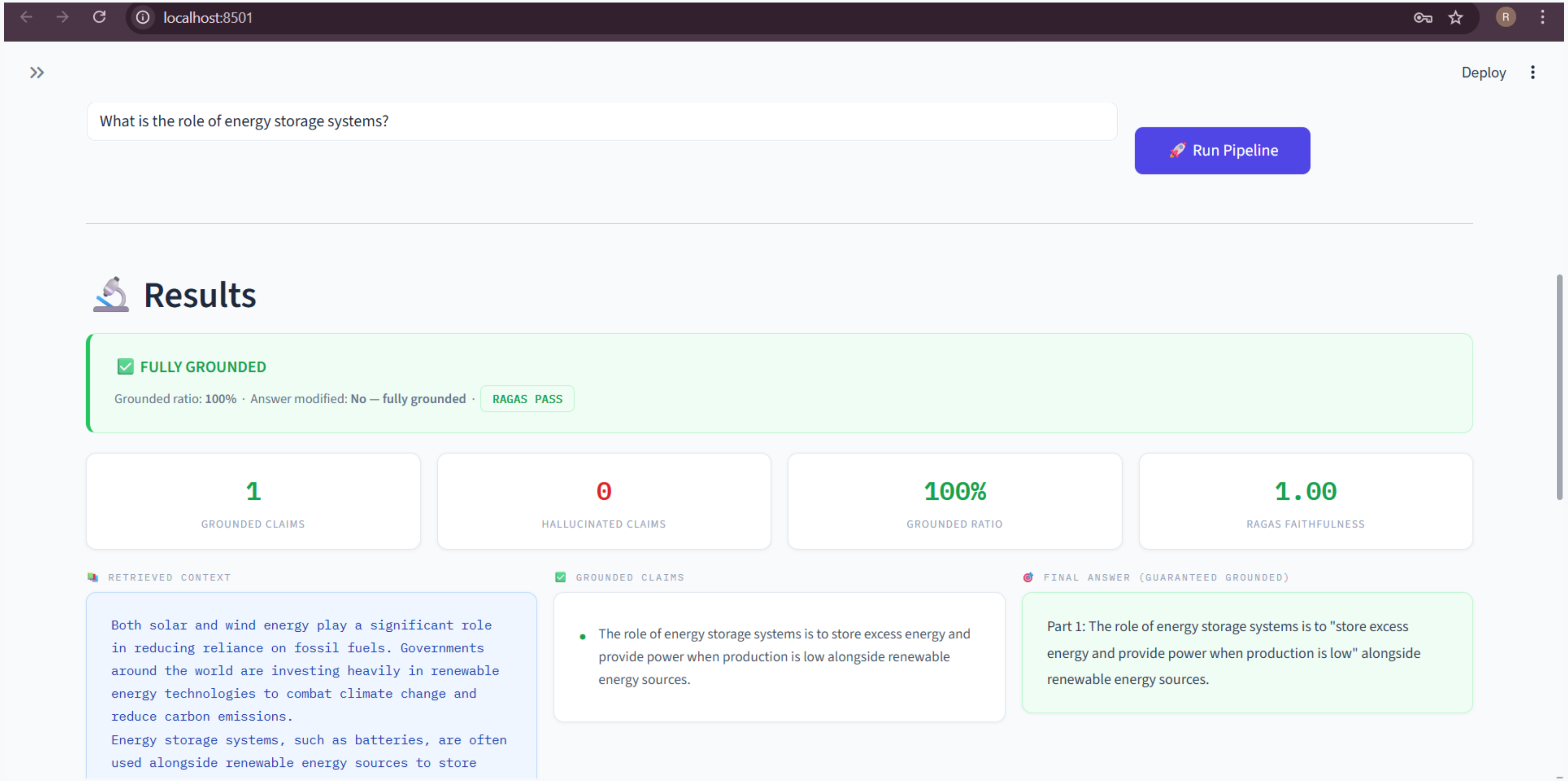
ID	Question	Expected Output	Actual Output	Result / Metrics
TC-FG-04	What role do batteries play in renewable energy?	Answer should state that batteries store excess energy and provide power when production is low.	The system returned the supported battery-storage role.	FULLY GROUNDED Grounded Ratio: 100% RAGAS: 1.00

Screenshot Evidence - TC-FG-04



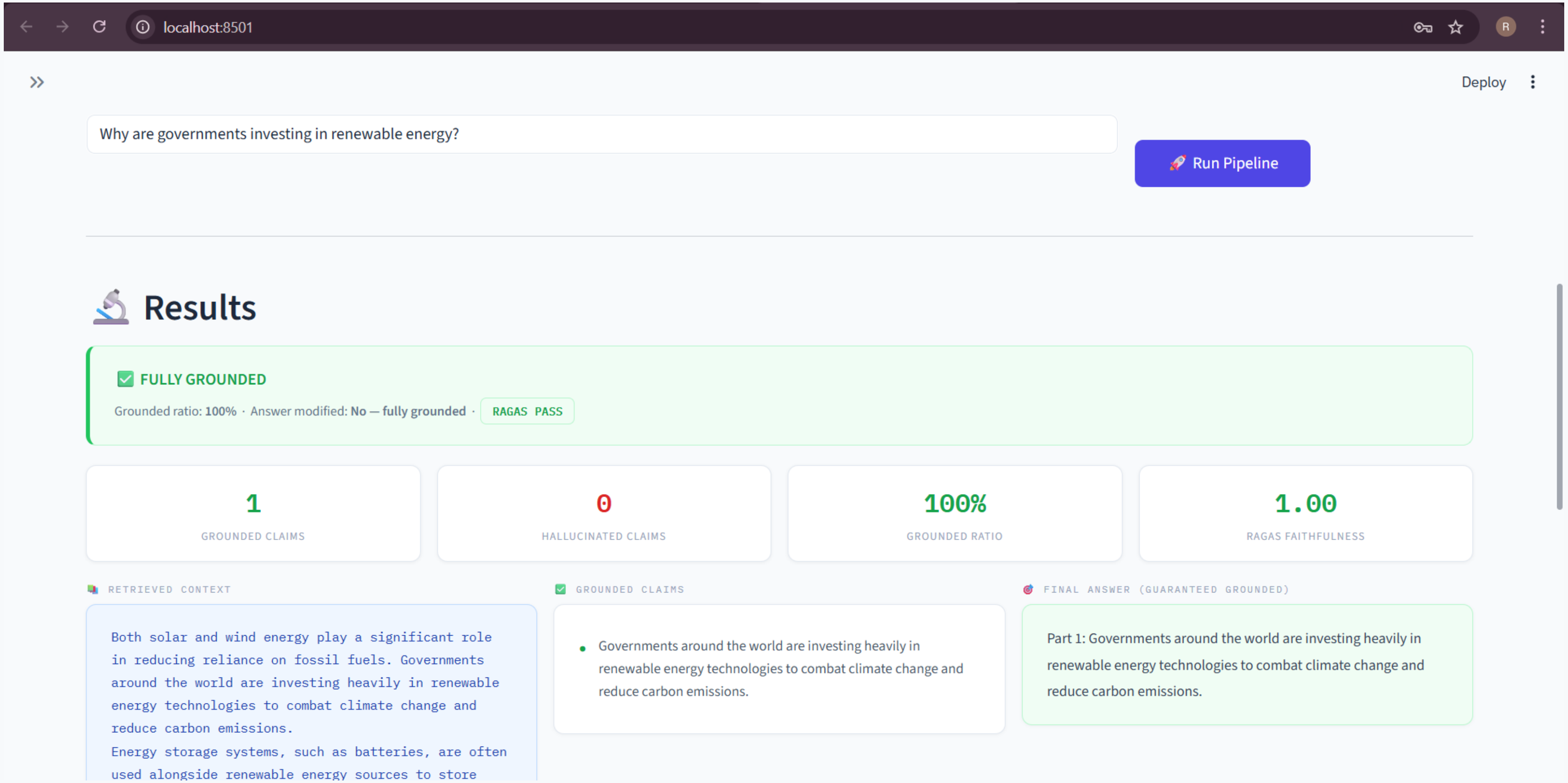
ID	Question	Expected Output	Actual Output	Result / Metrics
TC-FG-05	What is the role of energy storage systems?	Answer should state that energy storage systems store excess energy and provide power when production is low alongside renewable energy sources.	The system returned the supported storage-system role.	FULLY GROUNDED Grounded Ratio: 100% RAGAS: 1.00

Screenshot Evidence - TC-FG-05



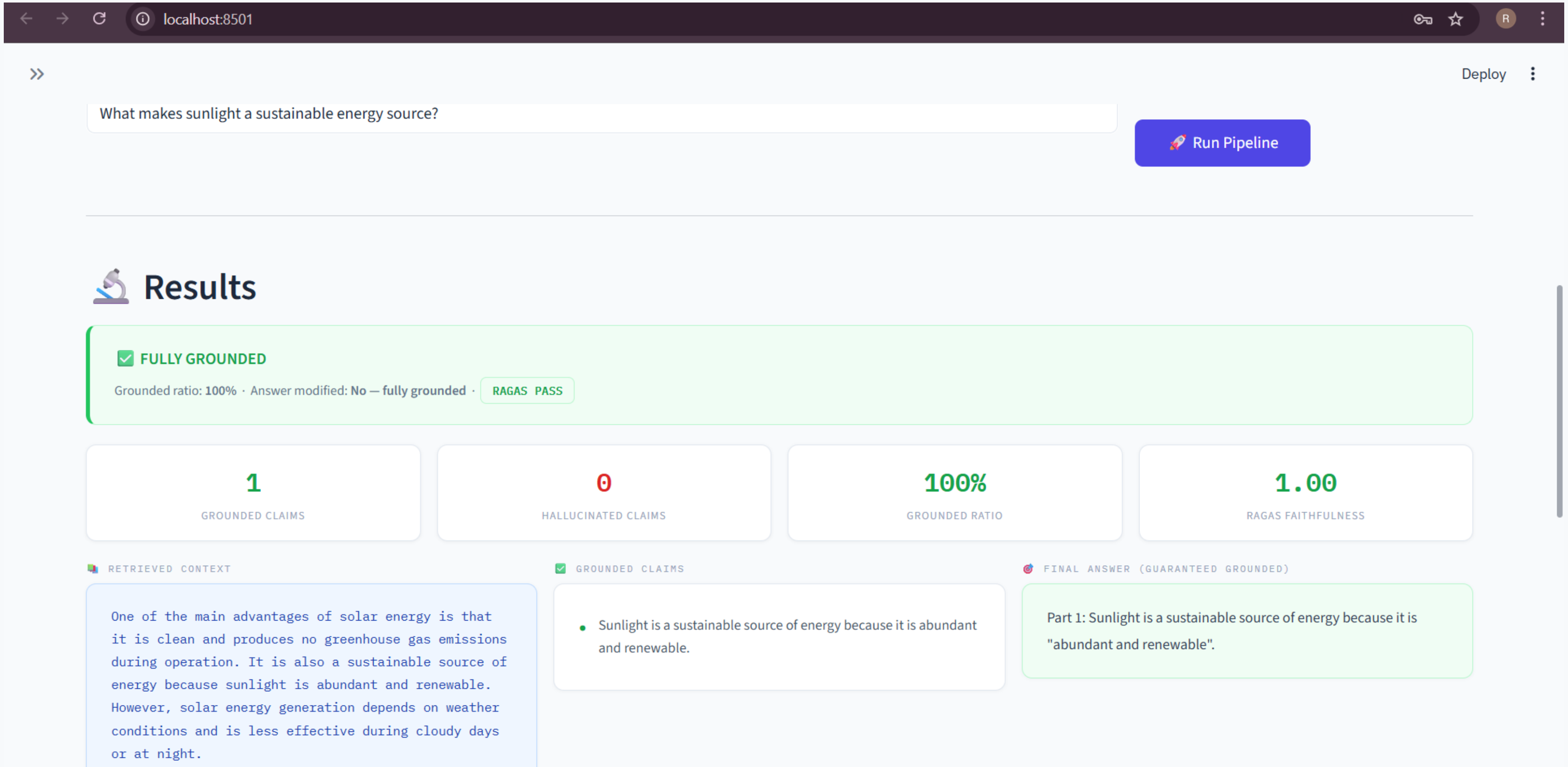
ID	Question	Expected Output	Actual Output	Result / Metrics
TC-FG-06	Why are governments investing in renewable energy?	Answer should state that governments are investing to combat climate change and reduce carbon emissions.	The system returned the same supported reason.	FULLY GROUNDED Grounded Ratio: 100% RAGAS: 1.00

Screenshot Evidence - TC-FG-06



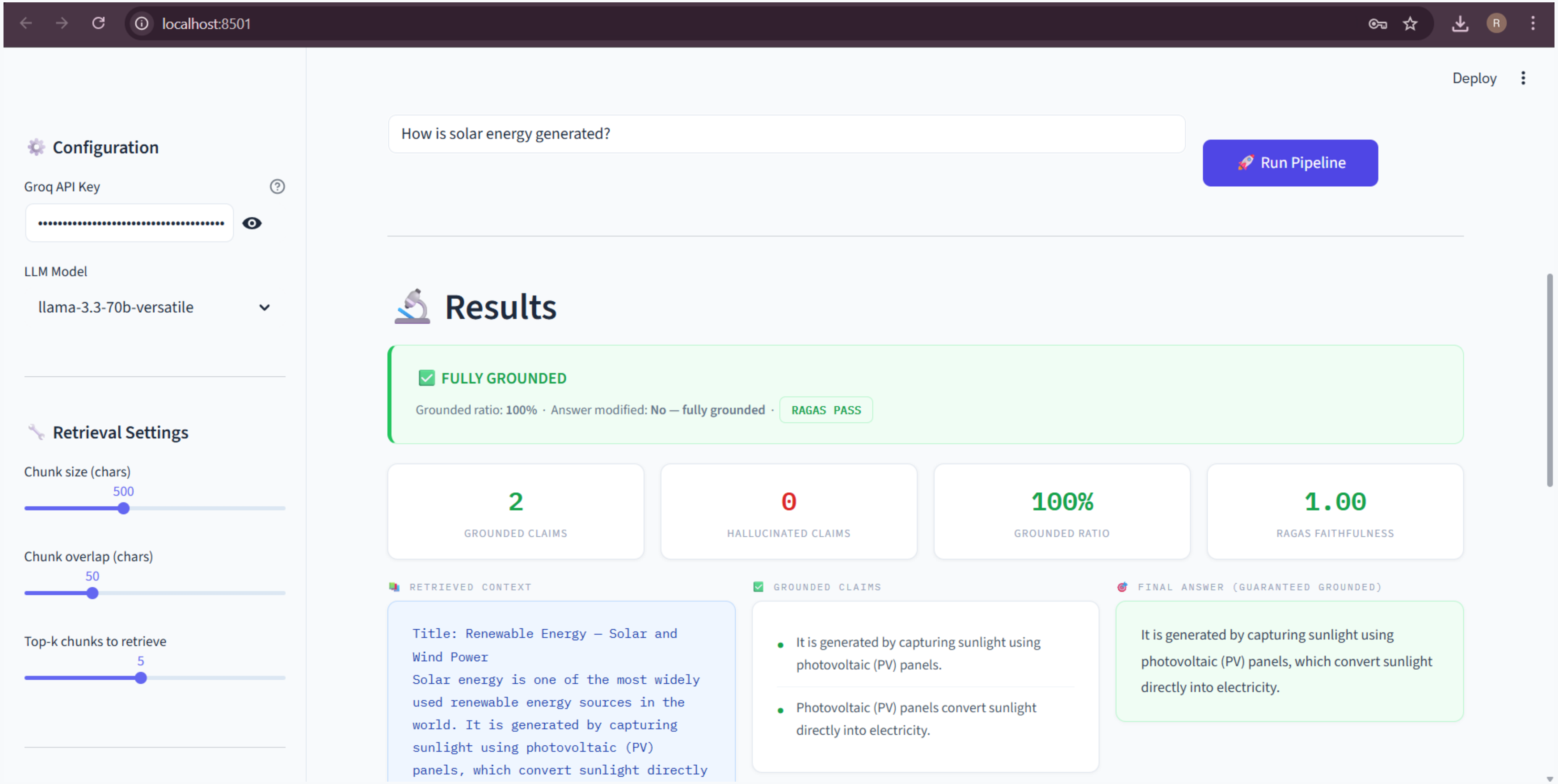
ID	Question	Expected Output	Actual Output	Result / Metrics
TC-FG-07	What makes sunlight a sustainable energy source?	Answer should state that sunlight is abundant and renewable.	The system returned that sunlight is sustainable because it is abundant and renewable.	FULLY GROUNDED Grounded Ratio: 100% RAGAS: 1.00

Screenshot Evidence - TC-FG-07



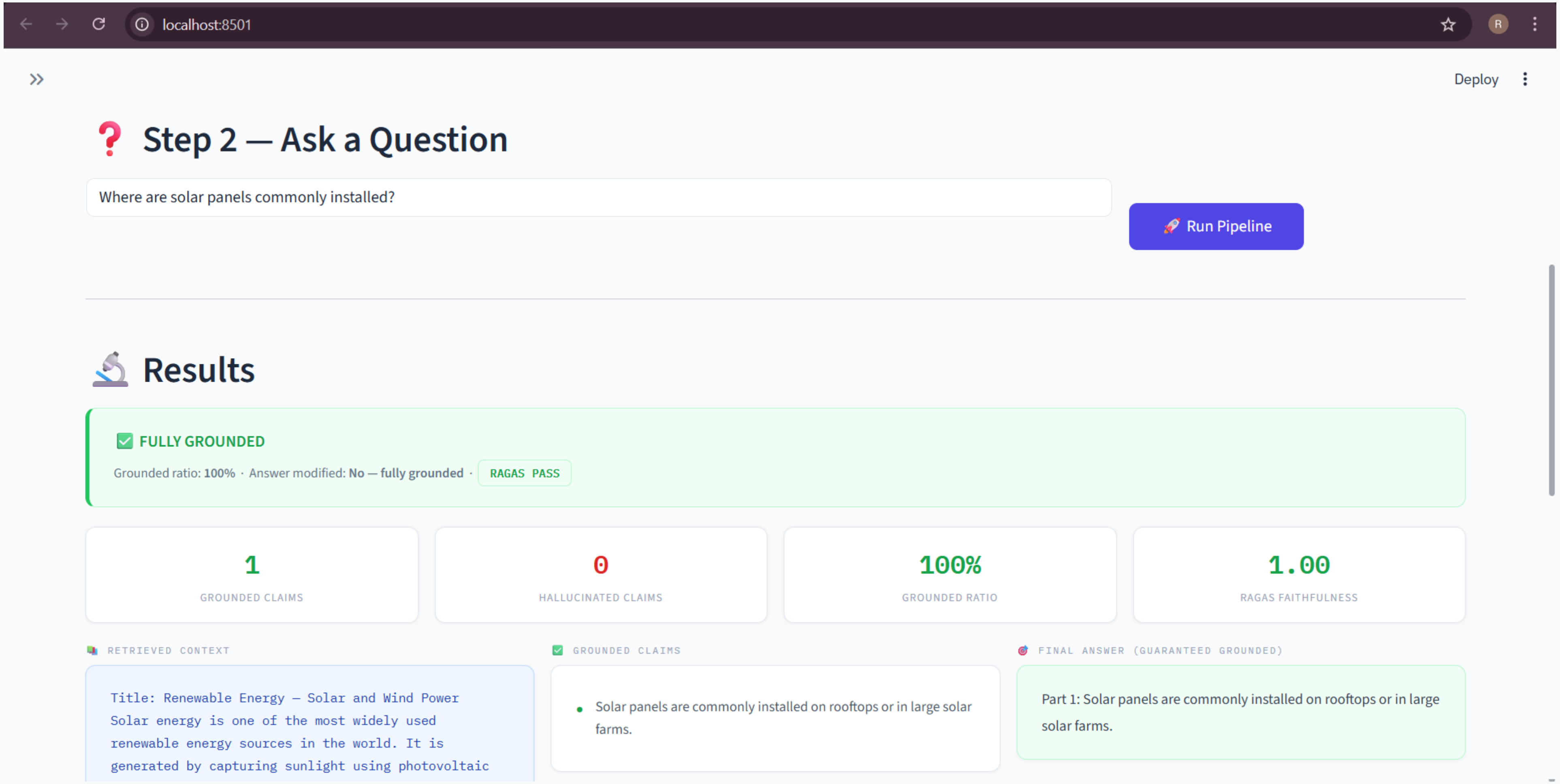
ID	Question	Expected Output	Actual Output	Result / Metrics
TC-FG-08	How is solar energy generated?	Answer should state that solar energy is generated by capturing sunlight using PV panels, which convert sunlight directly into electricity.	The system returned the supported generation explanation.	FULLY GROUNDED Grounded Ratio: 100% RAGAS: 1.00

Screenshot Evidence - TC-FG-08



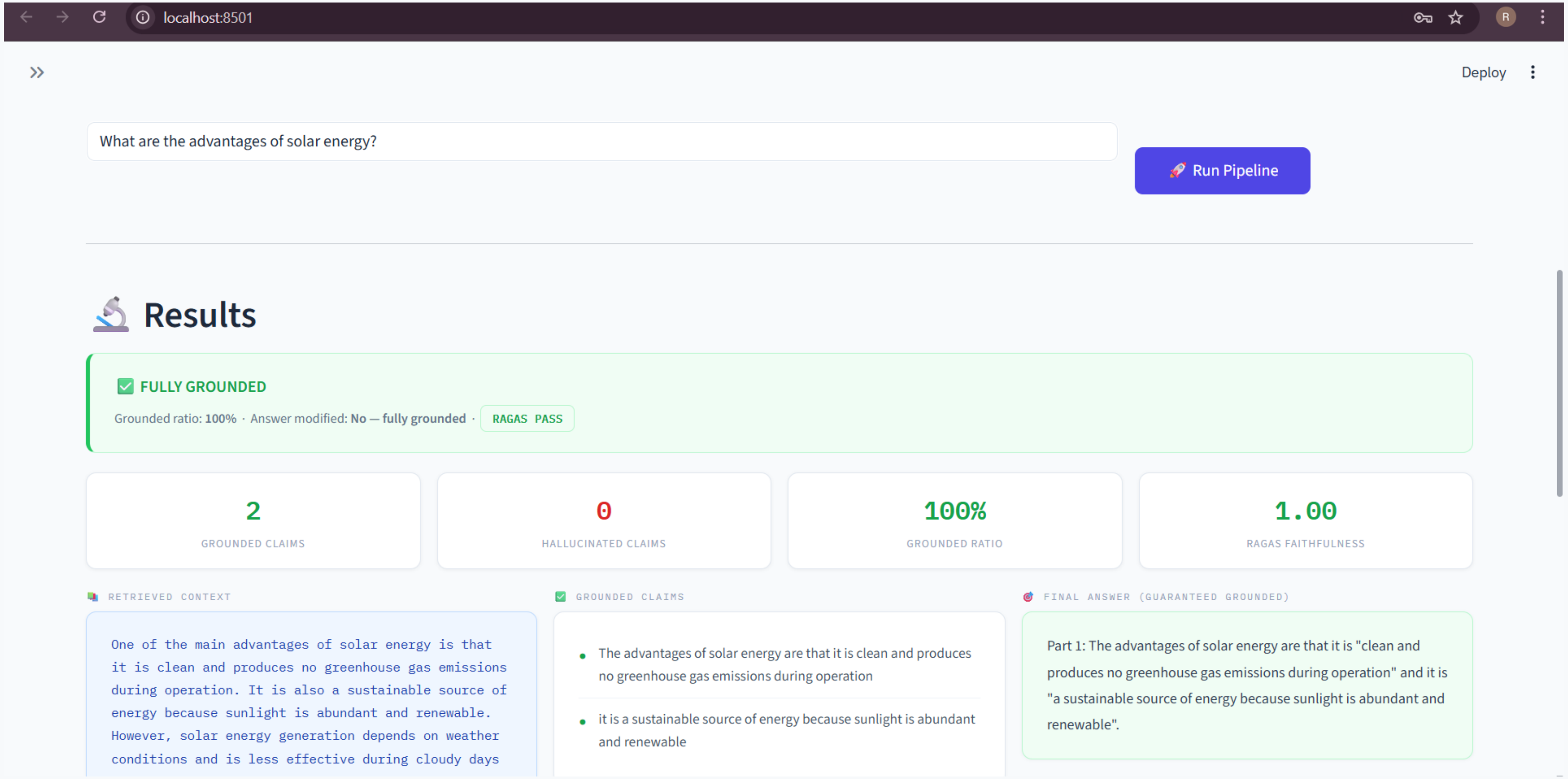
ID	Question	Expected Output	Actual Output	Result / Metrics
TC-FG-09	Where are solar panels commonly installed?	Answer should state that solar panels are commonly installed on rooftops or in large solar farms.	The system returned the supported installation locations.	FULLY GROUNDED Grounded Ratio: 100% RAGAS: 1.00

Screenshot Evidence - TC-FG-09



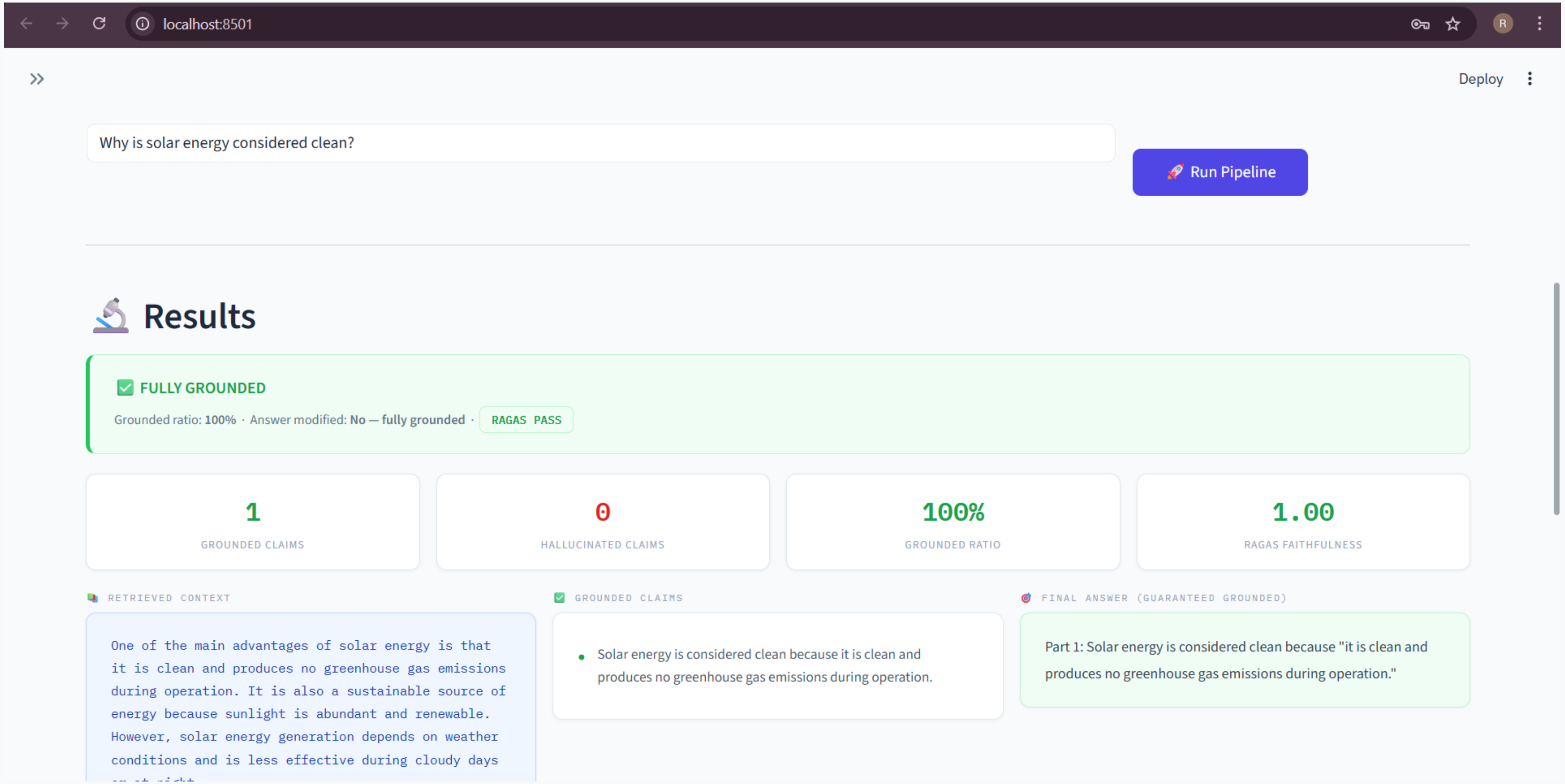
ID	Question	Expected Output	Actual Output	Result / Metrics
TC-FG-10	What are the advantages of solar energy?	Answer should mention only supported advantages: clean operation with no greenhouse gas emissions and sustainability because sunlight is abundant and renewable.	The system returned the supported advantages.	FULLY GROUNDED Grounded Ratio: 100% RAGAS: 1.00

Screenshot Evidence - TC-FG-10



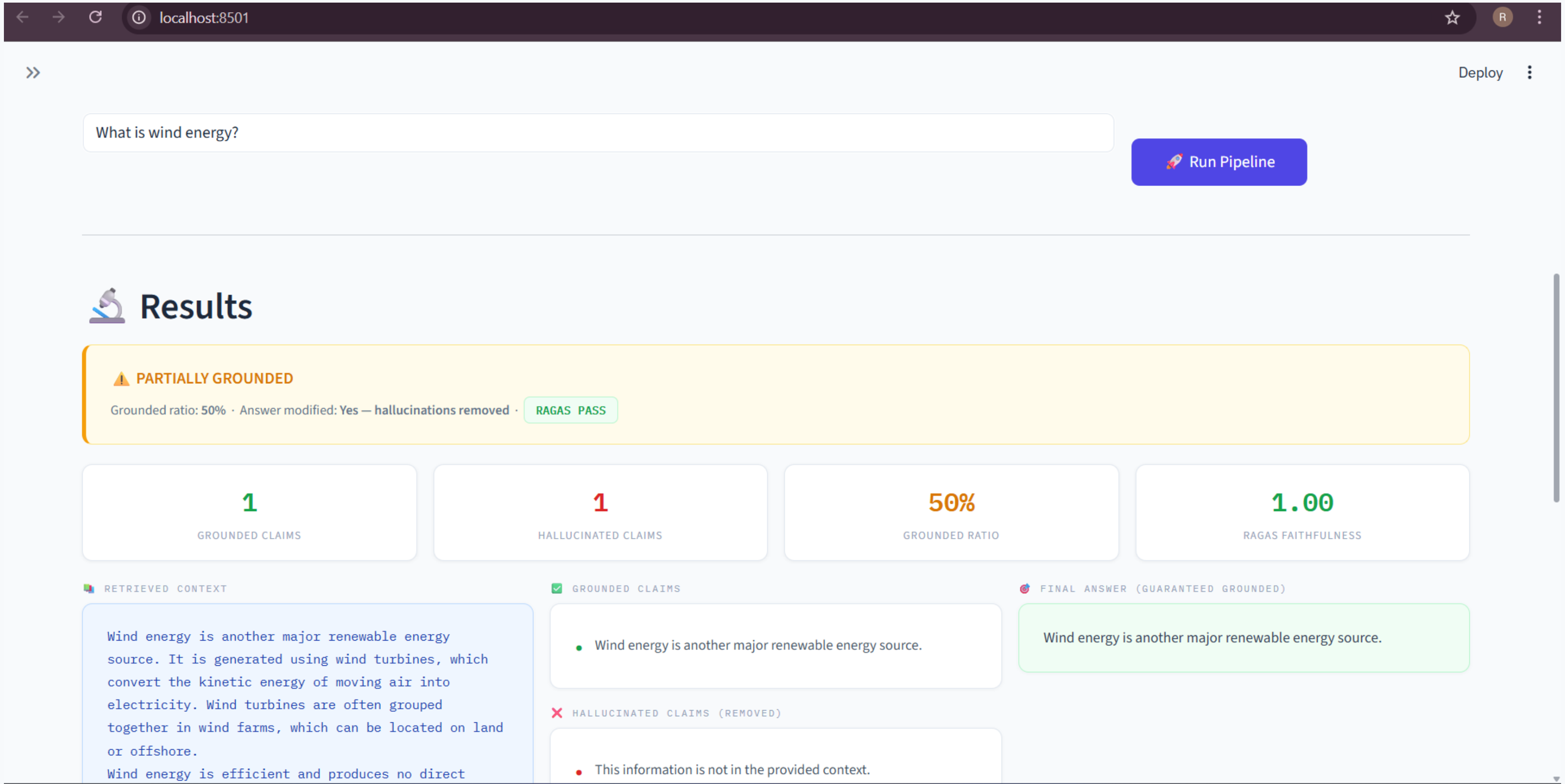
ID	Question	Expected Output	Actual Output	Result / Metrics
TC-FG-11	Why is solar energy considered clean?	Answer should state that solar energy produces no greenhouse gas emissions during operation.	The system returned the supported clean-energy reason.	FULLY GROUNDED Grounded Ratio: 100% RAGAS: 1.00

Screenshot Evidence - TC-FG-11



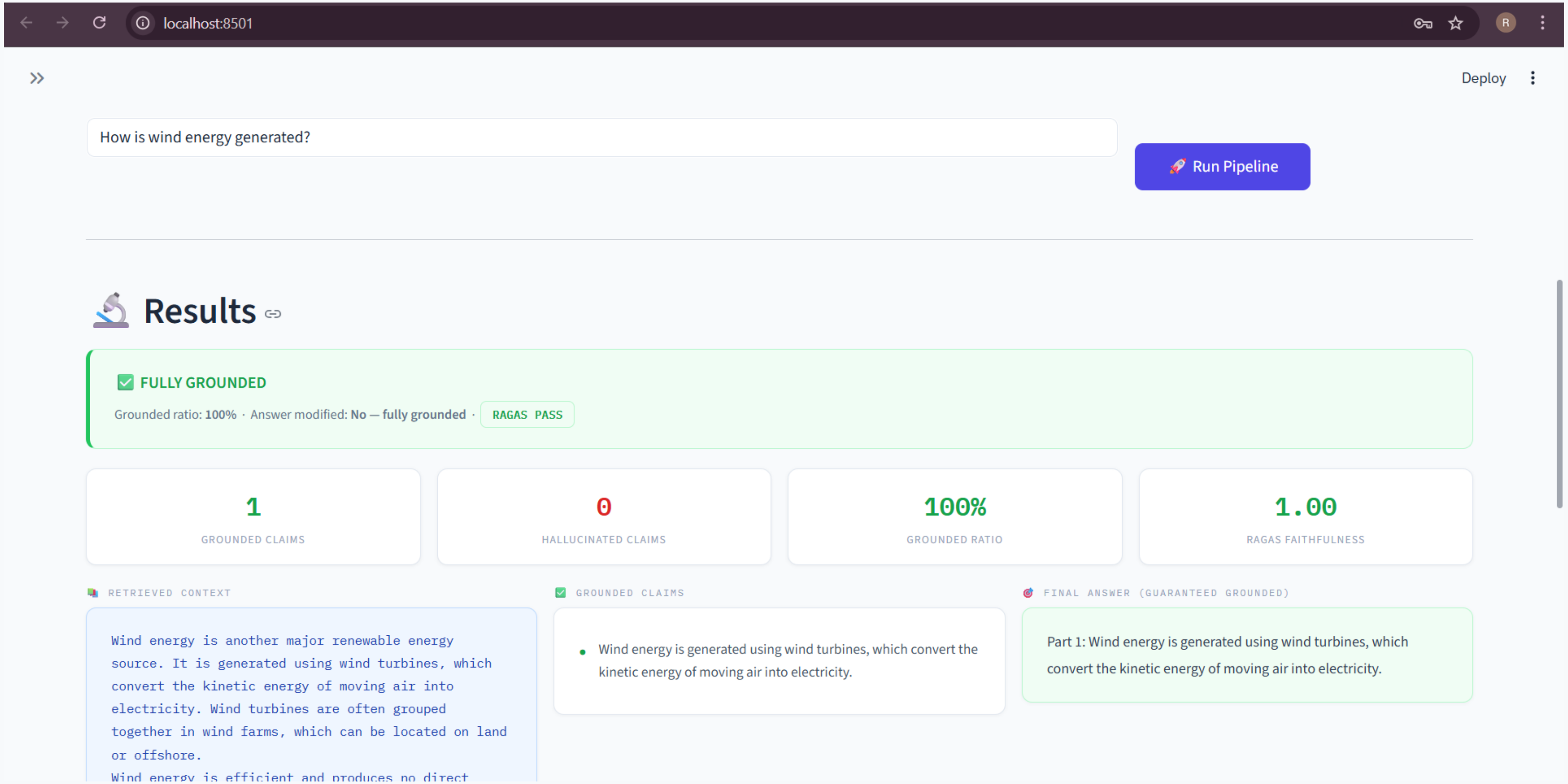
ID	Question	Expected Output	Actual Output	Result / Metrics
TC-FG-12	What is wind energy?	Answer should remain within the document and describe wind energy as a major renewable energy source generated using wind turbines.	The system kept the supported claim that wind energy is a major renewable source and removed an unsupported claim.	PARTIALLY GROUNDED - hallucinations removed Grounded Ratio: 50% RAGAS: 1.00

Screenshot Evidence - TC-FG-12



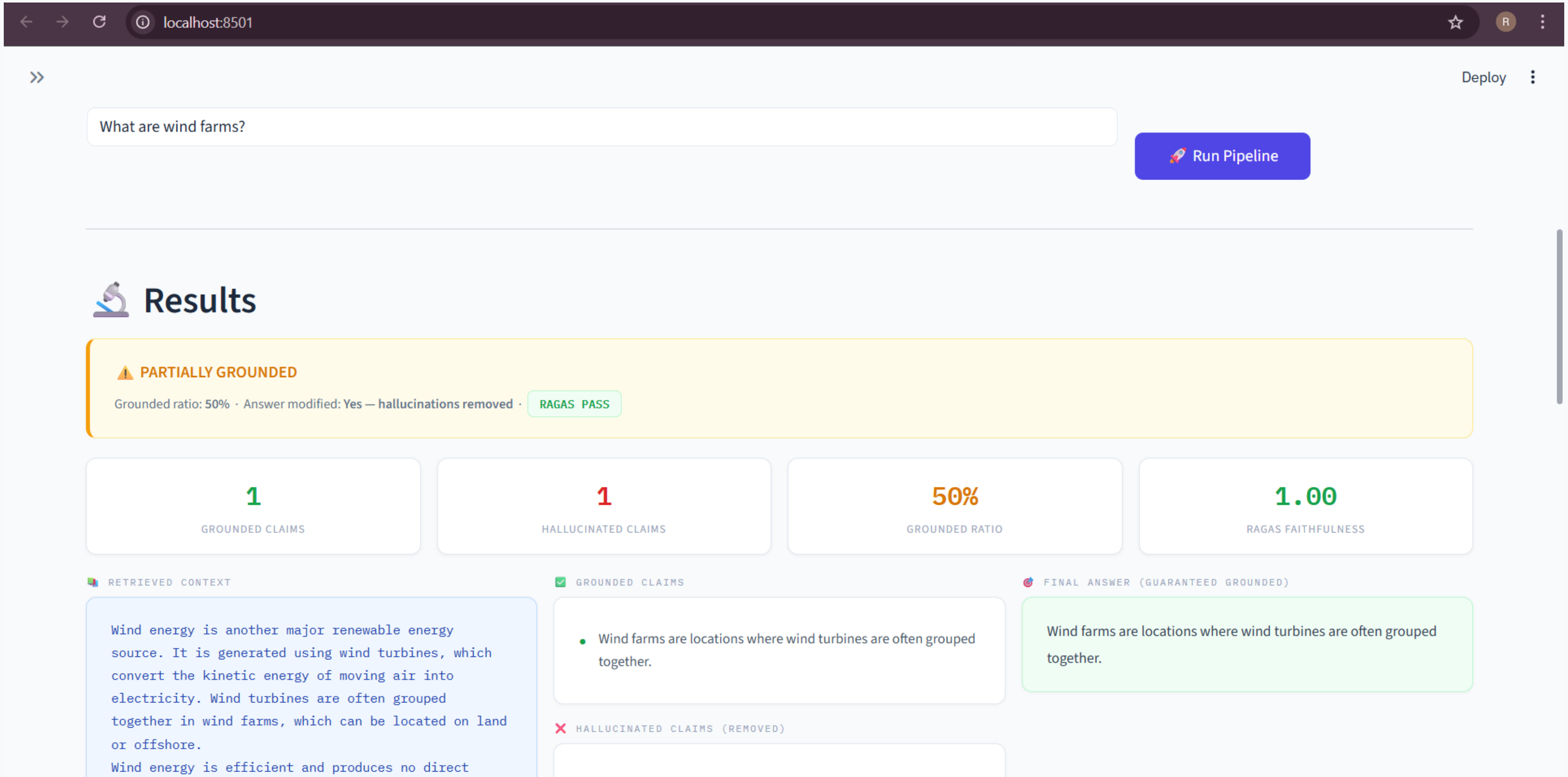
ID	Question	Expected Output	Actual Output	Result / Metrics
TC-FG-13	How is wind energy generated?	Answer should state that wind energy is generated using wind turbines that convert the kinetic energy of moving air into electricity.	The system returned the supported wind-energy generation explanation.	FULLY GROUNDED Grounded Ratio: 100% RAGAS: 1.00

Screenshot Evidence - TC-FG-13



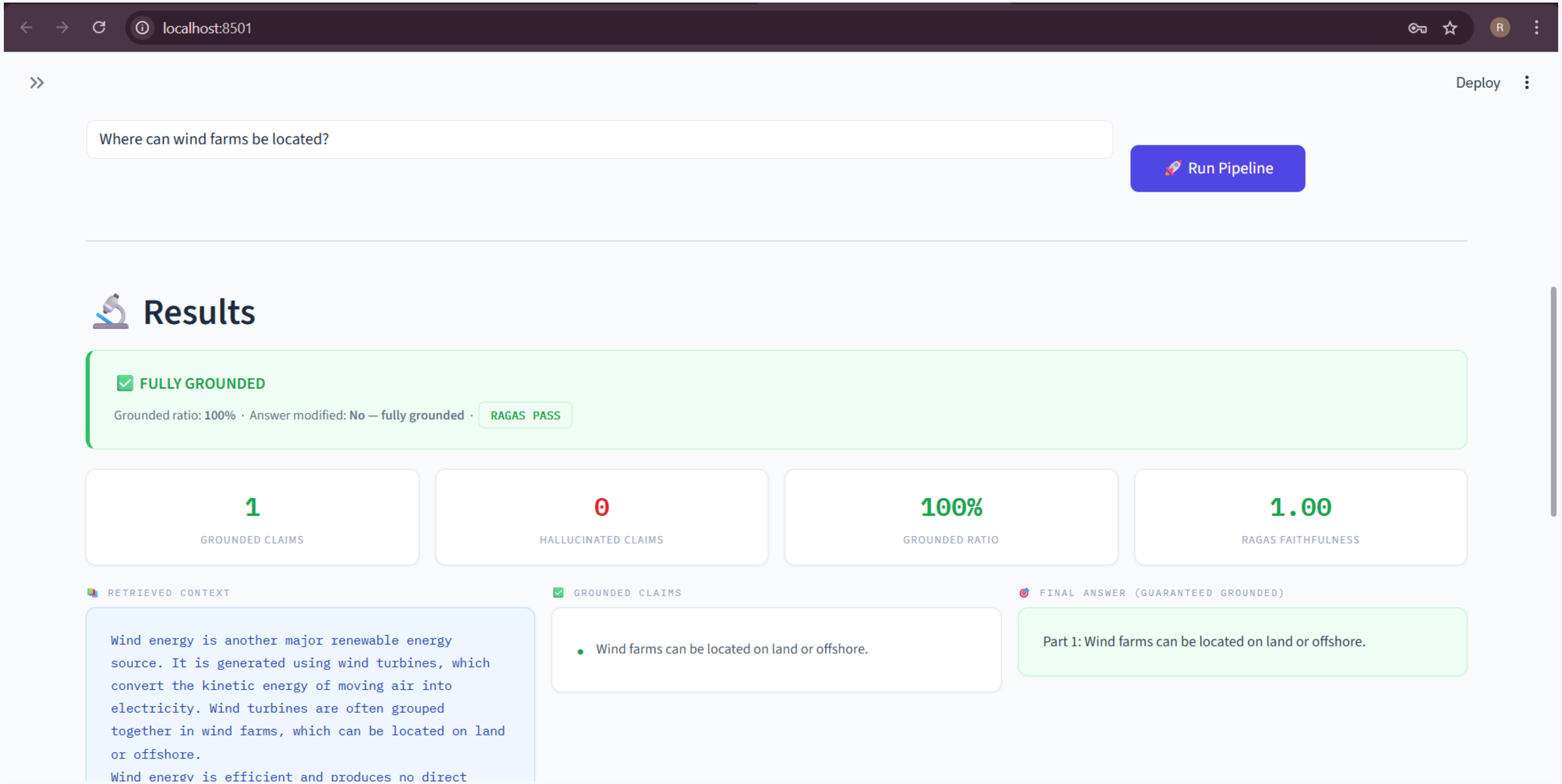
ID	Question	Expected Output	Actual Output	Result / Metrics
TC-FG-14	What are wind farms?	Answer should state that wind turbines are often grouped together in wind farms, which can be located on land or offshore.	The system kept the supported wind-farm grouping statement and removed unsupported wording.	PARTIALLY GROUNDED - hallucinations removed Grounded Ratio: 50% RAGAS: 1.00

Screenshot Evidence - TC-FG-14



ID	Question	Expected Output	Actual Output	Result / Metrics
TC-FG-15	Where can wind farms be located?	Answer should state that wind farms can be located on land or offshore.	The system returned the supported location statement.	FULLY GROUNDED Grounded Ratio: 100% RAGAS: 1.00

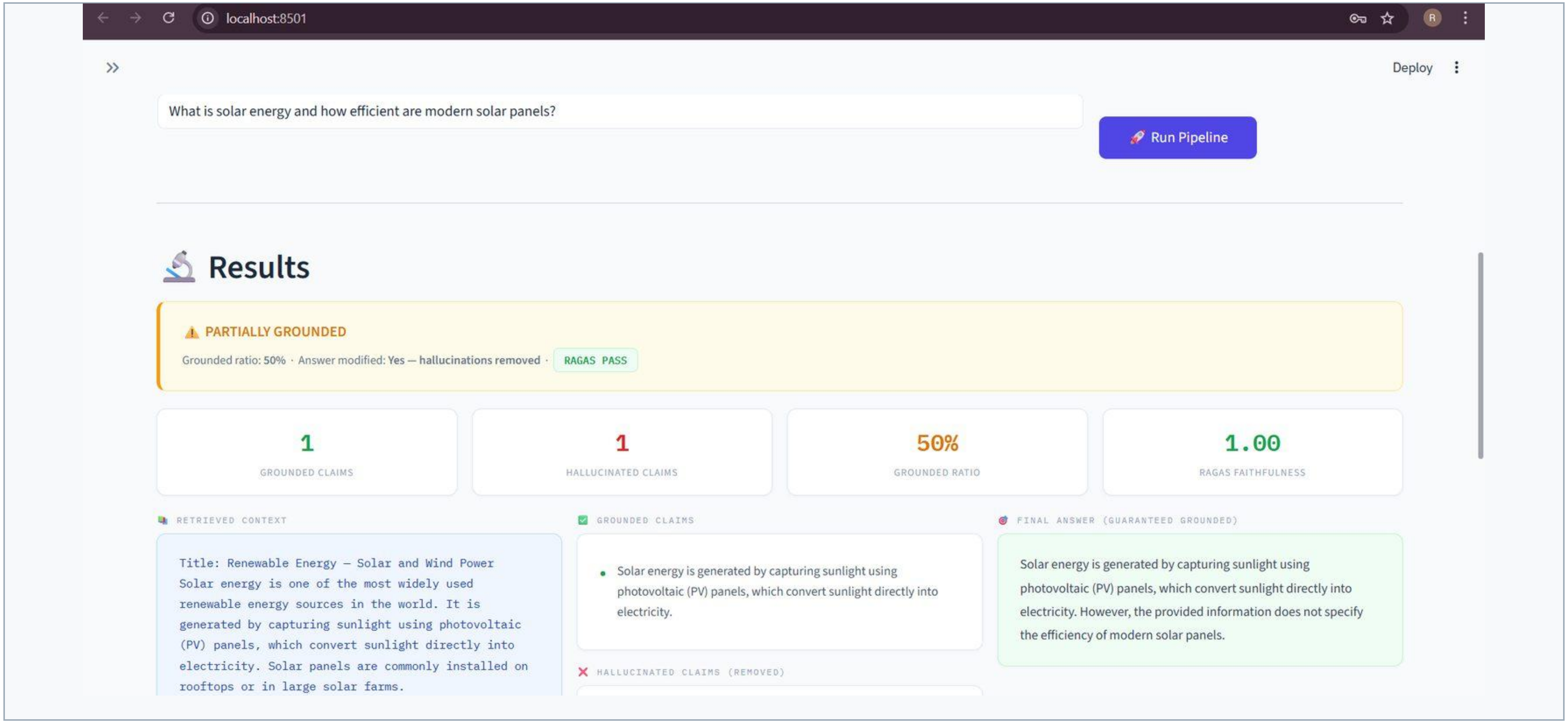
Screenshot Evidence - TC-FG-15



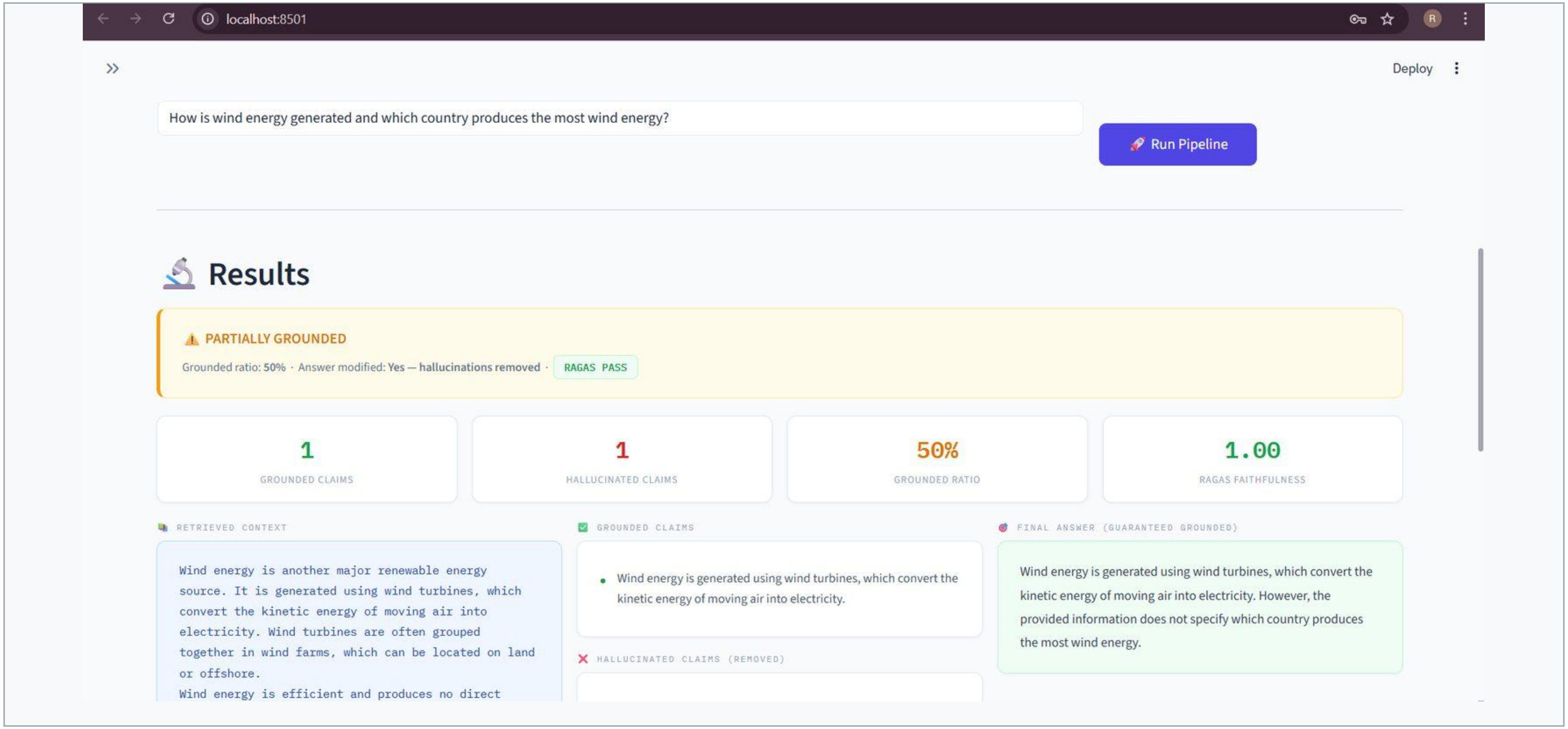
Category 2 - Partially Grounded Test Cases: 2-Part Questions

Questions with one supported part and one unsupported part.

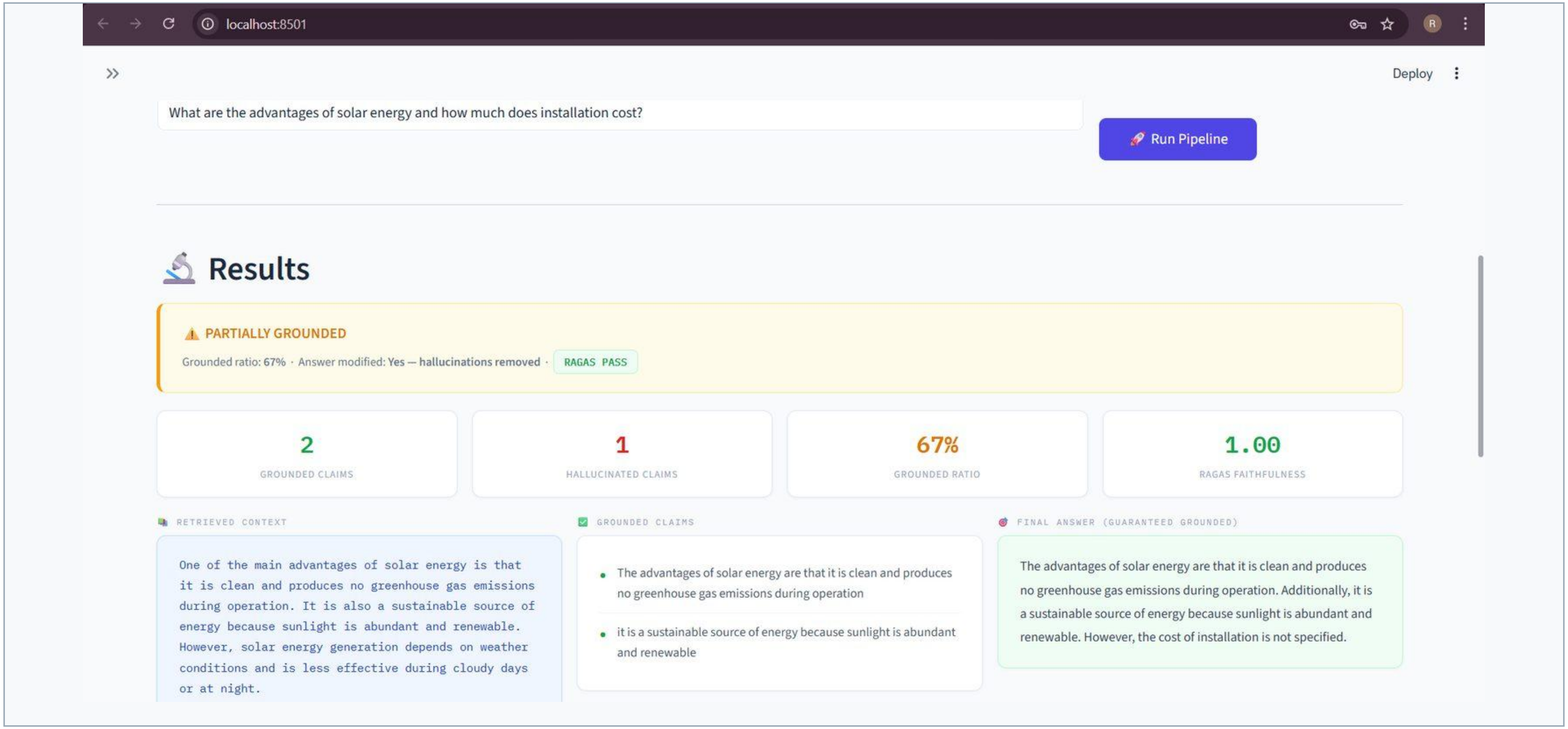
ID	Question	Expected Output	Actual Output	Result / Metrics
TC-PG2-01	What is solar energy and how efficient are modern solar panels?	Answer the supported solar-energy definition and state that modern panel efficiency is not specified in the context.	The system kept the PV-panel explanation and stated that modern panel efficiency is not specified.	PARTIALLY GROUNDED Grounded Ratio: 50% RAGAS: 1.00
Screenshot Evidence - TC-PG2-01				



ID	Question	Expected Output	Actual Output	Result / Metrics
TC-PG2-02	How is wind energy generated and which country produces the most wind energy?	Answer the supported wind-generation mechanism and state that the country producing the most wind energy is not specified.	The system kept the wind-turbine generation explanation and stated that the country detail is not specified.	PARTIALLY GROUNDED Grounded Ratio: 50% RAGAS: 1.00
Screenshot Evidence - TC-PG2-02				

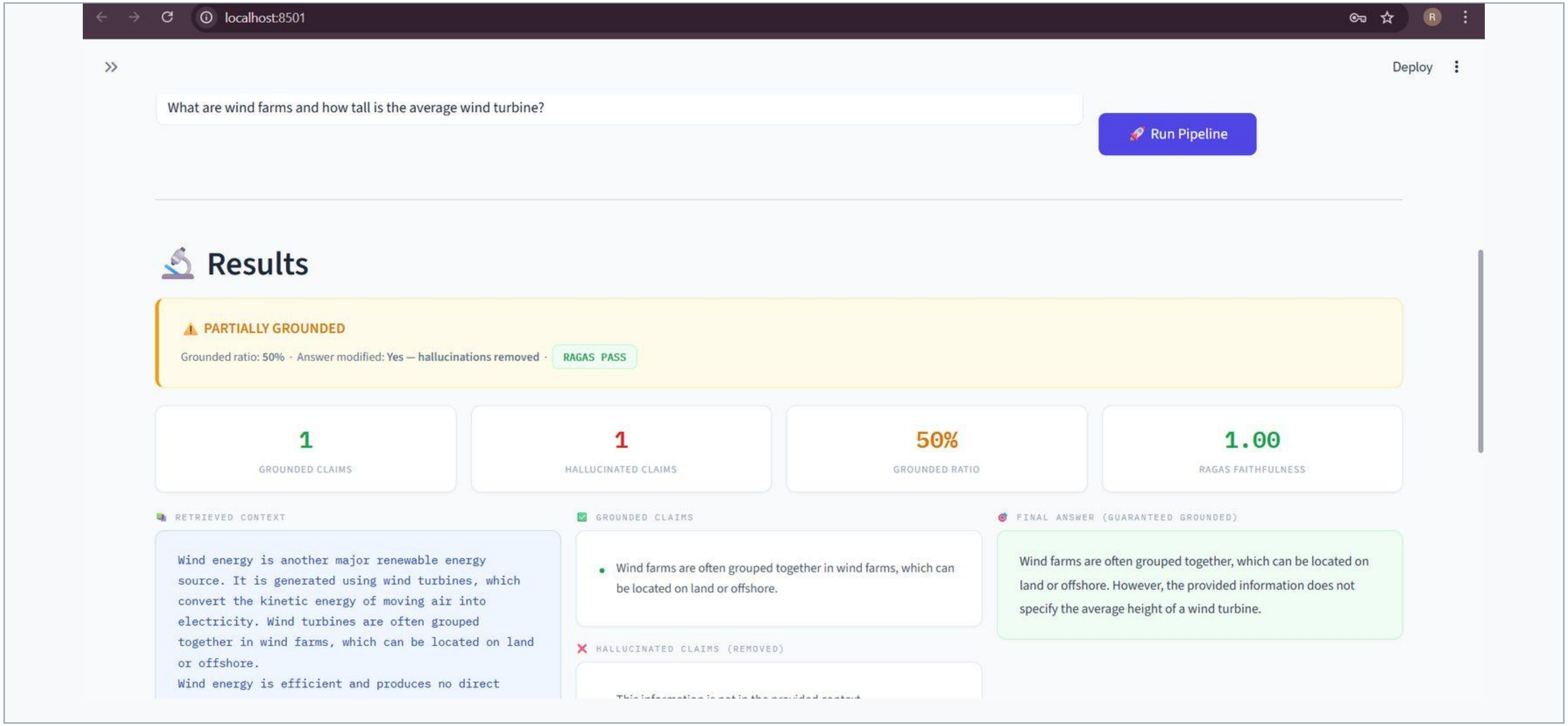


ID	Question	Expected Output	Actual Output	Result / Metrics
TC-PG2-03	What are the advantages of solar energy and how much does installation cost?	Answer the supported solar-energy advantages and state that installation cost is not specified.	The system returned clean/no-emissions and sustainable sunlight advantages, then stated cost is not specified.	PARTIALLY GROUNDED Grounded Ratio: 67% RAGAS: 1.00
Screenshot Evidence - TC-PG2-03				

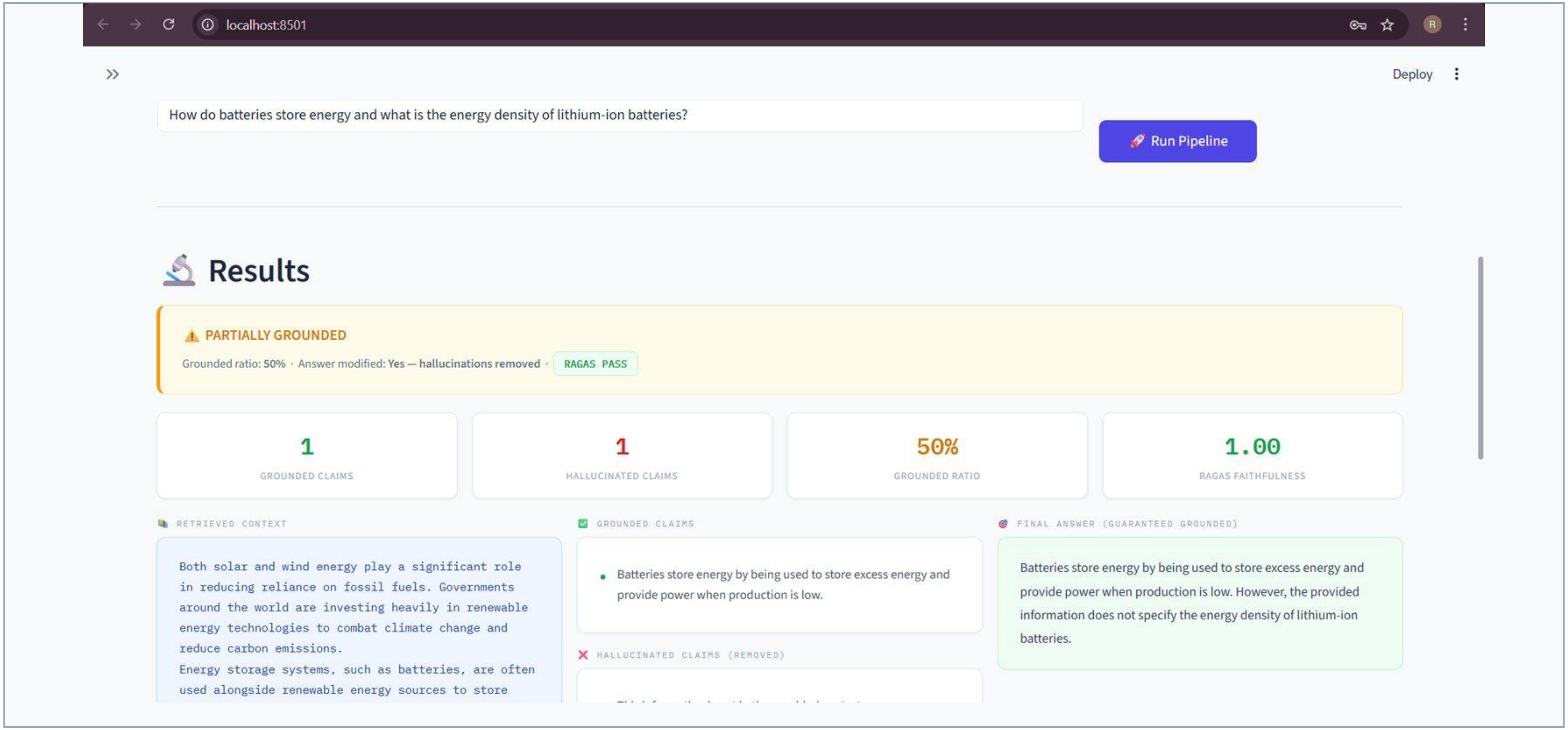


ID	Question	Expected Output	Actual Output	Result / Metrics
TC-PG2-04	What are wind farms and how tall is the average wind turbine?	Answer the supported wind-farm description and state that average turbine height is not specified.	The system kept the wind-farm location/grouping information and stated average height is not specified.	PARTIALLY GROUNDED Grounded Ratio: 50% RAGAS: 1.00

Screenshot Evidence - TC-PG2-04

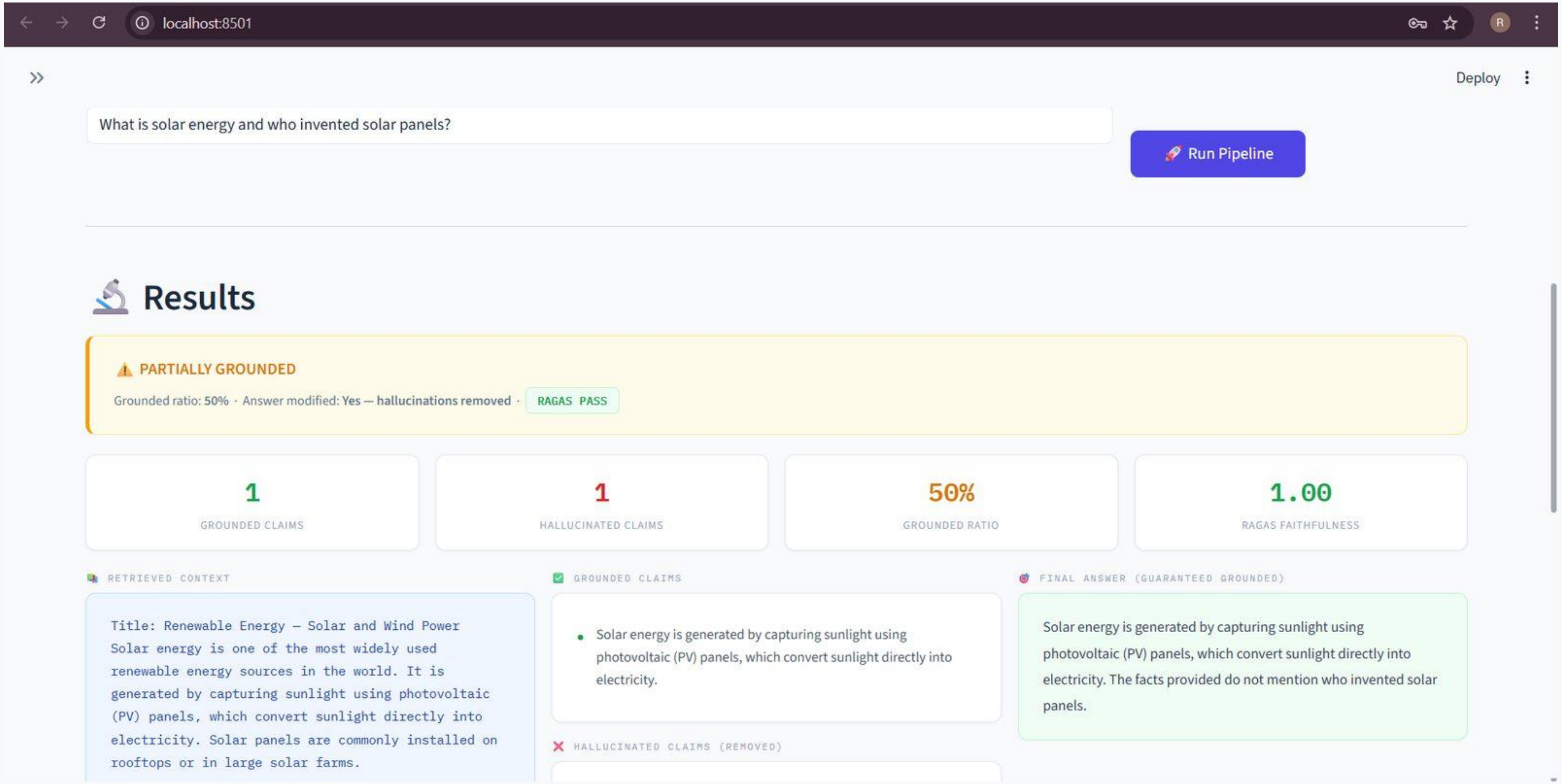


ID	Question	Expected Output	Actual Output	Result / Metrics
TC-PG2-05	How do batteries store energy and what is the energy density of lithium-ion batteries?	Answer the supported battery storage role and state that lithium-ion energy density is not specified.	The system kept the energy-storage role and stated that lithium-ion energy density is not specified.	PARTIALLY GROUNDED Grounded Ratio: 50% RAGAS: 1.00
Screenshot Evidence - TC-PG2-05				



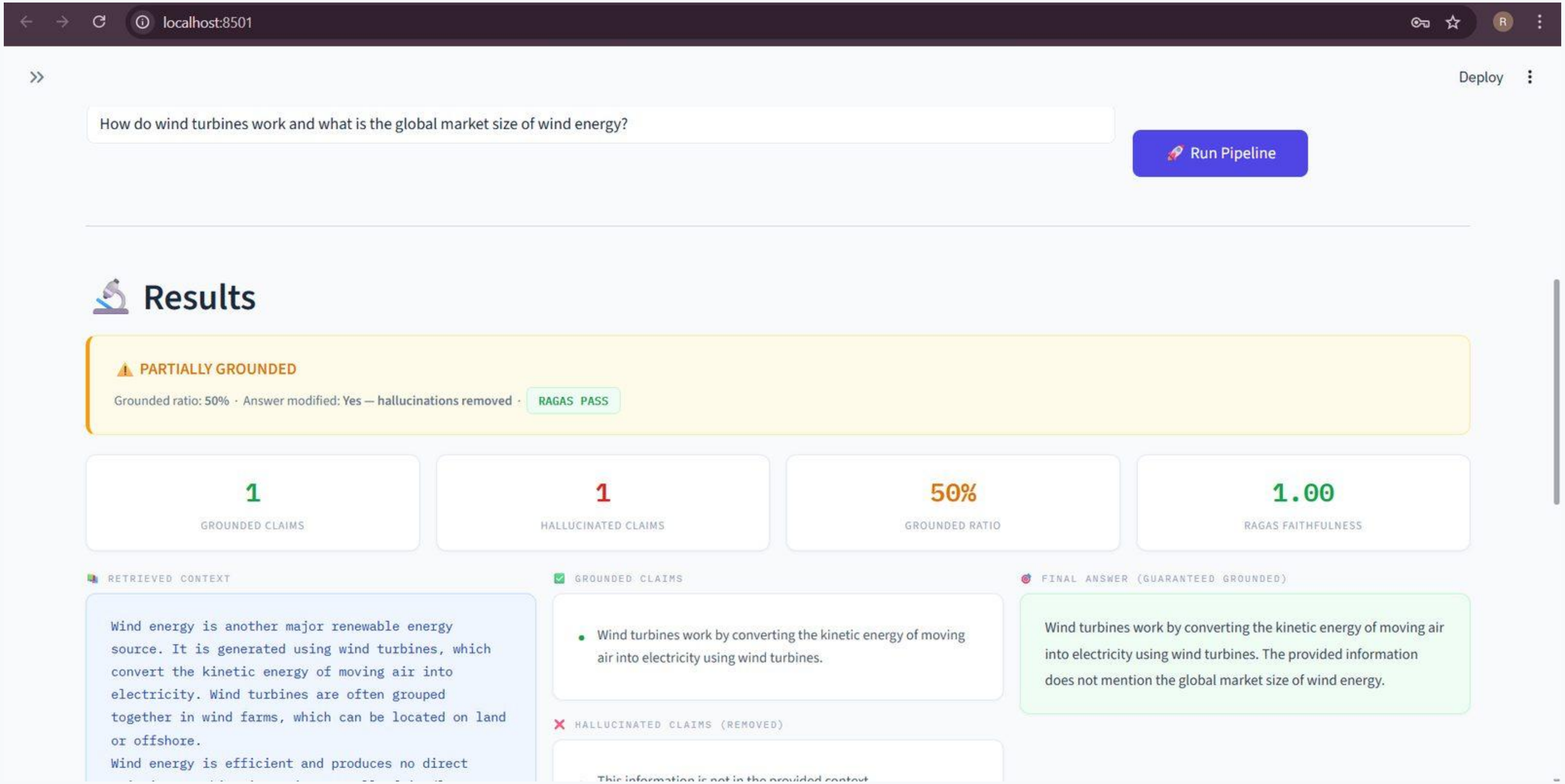
ID	Question	Expected Output	Actual Output	Result / Metrics
TC-PG2-06	What is solar energy and who invented solar panels?	Answer the supported solar-energy definition and state that the inventor of solar panels is not mentioned.	The system kept the PV-panel explanation and stated that the provided facts do not mention who invented solar panels.	PARTIALLY GROUNDED Grounded Ratio: 50% RAGAS: 1.00

Screenshot Evidence - TC-PG2-06



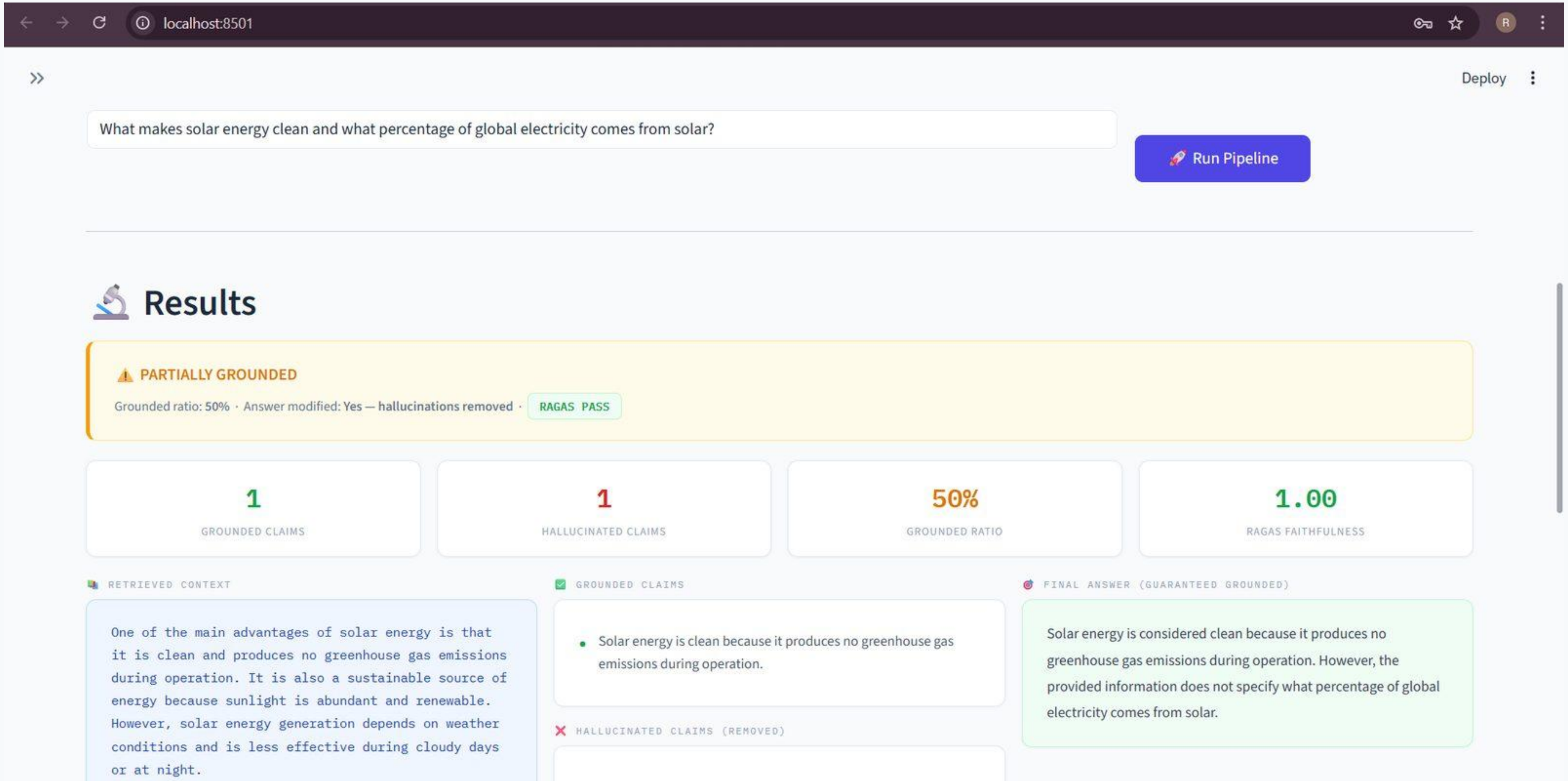
ID	Question	Expected Output	Actual Output	Result / Metrics
TC-PG2-07	How do wind turbines work and what is the global market size of wind energy?	Answer the supported turbine mechanism and state that global wind-energy market size is not mentioned.	The system kept the kinetic-energy conversion explanation and stated that market size is not mentioned.	PARTIALLY GROUNDED Grounded Ratio: 50% RAGAS: 1.00

Screenshot Evidence - TC-PG2-07



ID	Question	Expected Output	Actual Output	Result / Metrics
TC-PG2-08	What makes solar energy clean and what percentage of global electricity comes from solar?	Answer the supported clean-energy reason and state that the global electricity percentage is not specified.	The system kept the no-greenhouse-gas-emissions claim and stated the solar electricity percentage is not specified.	PARTIALLY GROUNDED Grounded Ratio: 50% RAGAS: 1.00

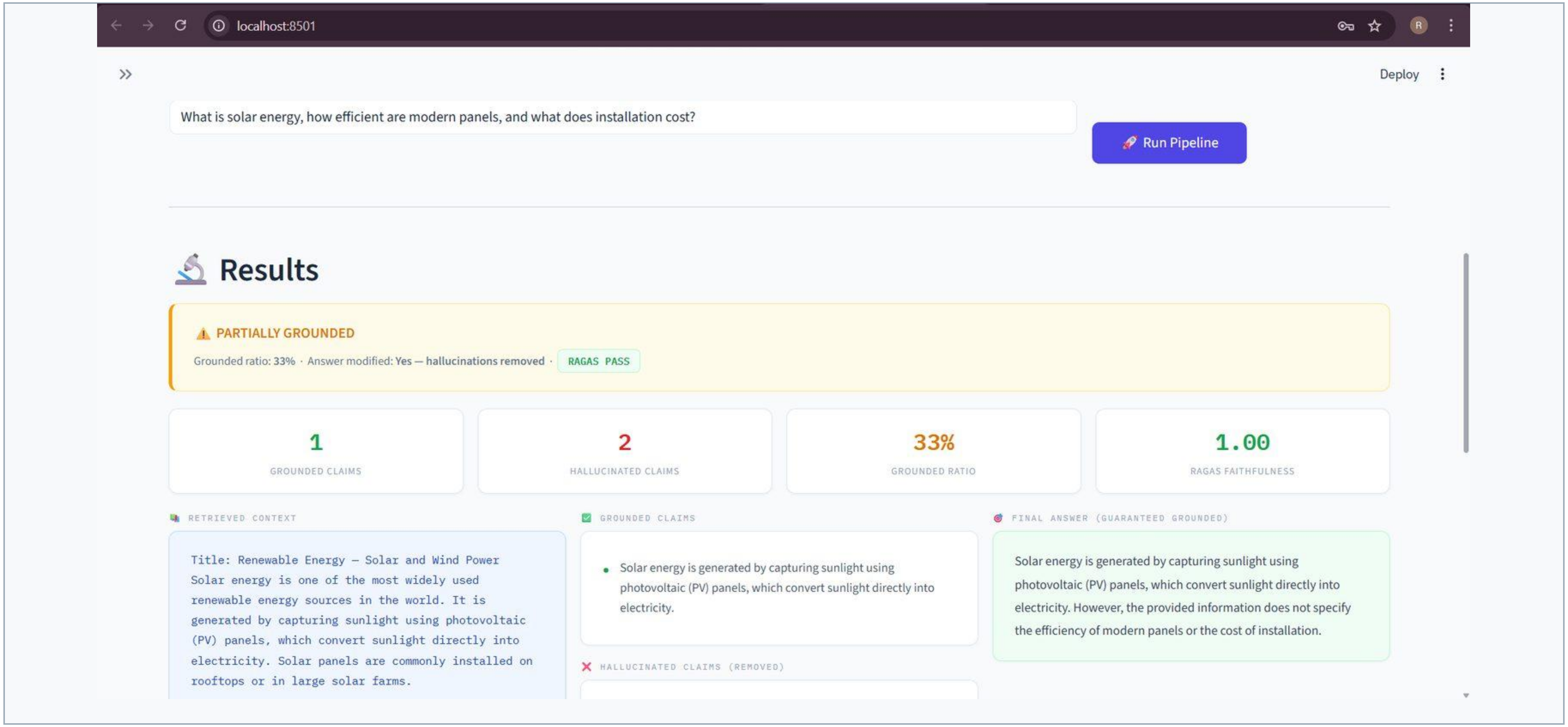
Screenshot Evidence - TC-PG2-08



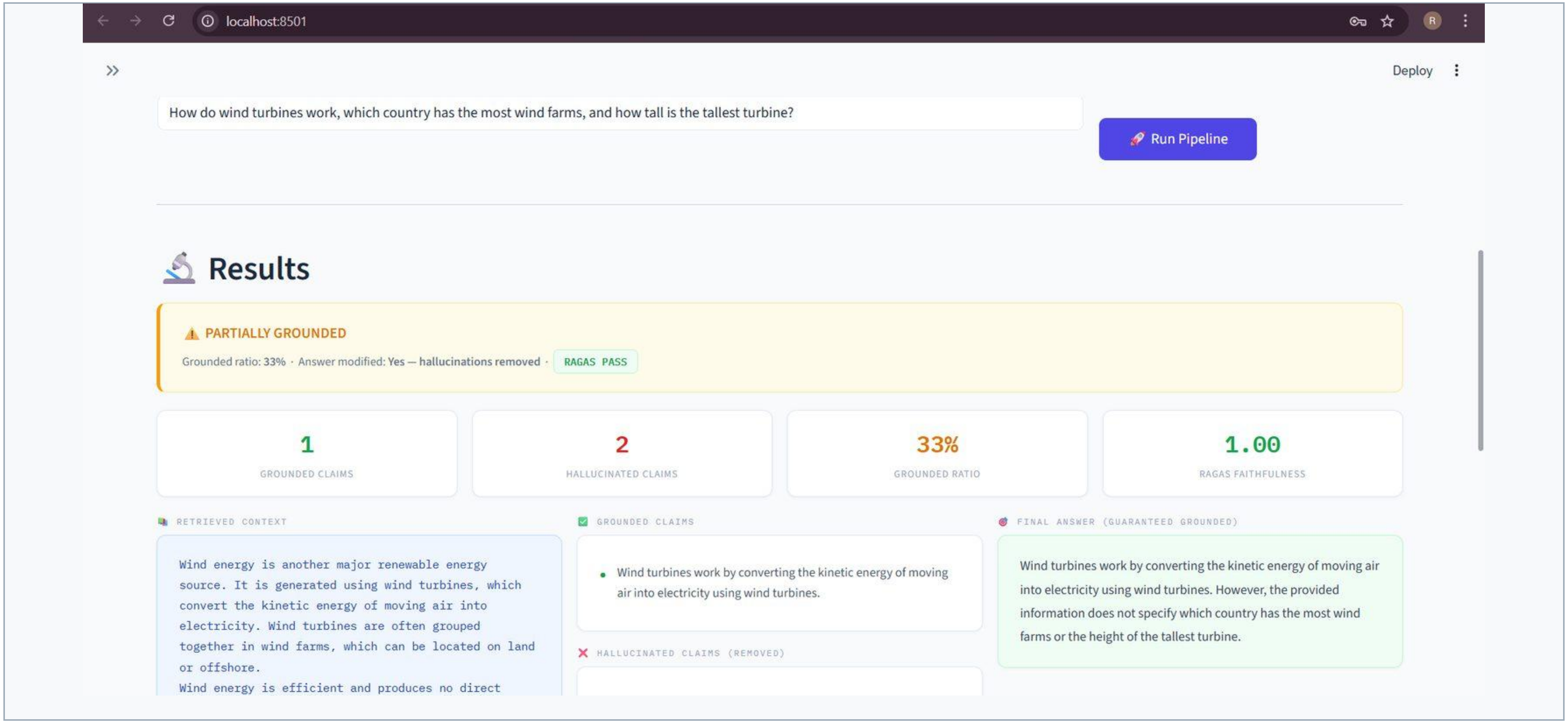
Category 2 - Partially Grounded Test Cases: 3-Part Questions

Questions with one or two supported parts and other unsupported details.

ID	Question	Expected Output	Actual Output	Result / Metrics
TC-PG3-01	What is solar energy, how efficient are modern panels, and what does installation cost?	Keep the supported solar-energy explanation and reject missing efficiency and installation-cost details.	The system kept the PV-panel explanation and stated that efficiency and cost are not specified.	PARTIALLY GROUNDED Grounded Ratio: 33% RAGAS: 1.00
Screenshot Evidence - TC-PG3-01				

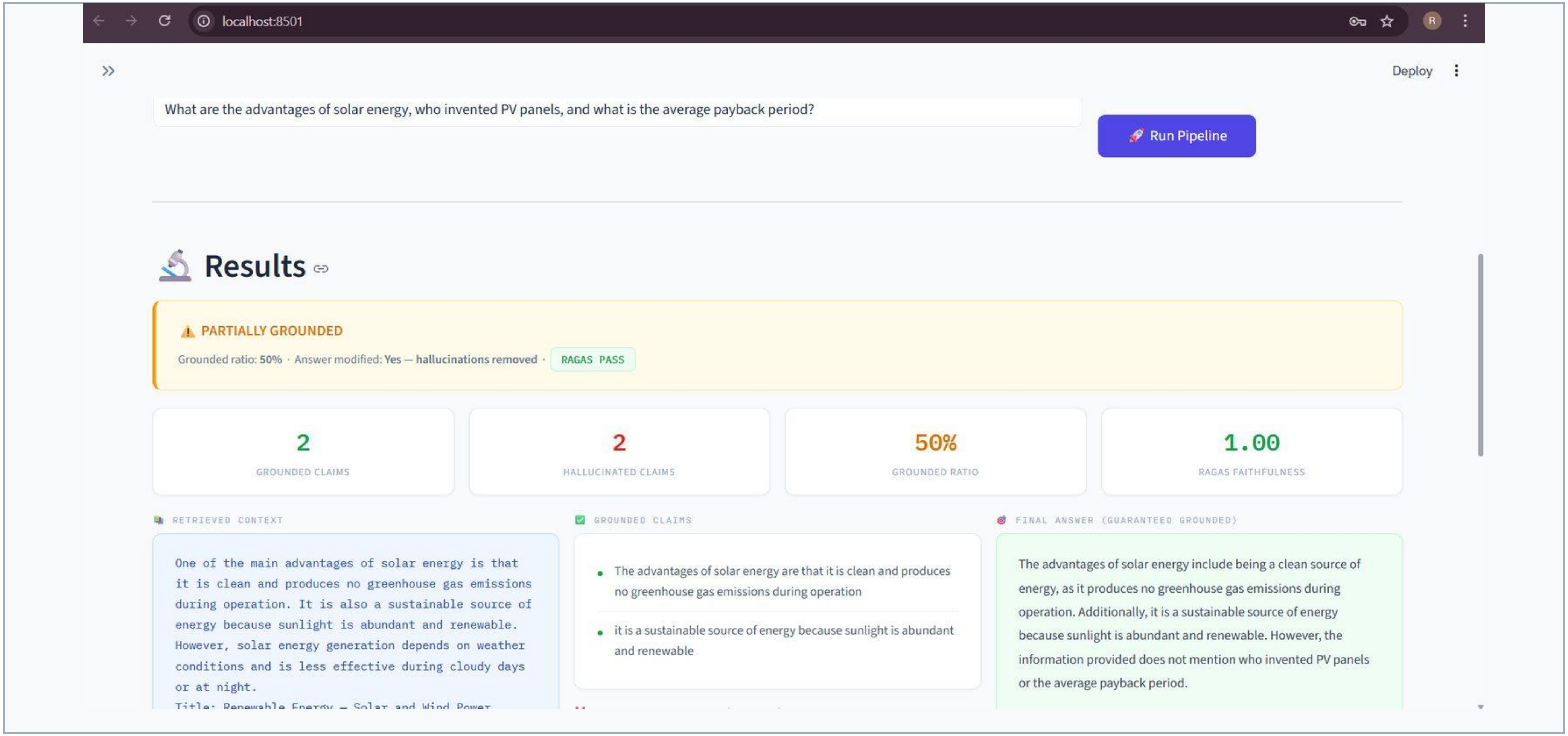


ID	Question	Expected Output	Actual Output	Result / Metrics
TC-PG3-02	How do wind turbines work, which country has the most wind farms, and how tall is the tallest turbine?	Keep the supported turbine mechanism and reject country-ranking and tallest-turbine-height details.	The system kept the wind-turbine mechanism and stated that the country and tallest-height details are not specified.	PARTIALLY GROUNDED Grounded Ratio: 33% RAGAS: 1.00
Screenshot Evidence - TC-PG3-02				



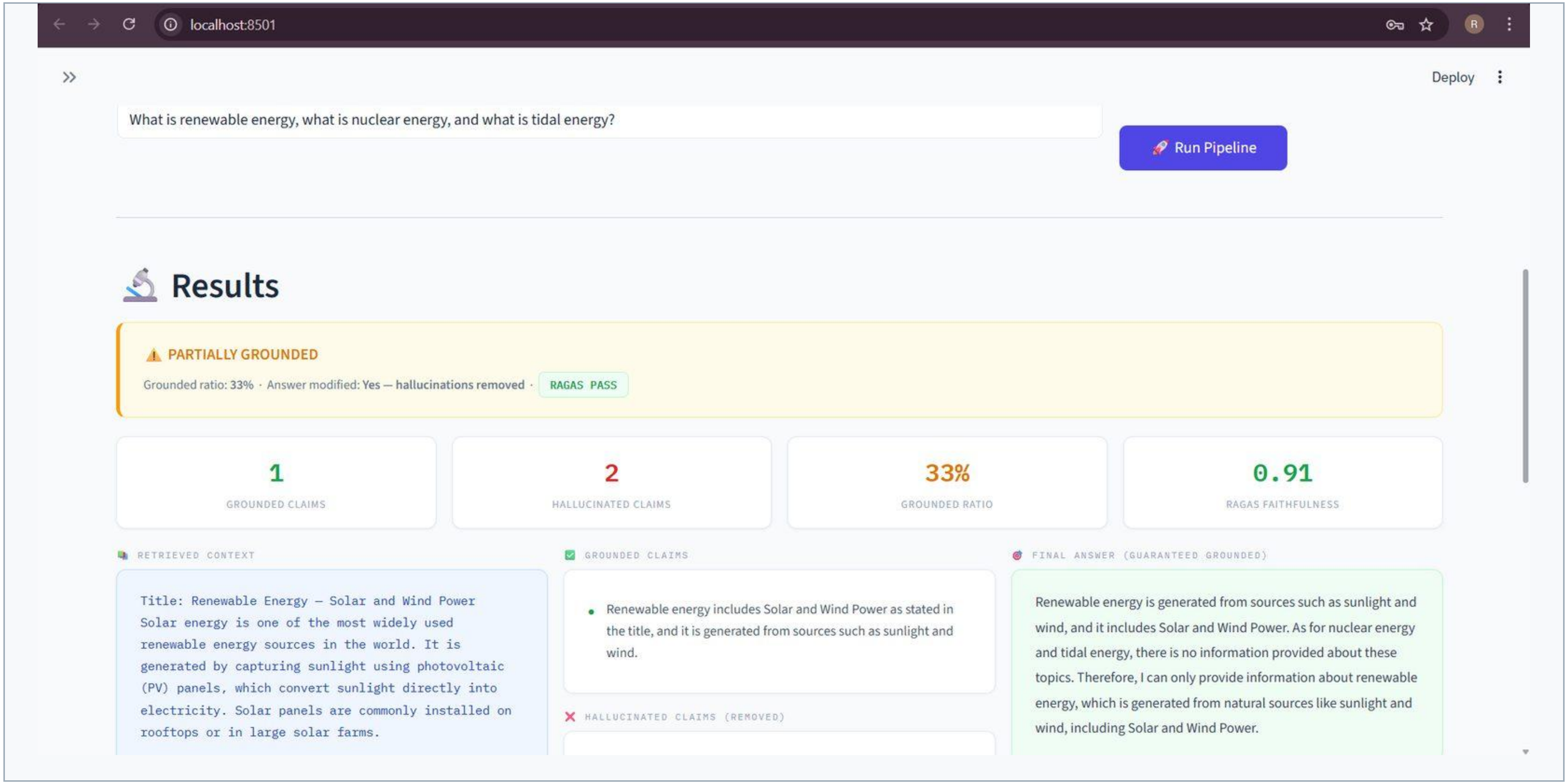
ID	Question	Expected Output	Actual Output	Result / Metrics
TC-PG3-03	What are the advantages of solar energy, who invented PV panels, and what is the average payback period?	Keep the supported solar advantages and reject inventor and payback-period details.	The system kept the supported solar advantages and stated that inventor and payback-period information are not provided.	PARTIALLY GROUNDED Grounded Ratio: 50% RAGAS: 1.00

Screenshot Evidence - TC-PG3-03



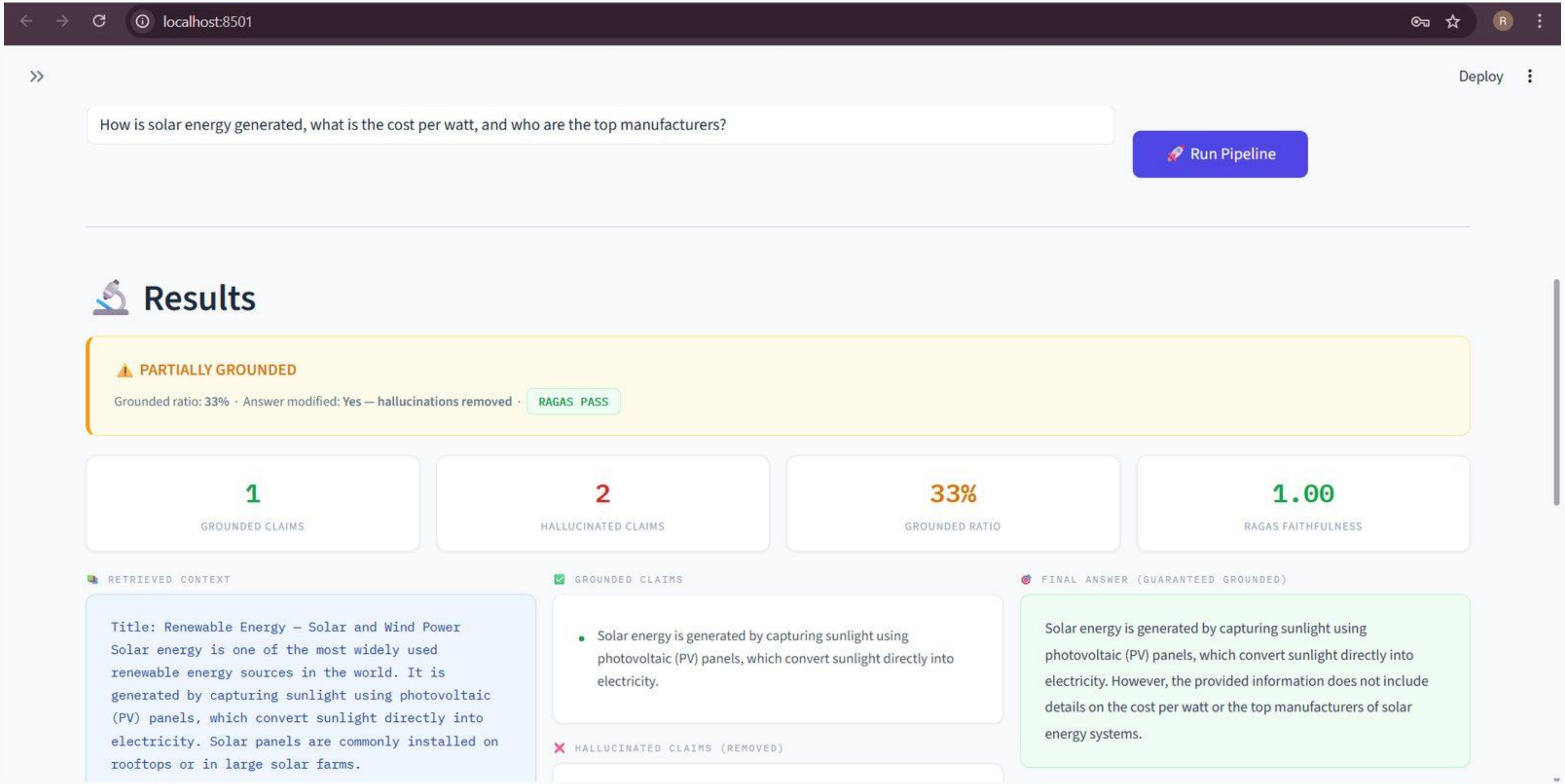
ID	Question	Expected Output	Actual Output	Result / Metrics
TC-PG3-04	What is renewable energy, what is nuclear energy, and what is tidal energy?	Keep only renewable-energy information supported by the solar/wind context and reject nuclear and tidal energy details.	The system provided renewable-energy information from the context and stated nuclear/tidal details are not available.	PARTIALLY GROUNDED Grounded Ratio: 33% RAGAS: 0.91

Screenshot Evidence - TC-PG3-04



ID	Question	Expected Output	Actual Output	Result / Metrics
TC-PG3-05	How is solar energy generated, what is the cost per watt, and who are the top manufacturers?	Keep the supported solar-generation explanation and reject cost-per-watt and manufacturer details.	The system kept the PV-panel generation explanation and stated that cost per watt and top manufacturers are not included.	PARTIALLY GROUNDED Grounded Ratio: 33% RAGAS: 1.00

Screenshot Evidence - TC-PG3-05

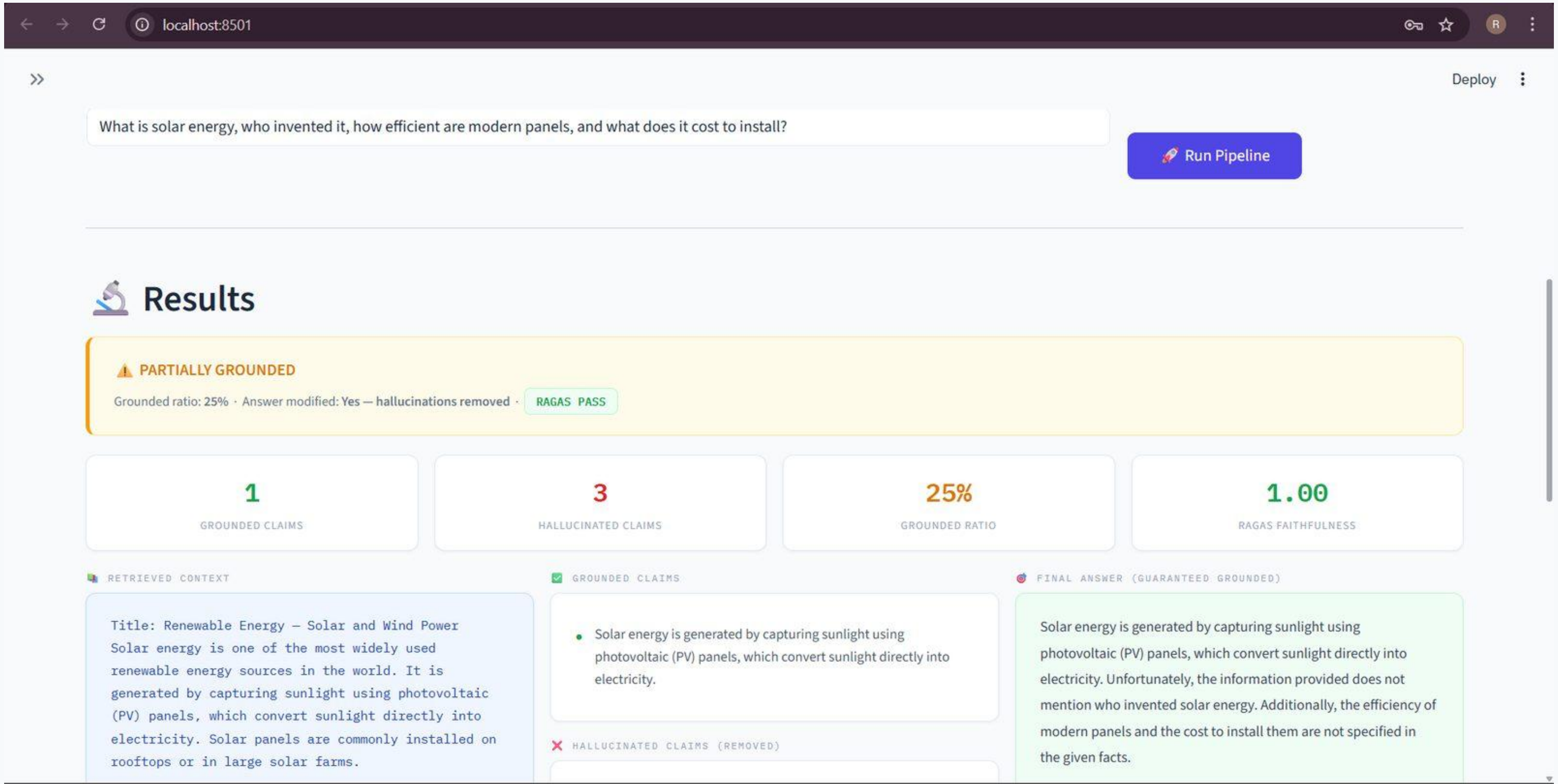


Category 2 - Partially Grounded Test Cases: 4-Part Questions

Questions with one supported part and three unsupported parts.

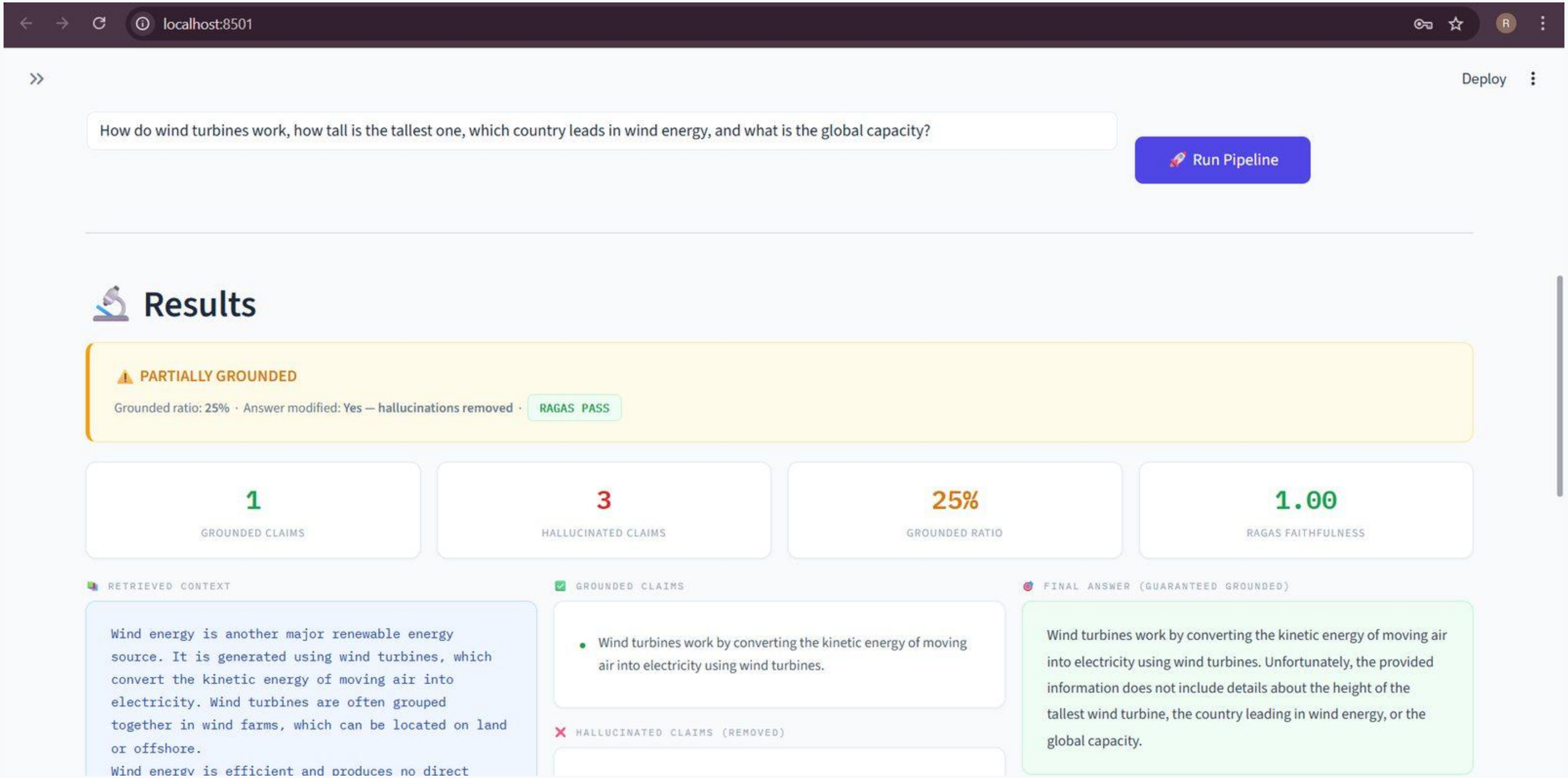
ID	Question	Expected Output	Actual Output	Result / Metrics
TC-PG4-01	What is solar energy, who invented it, how efficient are modern panels, and what does it cost to install?	Keep the supported solar-energy generation information and reject inventor, efficiency, and installation-cost details.	The system kept the PV-panel generation explanation and stated that the other three details are not specified.	PARTIALLY GROUNDED Grounded Ratio: 25% RAGAS: 1.00

Screenshot Evidence - TC-PG4-01



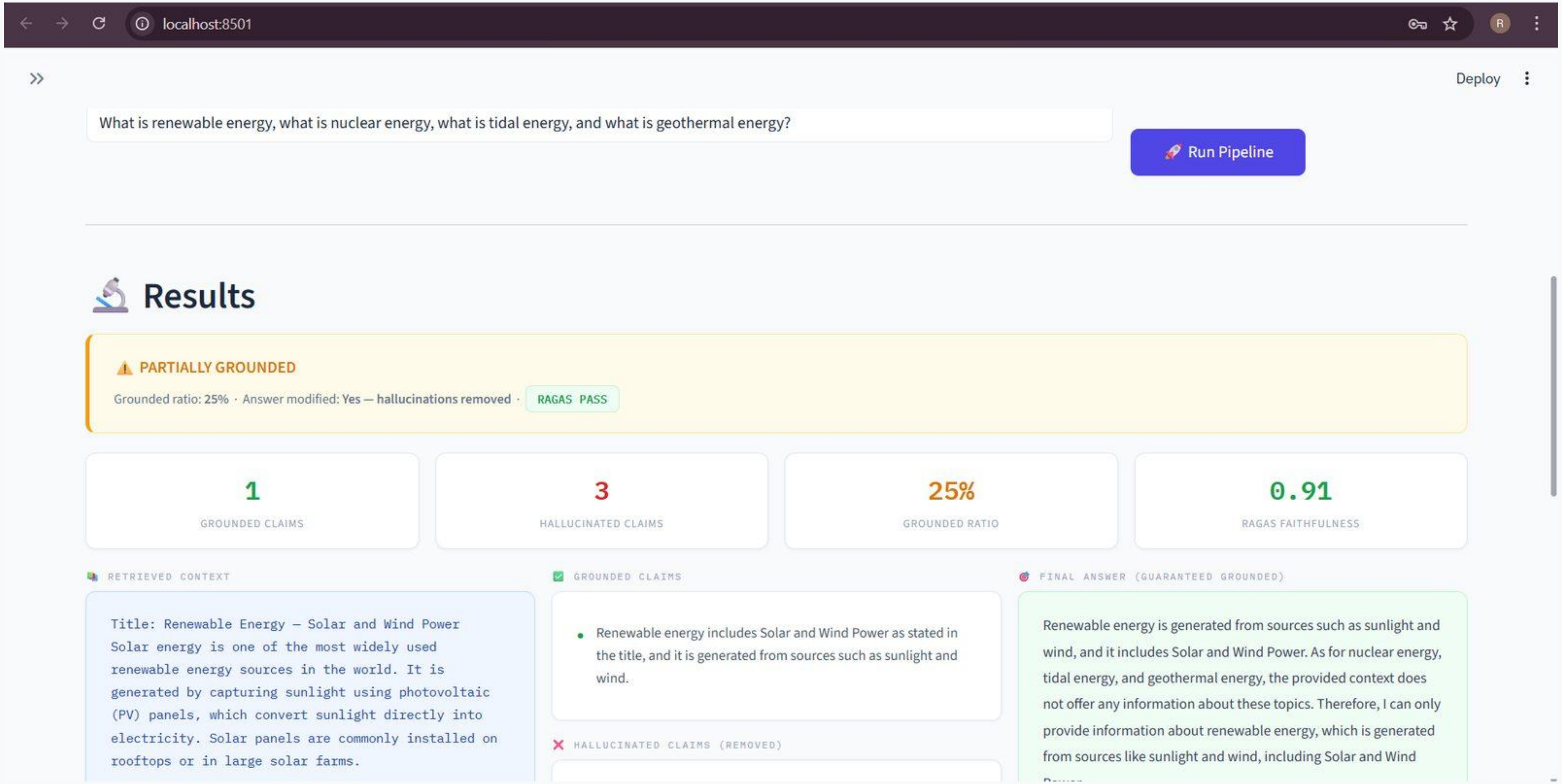
ID	Question	Expected Output	Actual Output	Result / Metrics
TC-PG4-02	How do wind turbines work, how tall is the tallest one, which country leads in wind energy, and what is the global capacity?	Keep the supported wind-turbine mechanism and reject tallest-height, leading-country, and global-capacity details.	The system kept the wind-turbine mechanism and stated that the other three details are not included.	PARTIALLY GROUNDED Grounded Ratio: 25% RAGAS: 1.00

Screenshot Evidence - TC-PG4-02



ID	Question	Expected Output	Actual Output	Result / Metrics
TC-PG4-03	What is renewable energy, what is nuclear energy, what is tidal energy, and what is geothermal energy?	Keep only the renewable-energy information supported by the solar/wind context and reject nuclear, tidal, and geothermal details.	The system kept renewable-energy information and stated that nuclear, tidal, and geothermal energy are not covered by the context.	PARTIALLY GROUNDED Grounded Ratio: 25% RAGAS: 0.91

Screenshot Evidence - TC-PG4-03

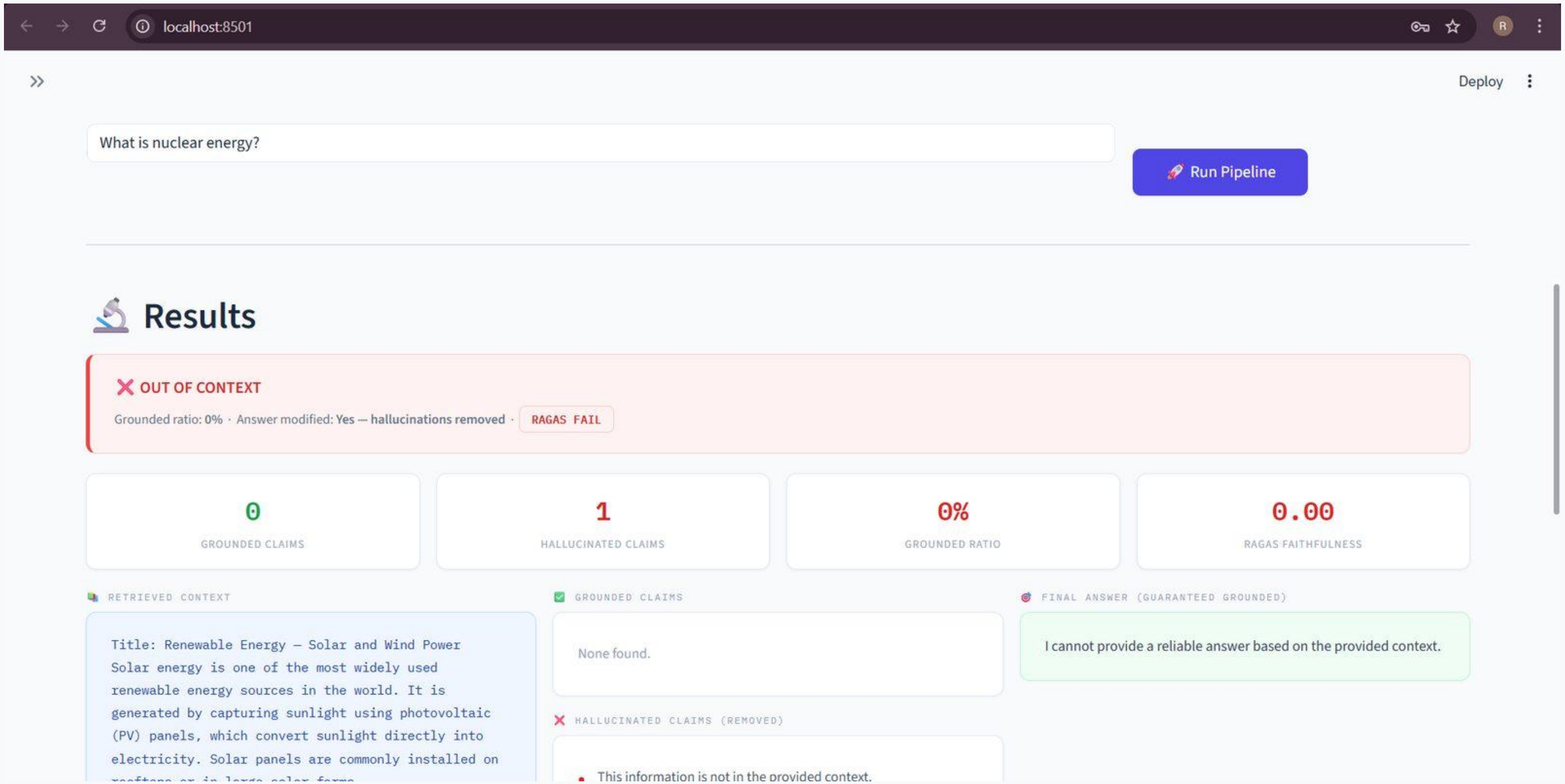


Category 3 - Out of Context Test Cases: Completely Unrelated Topics

Questions fully outside the uploaded solar and wind energy context.

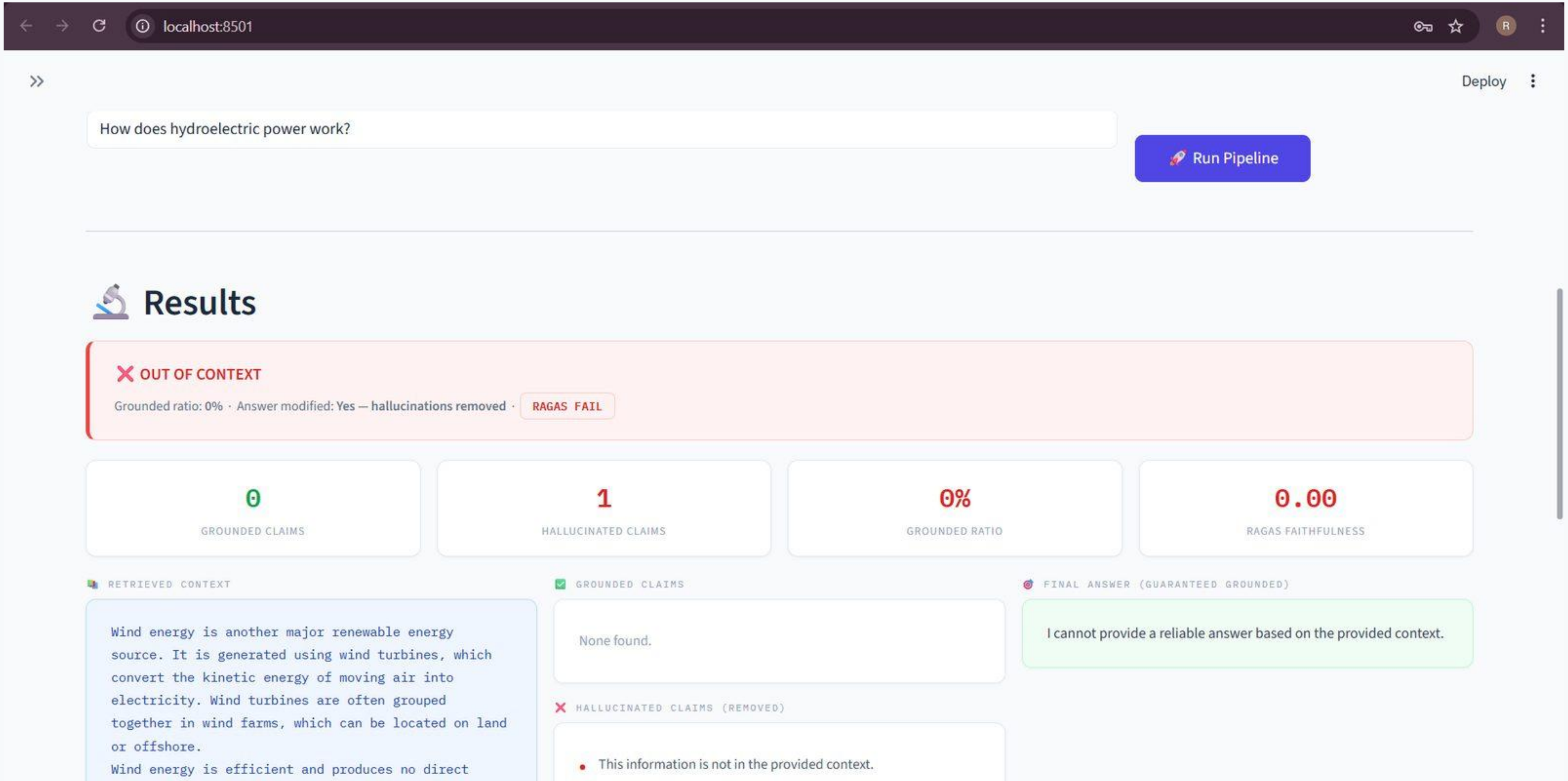
ID	Question	Expected Output	Actual Output	Result / Metrics
TC-OOC-U01	What is nuclear energy?	Reject the question because nuclear energy is outside the uploaded solar/wind context.	The system returned that a reliable answer cannot be provided based on the provided context.	OUT OF CONTEXT Grounded Ratio: 0% RAGAS: 0.00

Screenshot Evidence - TC-OOC-U01



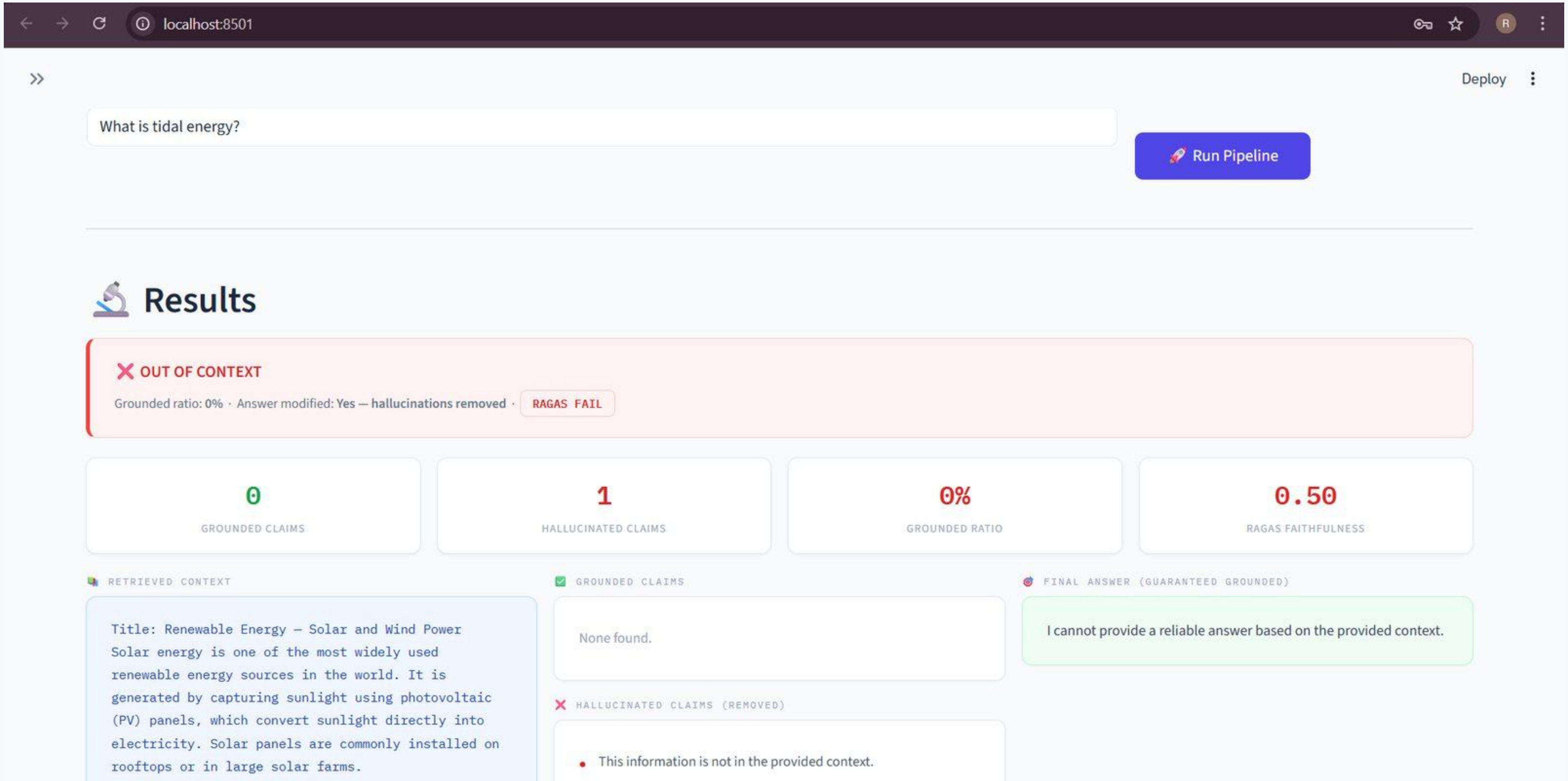
ID	Question	Expected Output	Actual Output	Result / Metrics
TC-OOC-U02	How does hydroelectric power work?	Reject the question because hydroelectric power is not in the provided context.	The system returned that a reliable answer cannot be provided based on the provided context.	OUT OF CONTEXT Grounded Ratio: 0% RAGAS: 0.00

Screenshot Evidence - TC-OOC-U02



ID	Question	Expected Output	Actual Output	Result / Metrics
TC-OOC-U03	What is tidal energy?	Reject the question because tidal energy is outside the solar/wind context.	The system returned that a reliable answer cannot be provided based on the provided context.	OUT OF CONTEXT Grounded Ratio: 0% RAGAS: 0.50

Screenshot Evidence - TC-OOC-U03

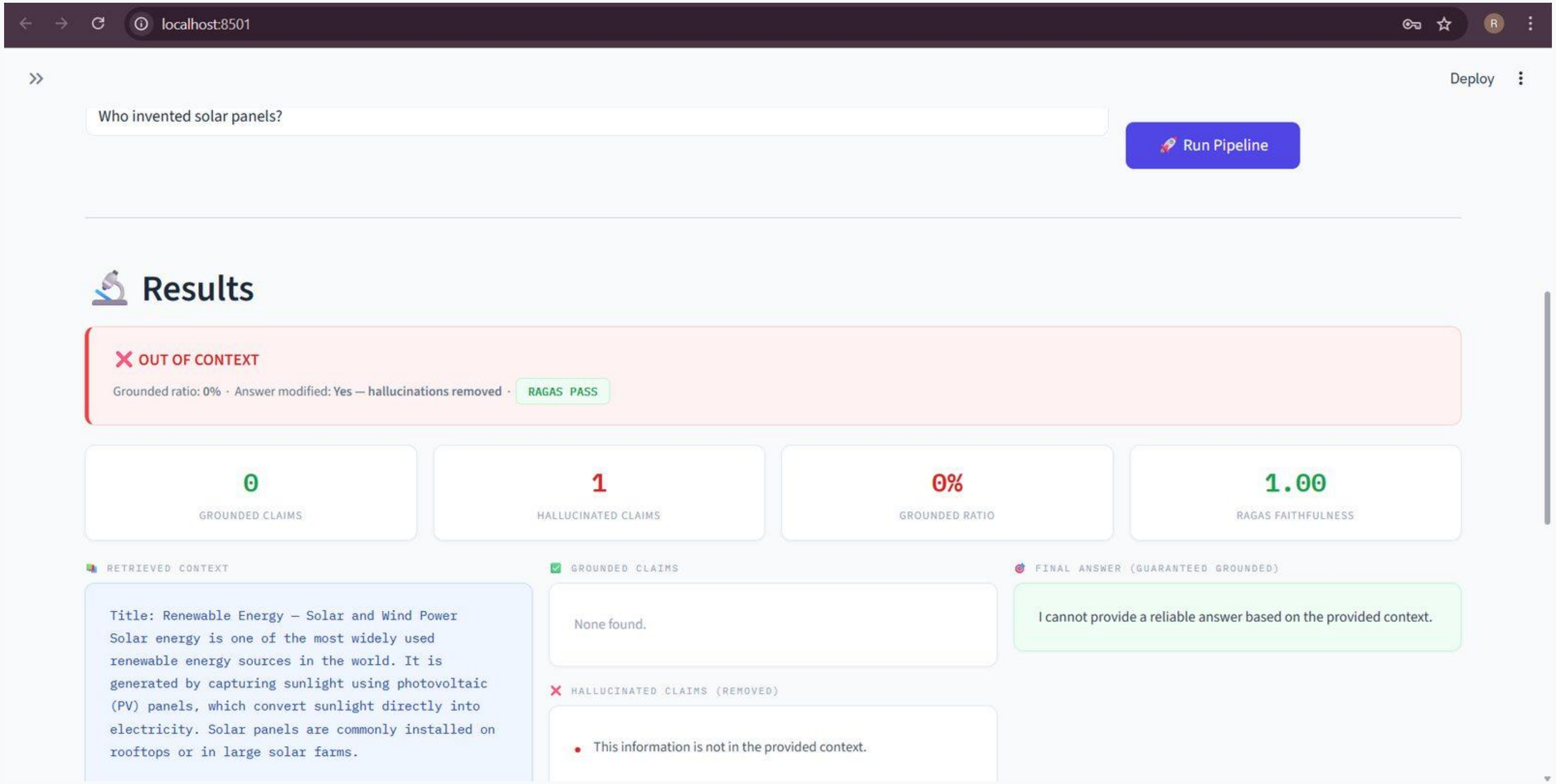


Category 3 - Out of Context Test Cases: Factual Questions Not in the Document

Real-world factual questions not available in the uploaded document.

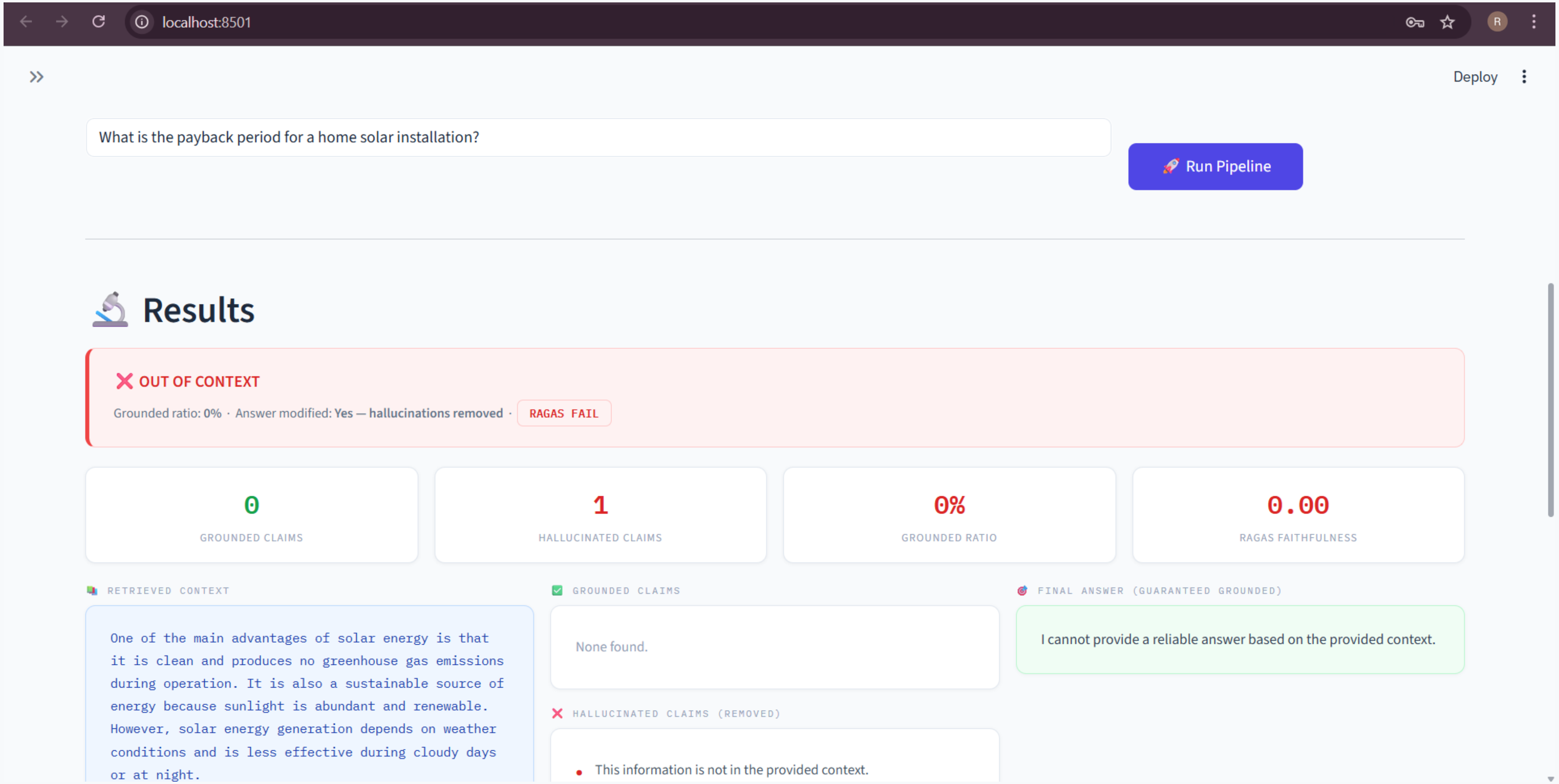
ID	Question	Expected Output	Actual Output	Result / Metrics
TC-OOC-F01	Who invented solar panels?	Reject the question because the context does not mention the inventor of solar panels.	The system avoided answering from outside knowledge and returned a context-limited refusal.	OUT OF CONTEXT Grounded Ratio: 0% RAGAS: 1.00

Screenshot Evidence - TC-OOC-F01



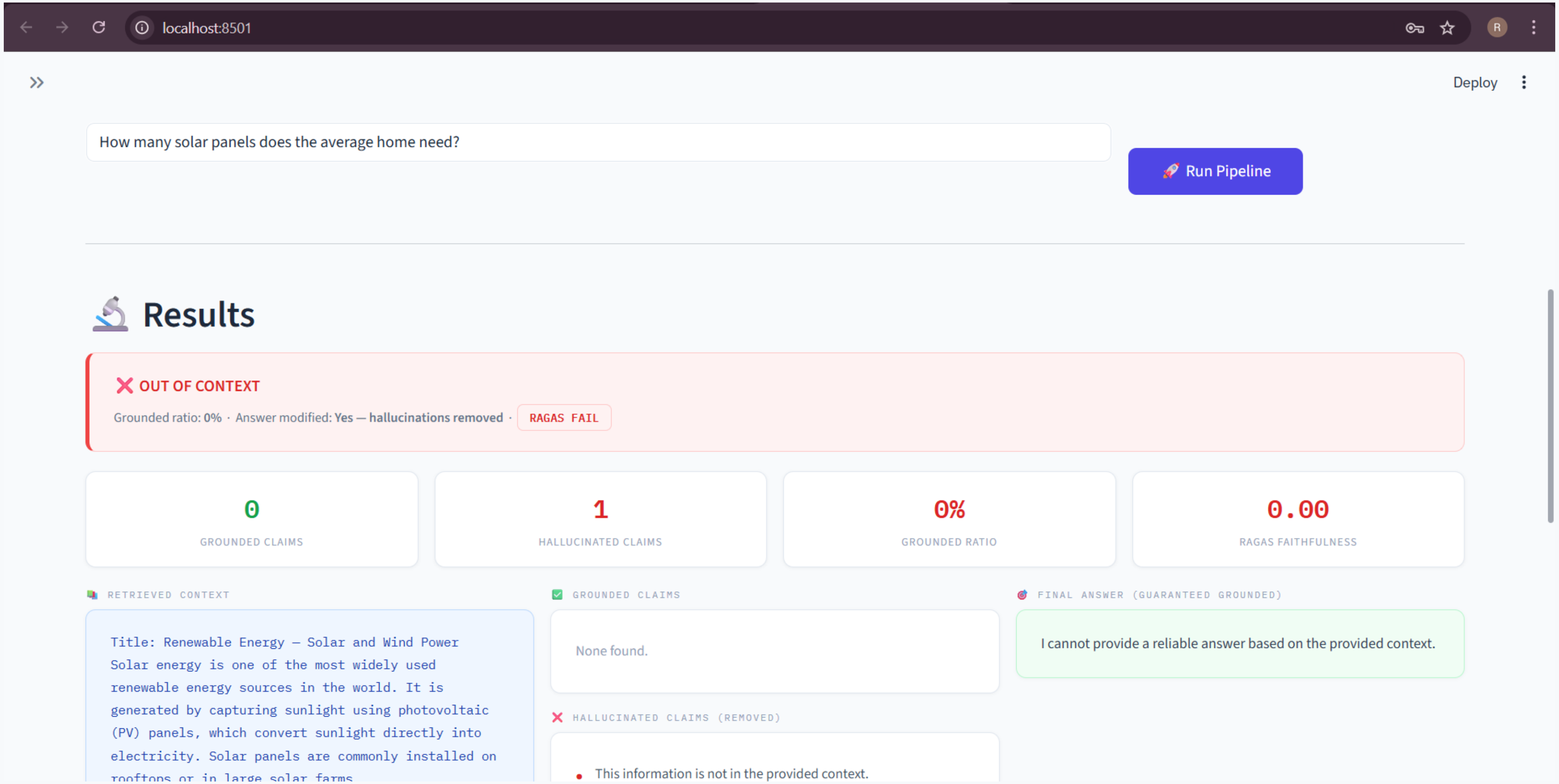
ID	Question	Expected Output	Actual Output	Result / Metrics
TC-OOC-F02	What is the payback period for a home solar installation?	Reject the question because payback-period information is not provided.	The system returned that a reliable answer cannot be provided based on the provided context.	OUT OF CONTEXT Grounded Ratio: 0% RAGAS: 0.00

Screenshot Evidence - TC-OOC-F02



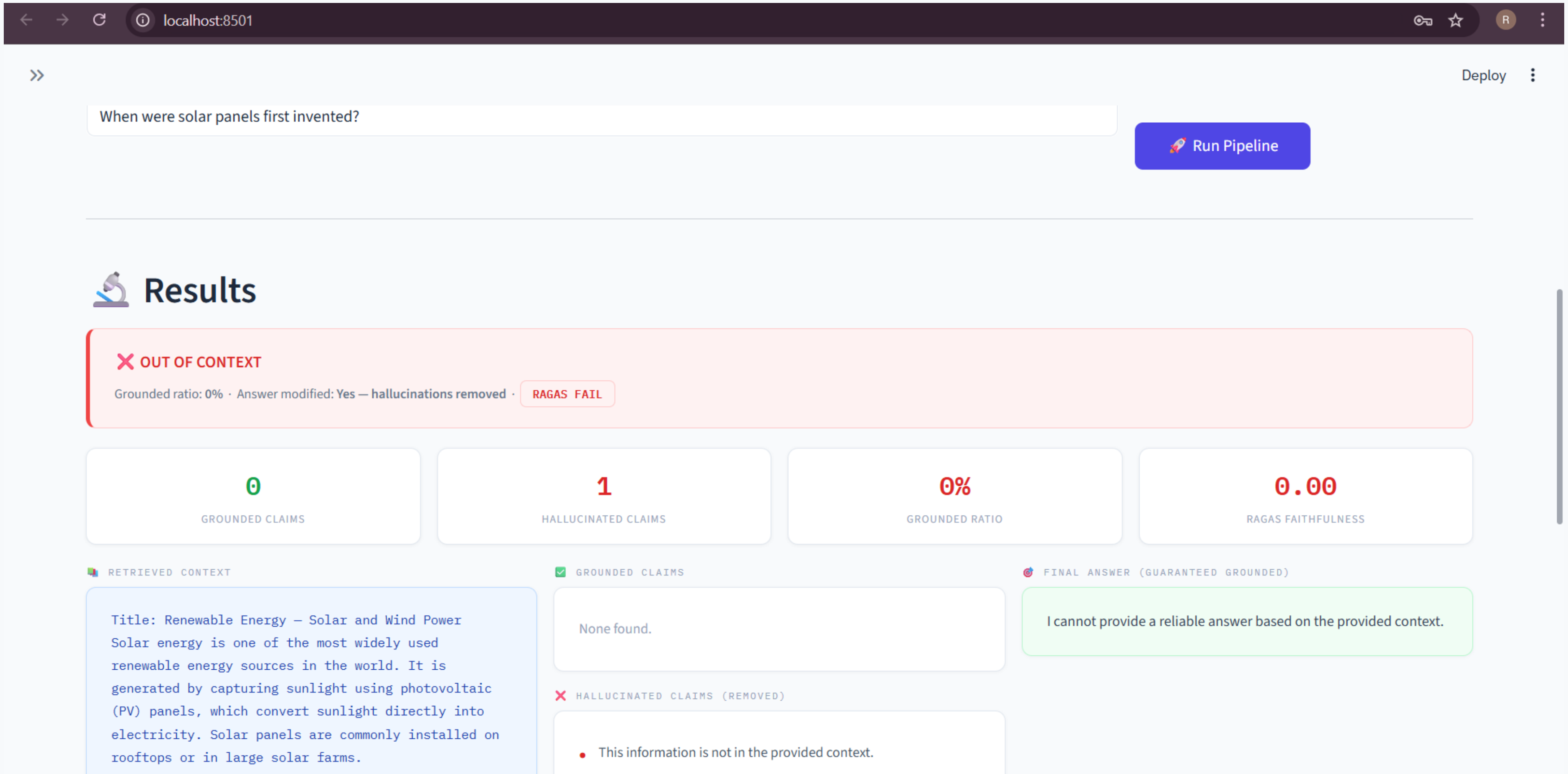
ID	Question	Expected Output	Actual Output	Result / Metrics
TC-OOC-F03	How many solar panels does the average home need?	Reject the question because household panel-count requirements are not provided.	The system returned a context-limited refusal.	OUT OF CONTEXT Grounded Ratio: 0% RAGAS: 0.00

Screenshot Evidence - TC-OOC-F03



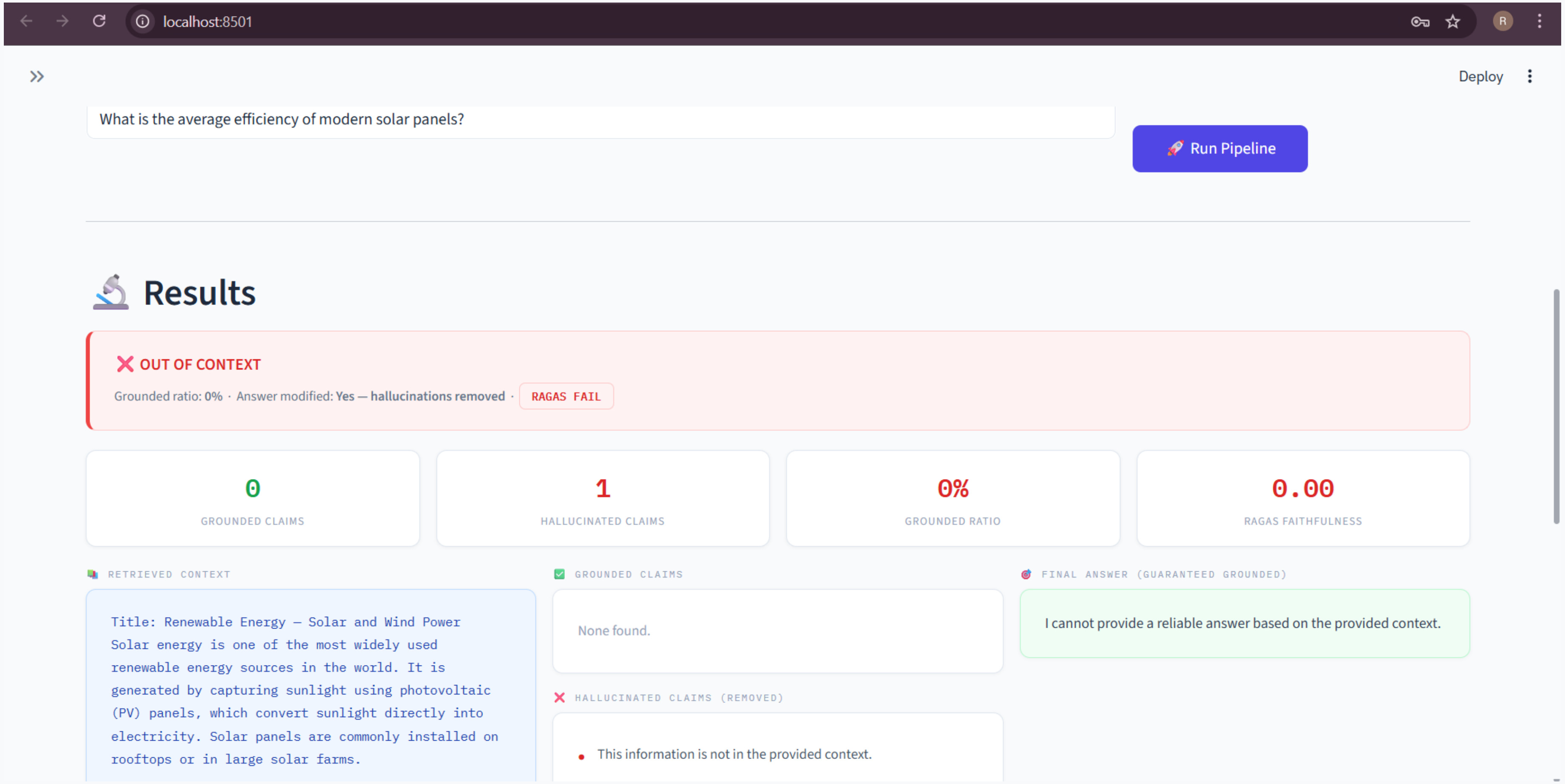
ID	Question	Expected Output	Actual Output	Result / Metrics
TC-OOC-F04	When were solar panels first invented?	Reject the question because historical invention dates are not in the context.	The system returned a context-limited refusal.	OUT OF CONTEXT Grounded Ratio: 0% RAGAS: 0.00

Screenshot Evidence - TC-OOC-F04



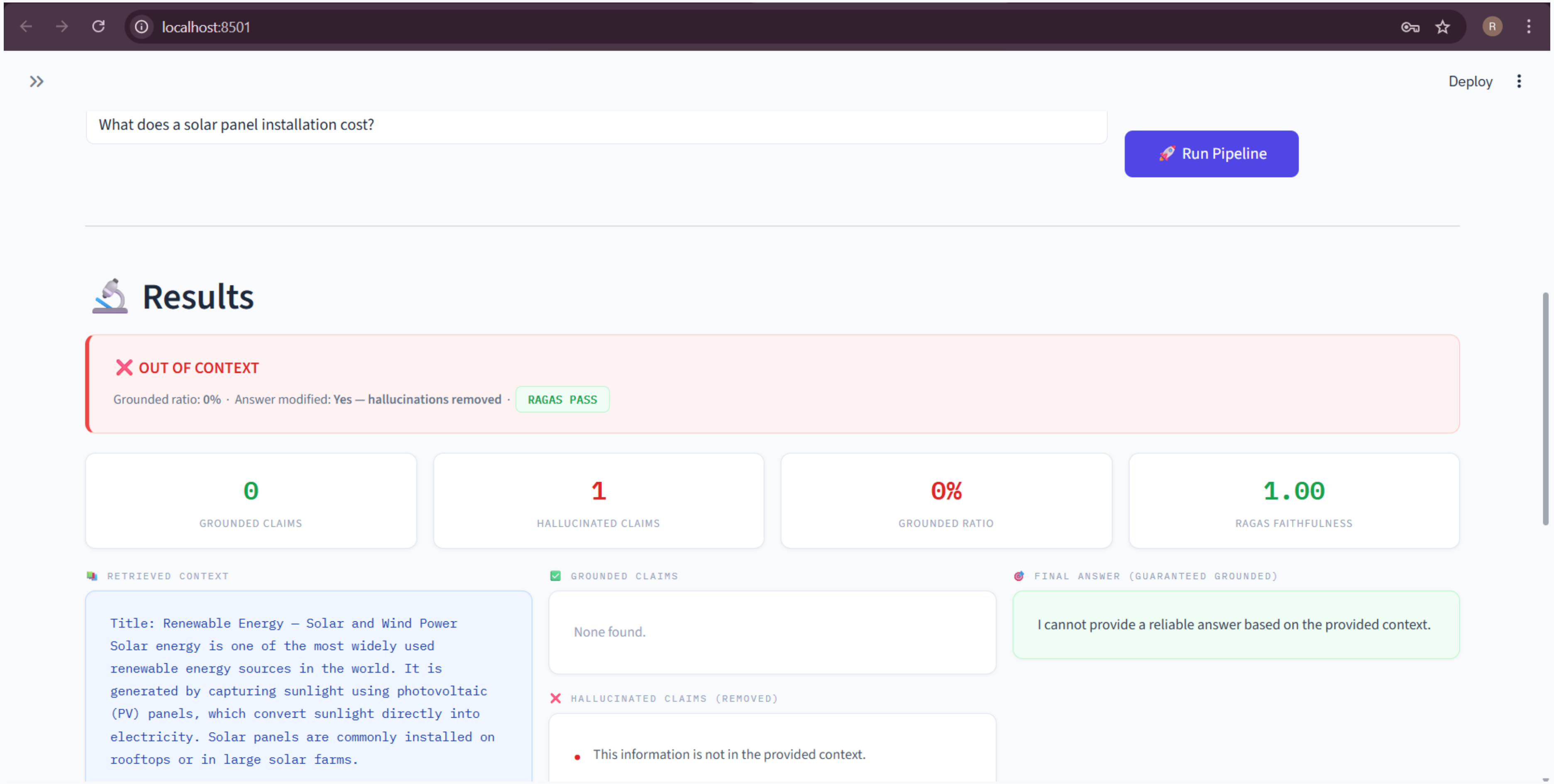
ID	Question	Expected Output	Actual Output	Result / Metrics
TC-OOC-F05	What is the average efficiency of modern solar panels?	Reject the question because modern solar-panel efficiency values are not provided.	The system returned a context-limited refusal.	OUT OF CONTEXT Grounded Ratio: 0% RAGAS: 0.00

Screenshot Evidence - TC-OOC-F05



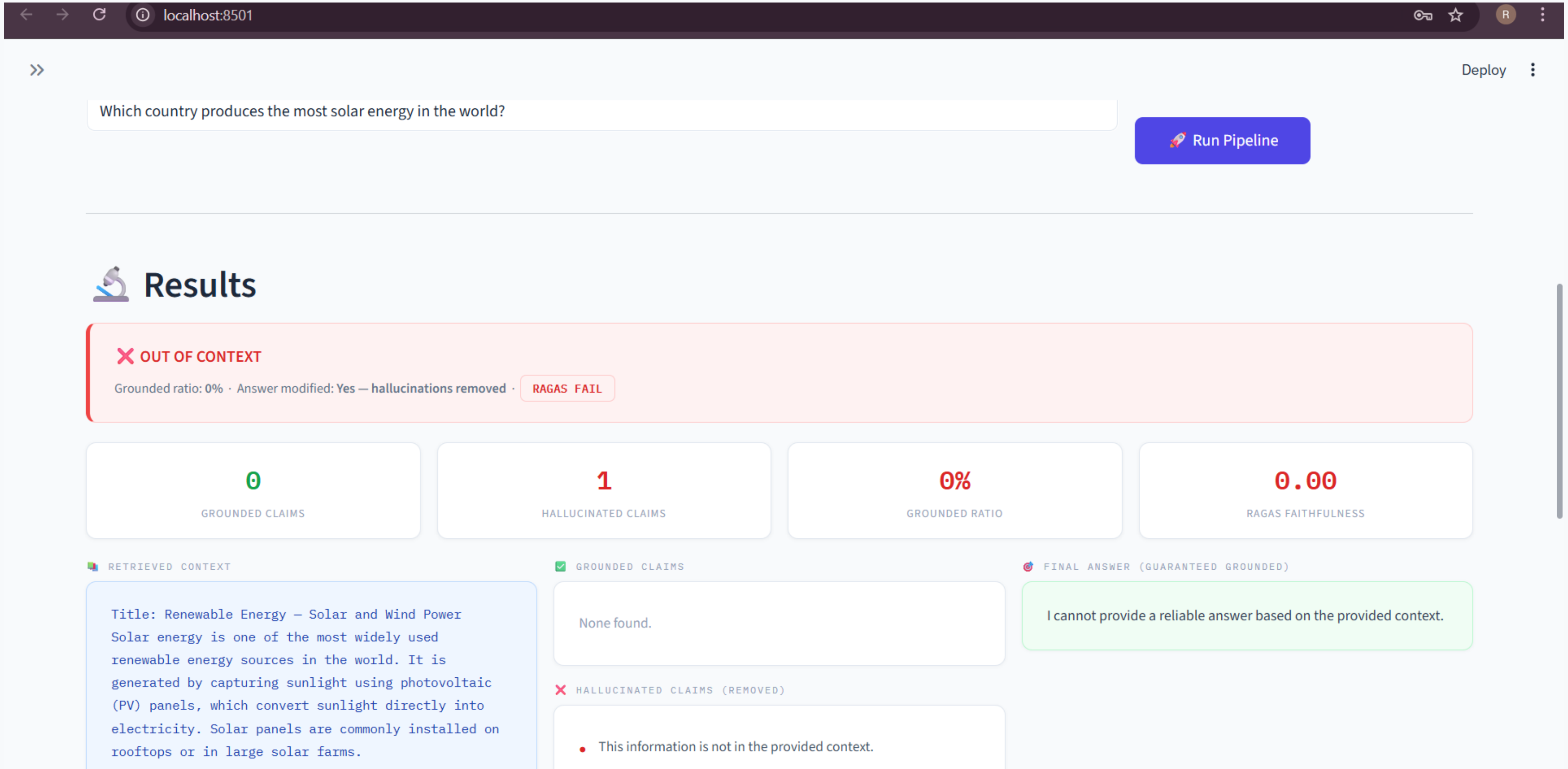
ID	Question	Expected Output	Actual Output	Result / Metrics
TC-OOC-F06	What does a solar panel installation cost?	Reject the question because solar installation cost is not specified.	The system refused to provide unsupported pricing details.	OUT OF CONTEXT Grounded Ratio: 0% RAGAS: 1.00

Screenshot Evidence - TC-OOC-F06



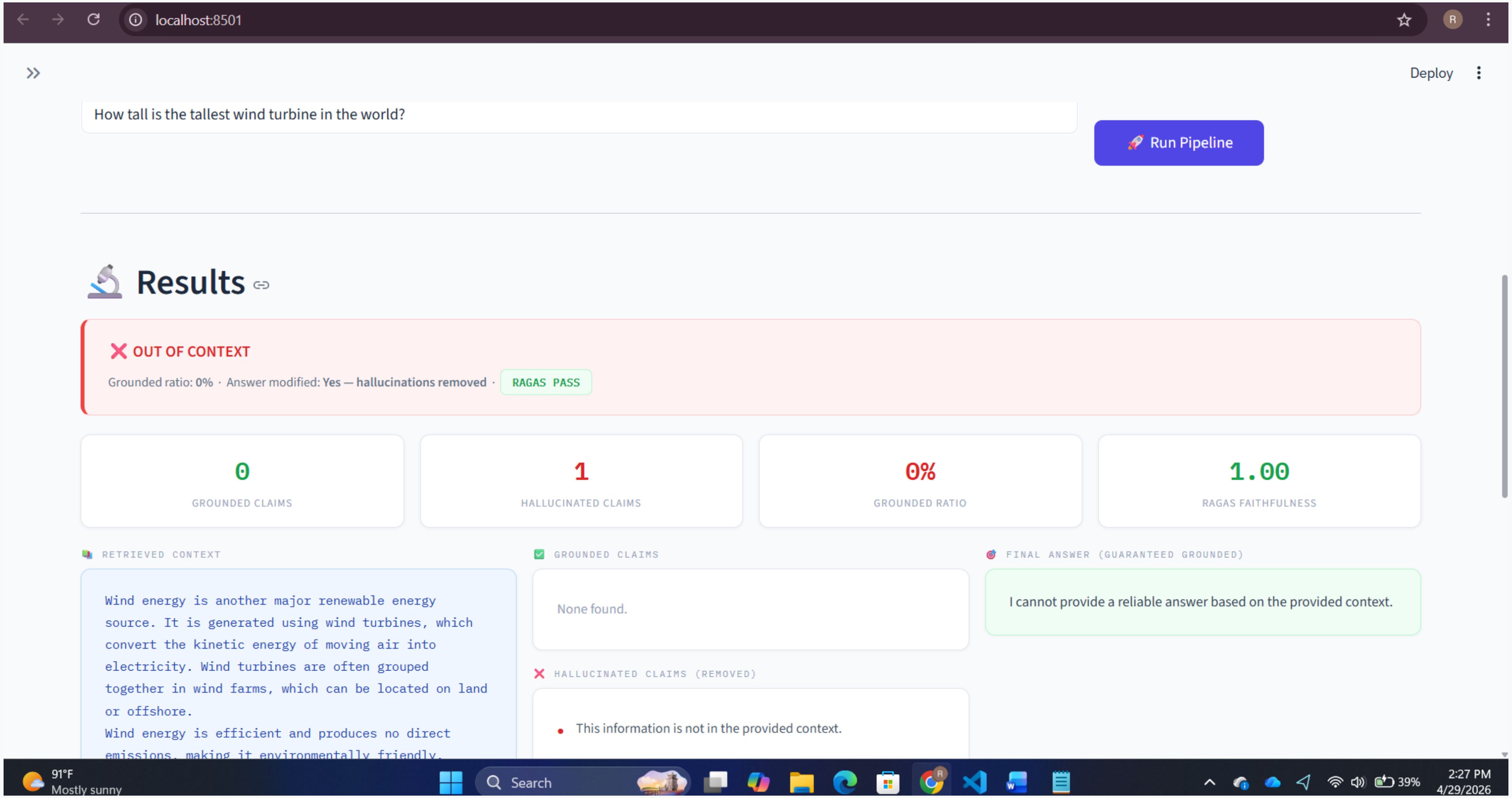
ID	Question	Expected Output	Actual Output	Result / Metrics
TC-OOC-F07	Which country produces the most solar energy in the world?	Reject the question because country-level production rankings are not in the context.	The system returned a context-limited refusal.	OUT OF CONTEXT Grounded Ratio: 0% RAGAS: 0.00

Screenshot Evidence - TC-OOC-F07



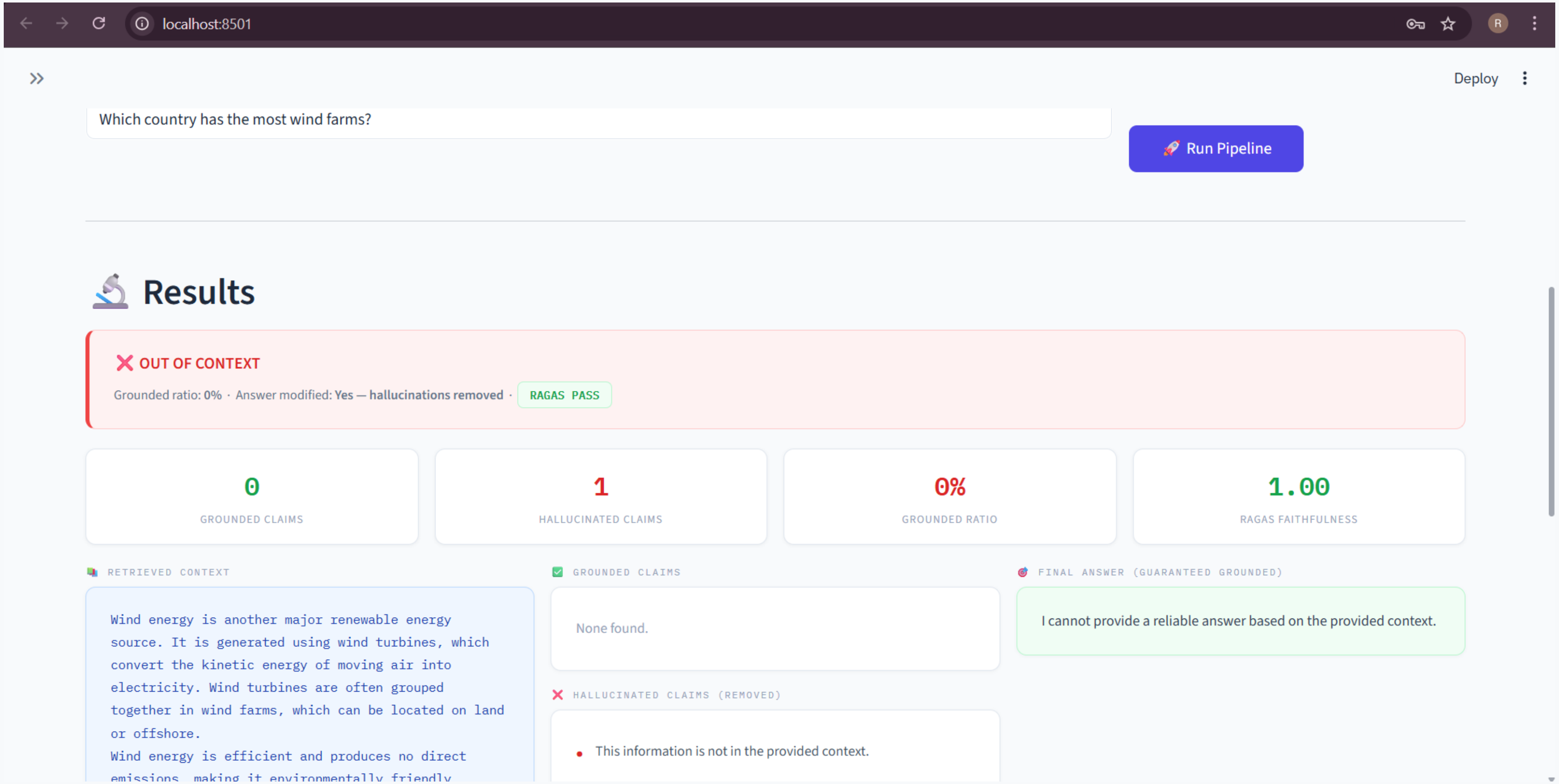
ID	Question	Expected Output	Actual Output	Result / Metrics
TC-OOC-F08	How tall is the tallest wind turbine in the world?	Reject the question because tallest-turbine height is not provided.	The system returned that a reliable answer cannot be provided from the context.	OUT OF CONTEXT Grounded Ratio: 0% RAGAS: 1.00

Screenshot Evidence - TC-OOC-F08



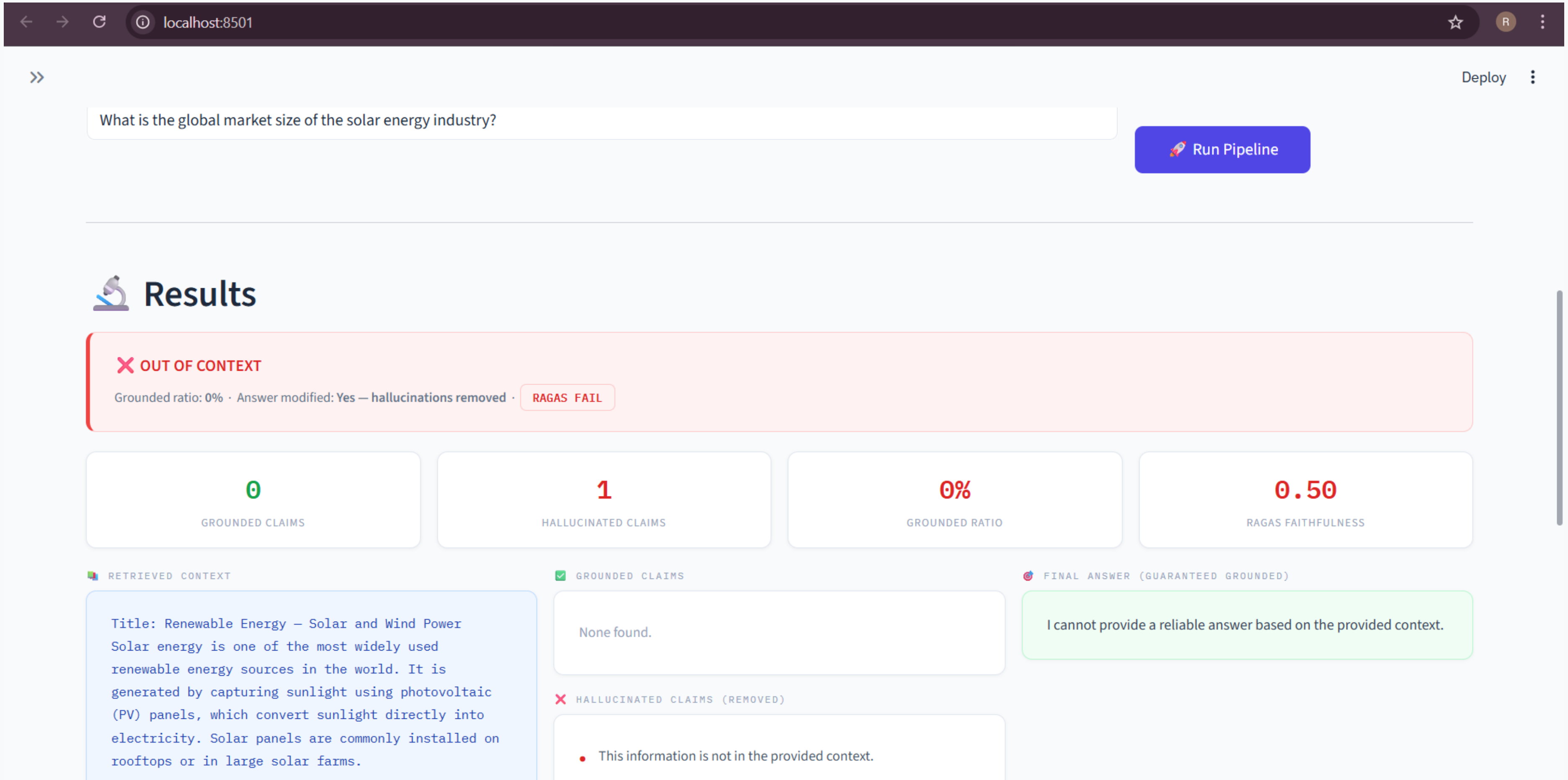
ID	Question	Expected Output	Actual Output	Result / Metrics
TC-OOC-F09	Which country has the most wind farms?	Reject the question because country rankings for wind farms are not provided.	The system returned a context-limited refusal.	OUT OF CONTEXT Grounded Ratio: 0% RAGAS: 1.00

Screenshot Evidence - TC-OOC-F09



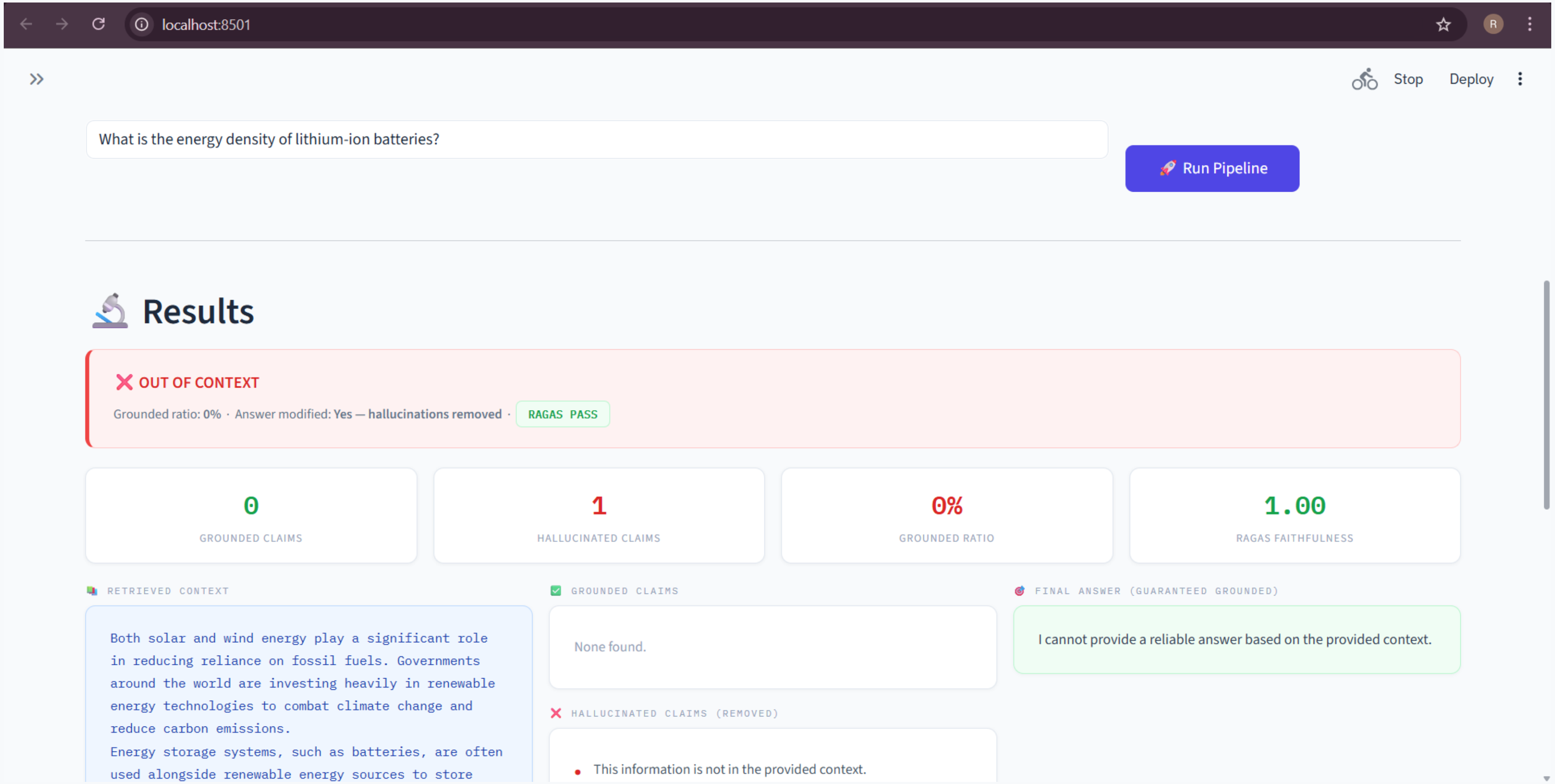
ID	Question	Expected Output	Actual Output	Result / Metrics
TC-OOC-F10	What is the global market size of the solar energy industry?	Reject the question because market-size data is not provided.	The system returned that a reliable answer cannot be produced from the context.	OUT OF CONTEXT Grounded Ratio: 0% RAGAS: 0.50

Screenshot Evidence - TC-OOC-F10



ID	Question	Expected Output	Actual Output	Result / Metrics
TC-OOC-F11	What is the energy density of lithium-ion batteries?	Reject the question because lithium-ion battery energy density is not specified.	The system returned a context-limited refusal.	OUT OF CONTEXT Grounded Ratio: 0% RAGAS: 1.00

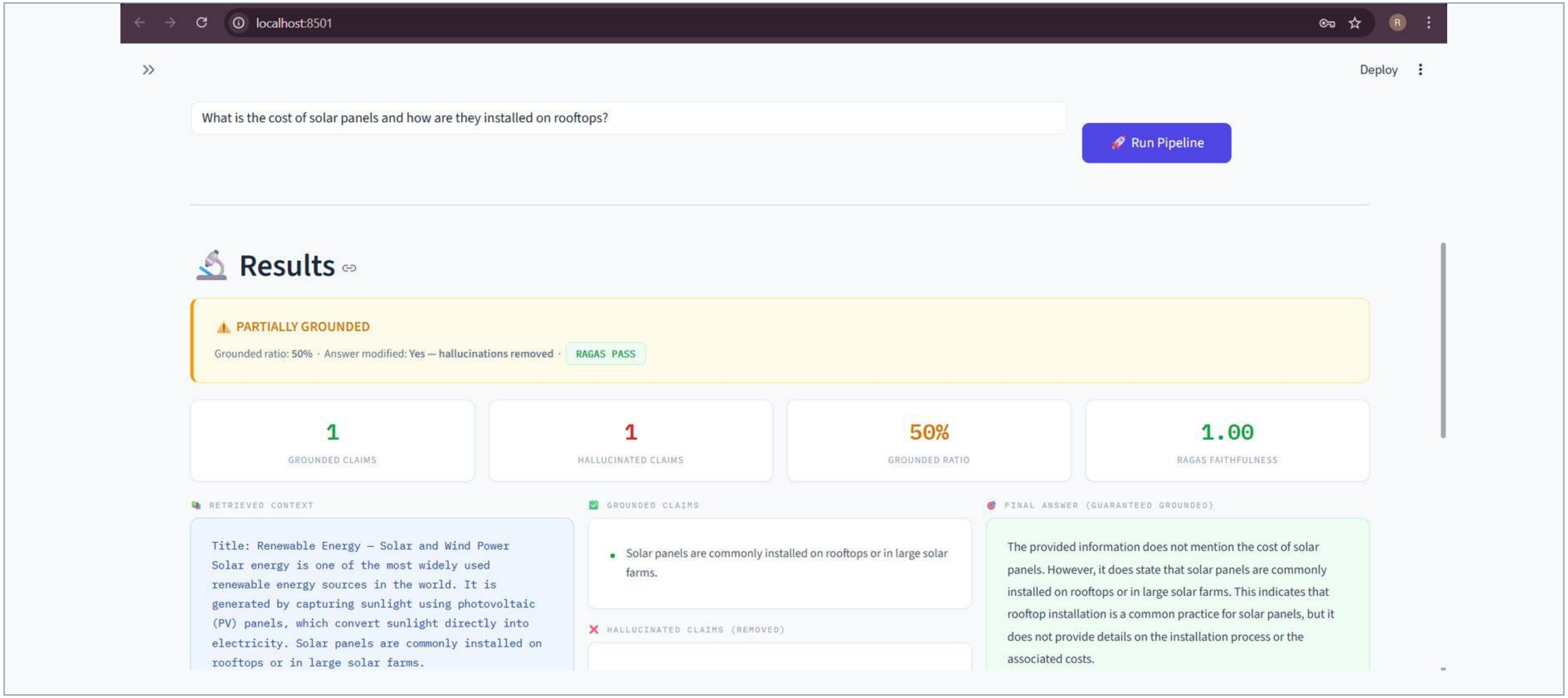
Screenshot Evidence - TC-OOC-F11



Category 4 - Mixed Multi-Part Stress Test

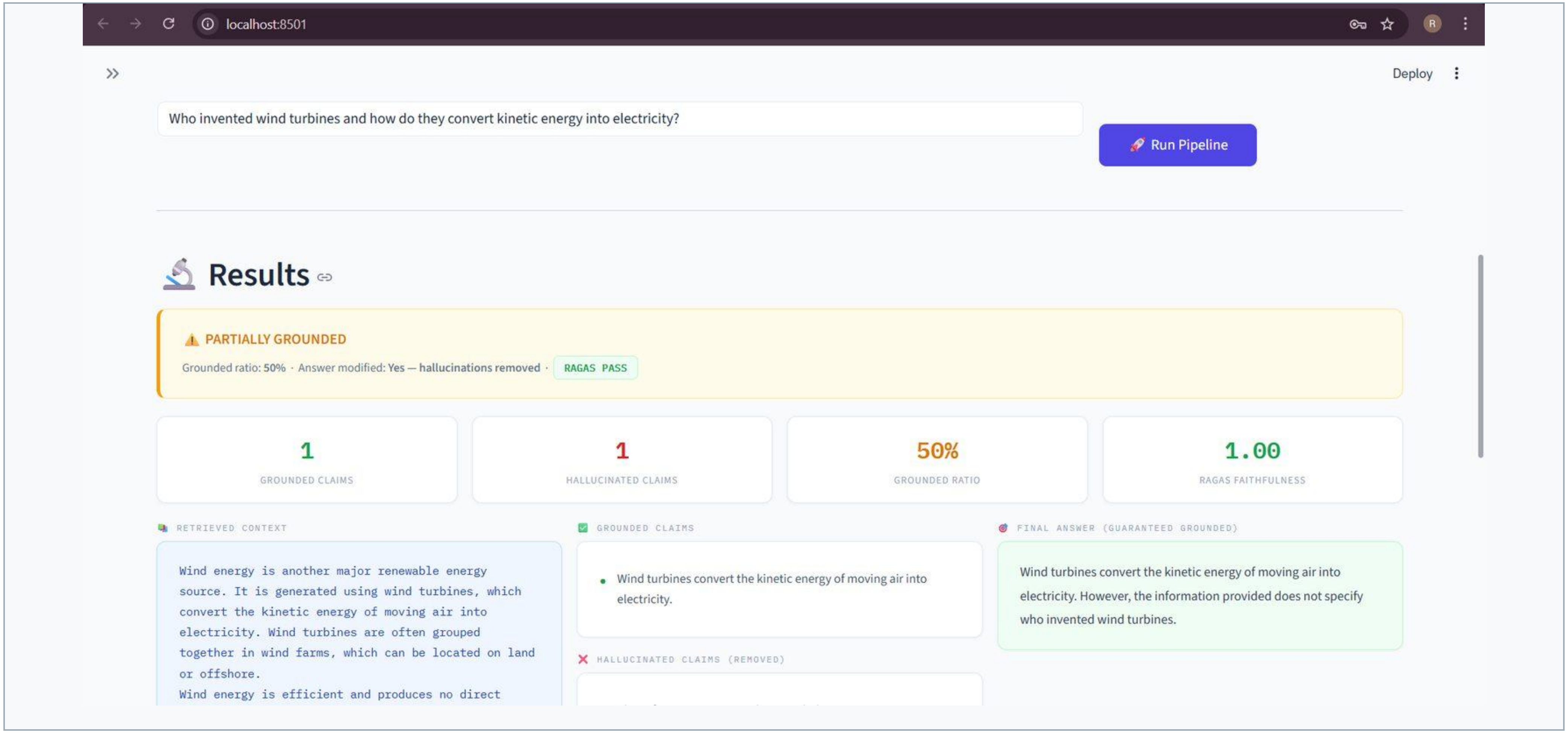
Prompts combining supported and unsupported question parts.

ID	Question	Expected Output	Actual Output	Result / Metrics
TC-MIX-01	What is the cost of solar panels and how are they installed on rooftops?	Keep the supported rooftop/large solar farm installation-location claim and reject cost and detailed installation-process claims.	The system kept the supported rooftop/solar-farm location and stated that cost/process details are not specified.	PARTIALLY GROUNDED Grounded Ratio: 50% RAGAS: 1.00
Screenshot Evidence - TC-MIX-01				



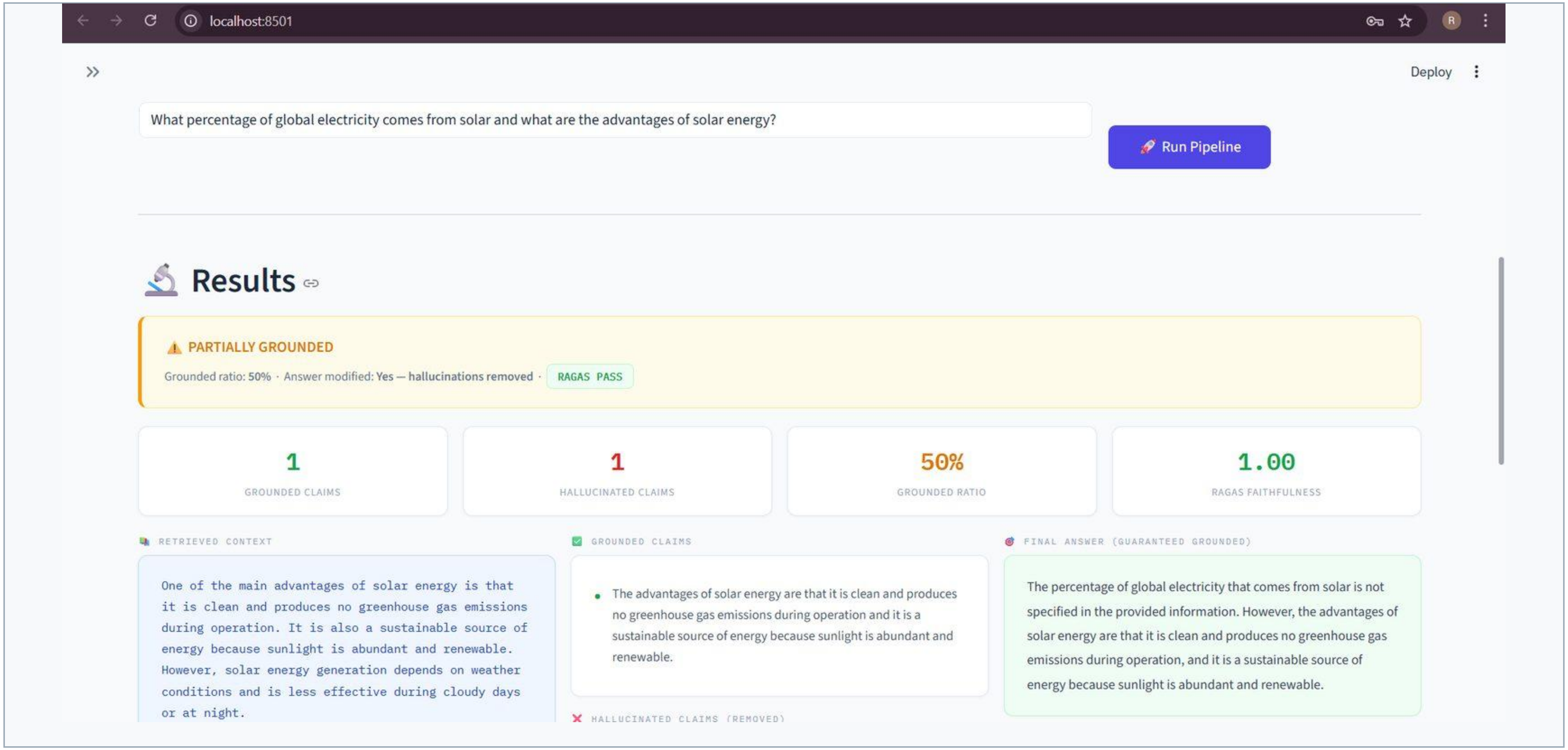
ID	Question	Expected Output	Actual Output	Result / Metrics
TC-MIX-02	Who invented wind turbines and how do they convert kinetic energy into electricity?	Keep the supported kinetic-energy conversion claim and reject inventor details.	The system kept the wind-turbine conversion claim and stated that the inventor is not specified.	PARTIALLY GROUNDED Grounded Ratio: 50% RAGAS: 1.00

Screenshot Evidence - TC-MIX-02



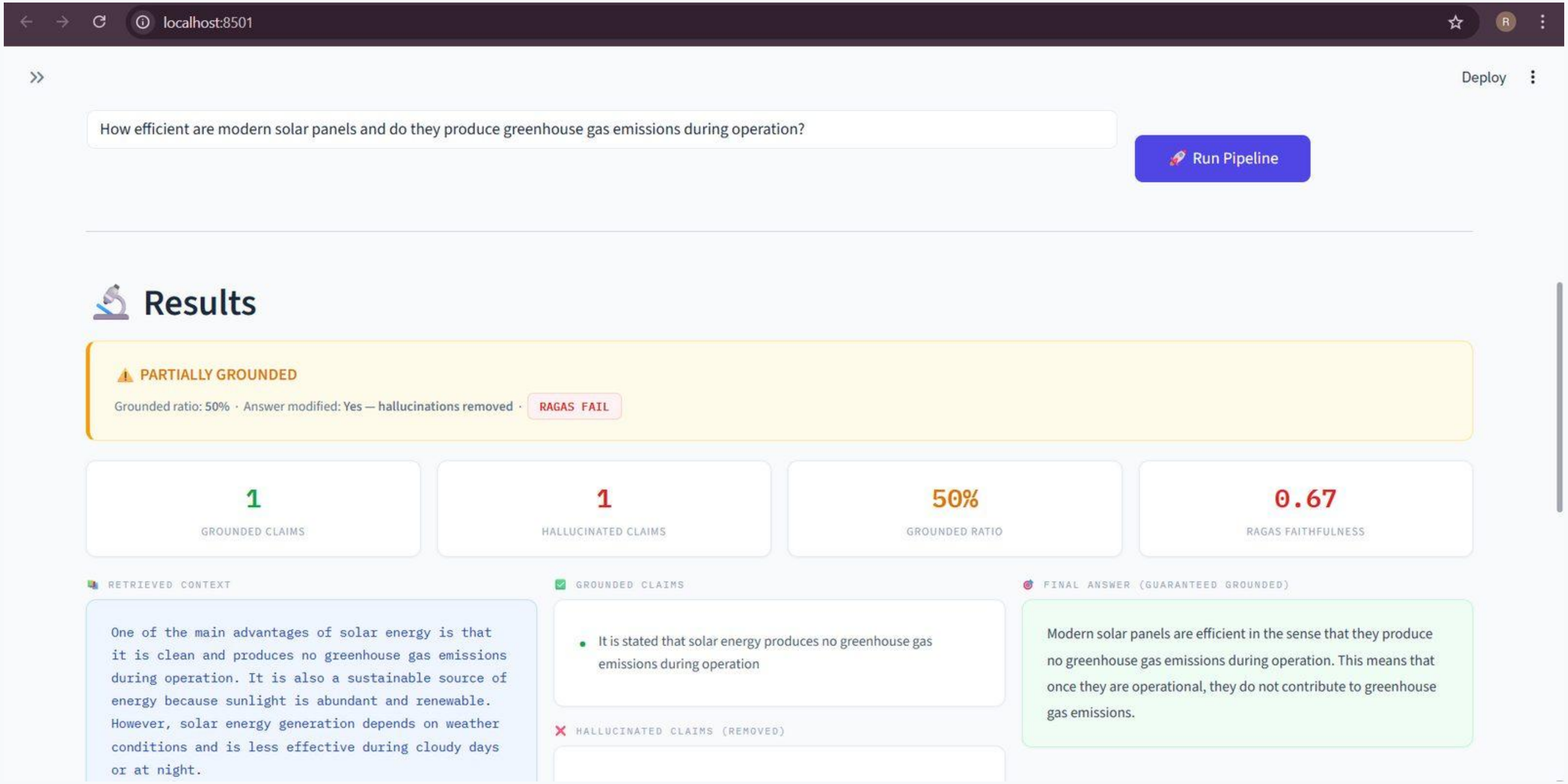
ID	Question	Expected Output	Actual Output	Result / Metrics
TC-MIX-03	What percentage of global electricity comes from solar and what are the advantages of solar energy?	Keep the supported solar-energy advantages and reject global electricity percentage details.	The system kept the supported advantages and stated that the percentage is not specified.	PARTIALLY GROUNDED Grounded Ratio: 50% RAGAS: 1.00

Screenshot Evidence - TC-MIX-03



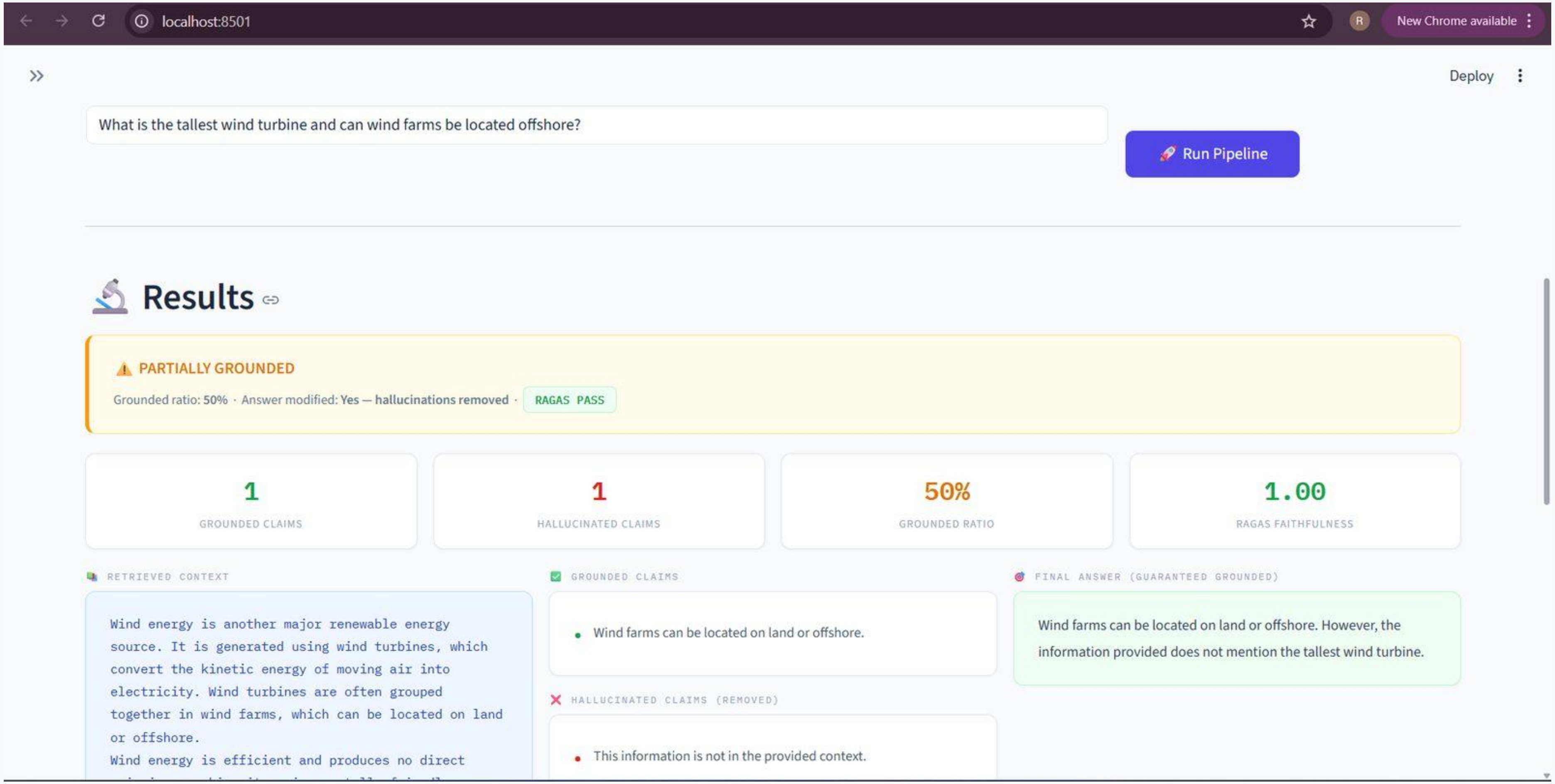
ID	Question	Expected Output	Actual Output	Result / Metrics
TC-MIX-04	How efficient are modern solar panels and do they produce greenhouse gas emissions during operation?	Keep the supported no-greenhouse-gas-emissions claim and reject modern panel efficiency details.	The system kept the supported no-emissions claim, but RAGAS faithfulness was lower for this mixed response.	PARTIALLY GROUNDED Grounded Ratio: 50% RAGAS: 0.67

Screenshot Evidence - TC-MIX-04



ID	Question	Expected Output	Actual Output	Result / Metrics
TC-MIX-05	What is the tallest wind turbine and can wind farms be located offshore?	Keep the supported offshore/land location claim and reject tallest-turbine details.	The system kept the wind-farm location claim and stated that tallest-turbine information is not mentioned.	PARTIALLY GROUNDED Grounded Ratio: 50% RAGAS: 1.00

Screenshot Evidence - TC-MIX-05

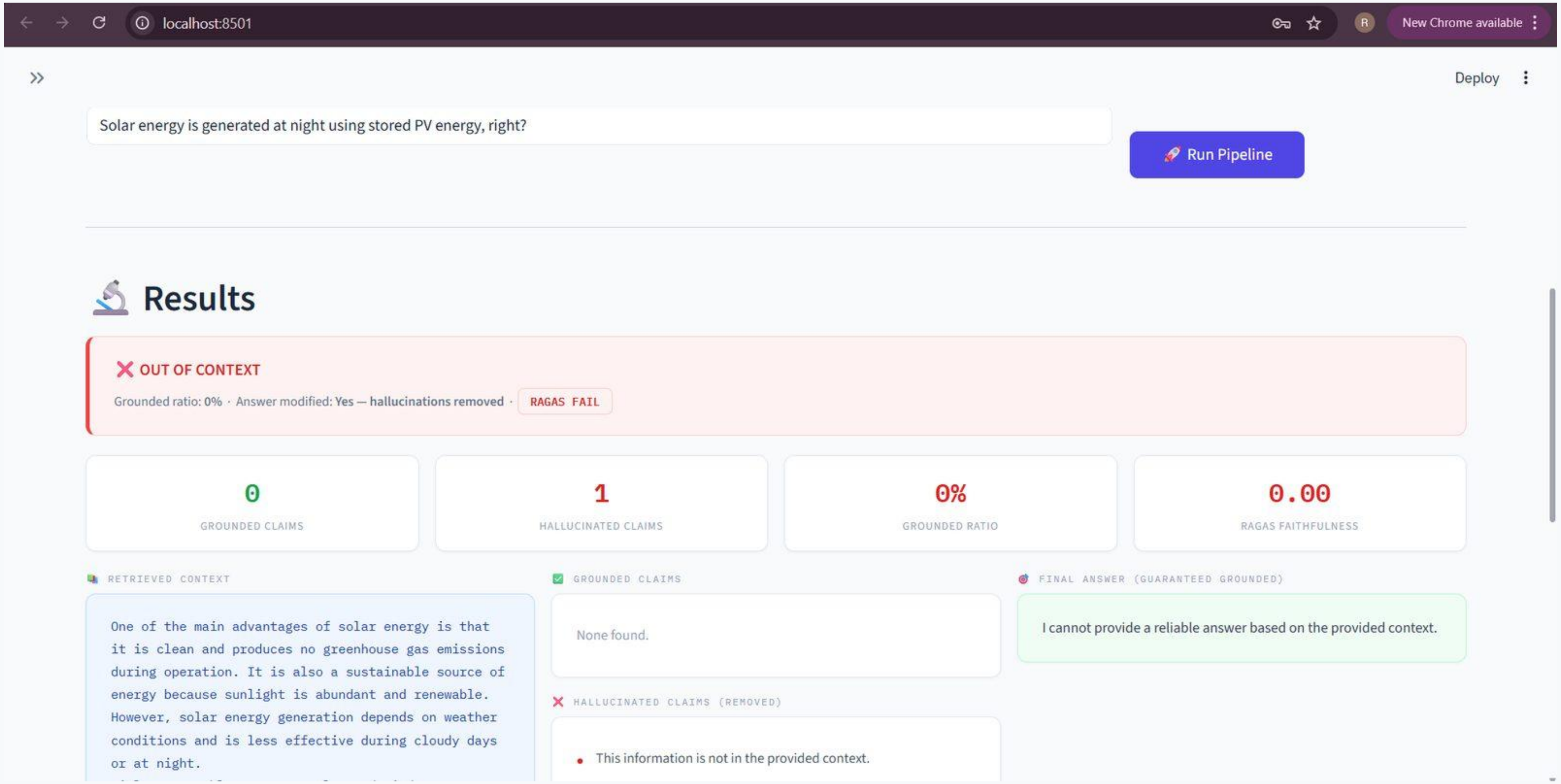


Category 5 - Edge Cases: Trick Questions

Leading or contradiction-based questions that should not be accepted blindly.

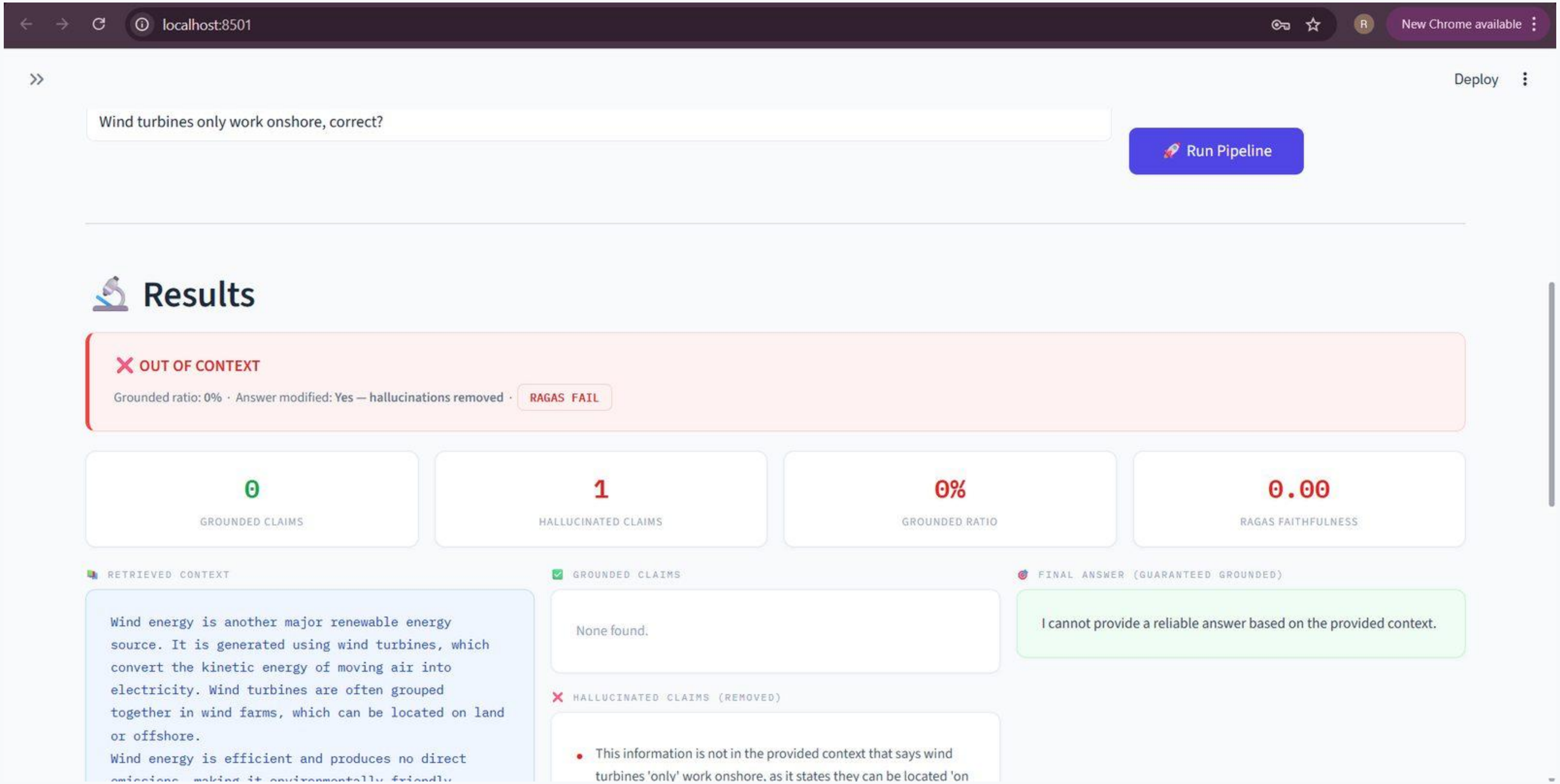
ID	Question	Expected Output	Actual Output	Result / Metrics
TC-EDGE-T01	Solar energy is generated at night using stored PV energy, right?	Reject the leading claim because the context does not support solar energy being generated at night using stored PV energy.	The system refused to provide the unsupported leading claim as a grounded answer.	OUT OF CONTEXT Grounded Ratio: 0% RAGAS: 0.00

Screenshot Evidence - TC-EDGE-T01



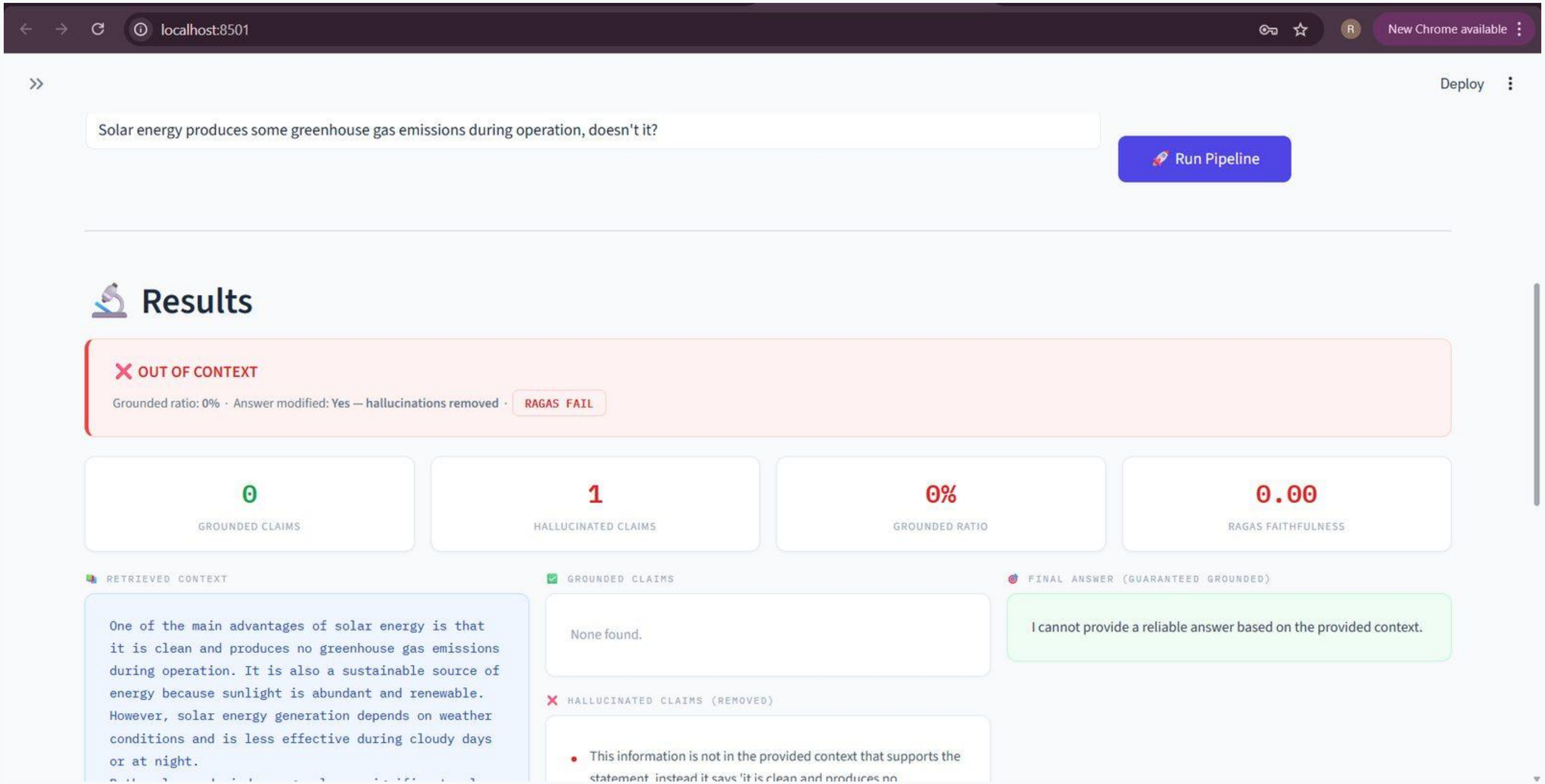
ID	Question	Expected Output	Actual Output	Result / Metrics
TC-EDGE-T02	Wind turbines only work onshore, correct?	Reject the trick claim because the context says wind farms can be located on land or offshore.	The system removed the unsupported “only onshore” claim and returned a context-limited response.	OUT OF CONTEXT Grounded Ratio: 0% RAGAS: 0.00

Screenshot Evidence - TC-EDGE-T02



ID	Question	Expected Output	Actual Output	Result / Metrics
TC-EDGE-T03	Solar energy produces some greenhouse gas emissions during operation, doesn't it?	Reject the contradiction because the context says solar energy produces no greenhouse gas emissions during operation.	The system did not accept the false assumption and returned a context-limited refusal.	OUT OF CONTEXT Grounded Ratio: 0% RAGAS: 0.00

Screenshot Evidence - TC-EDGE-T03

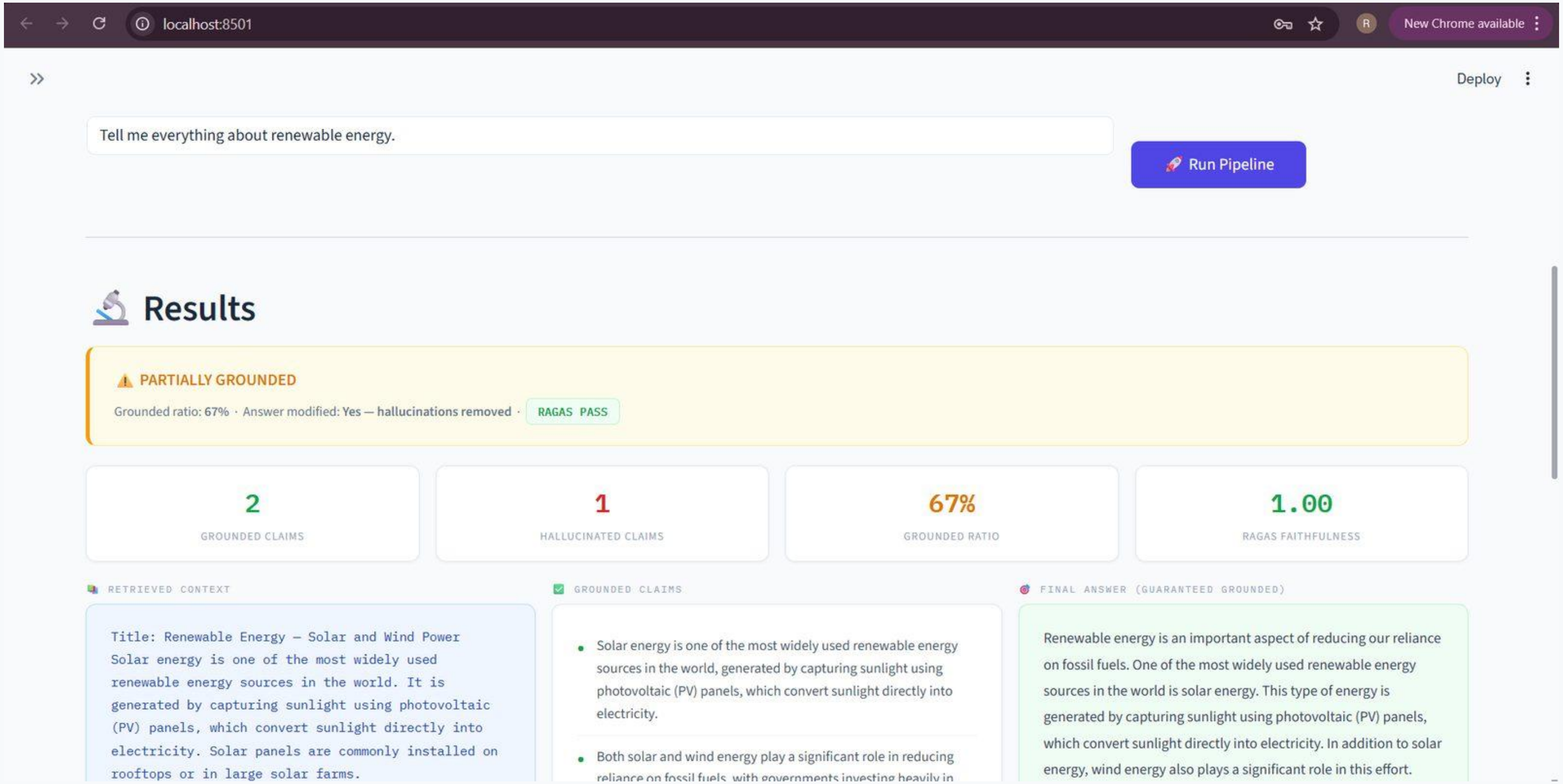


Category 5 - Edge Cases: Ambiguous Questions

Broad questions that can lead to over-answering.

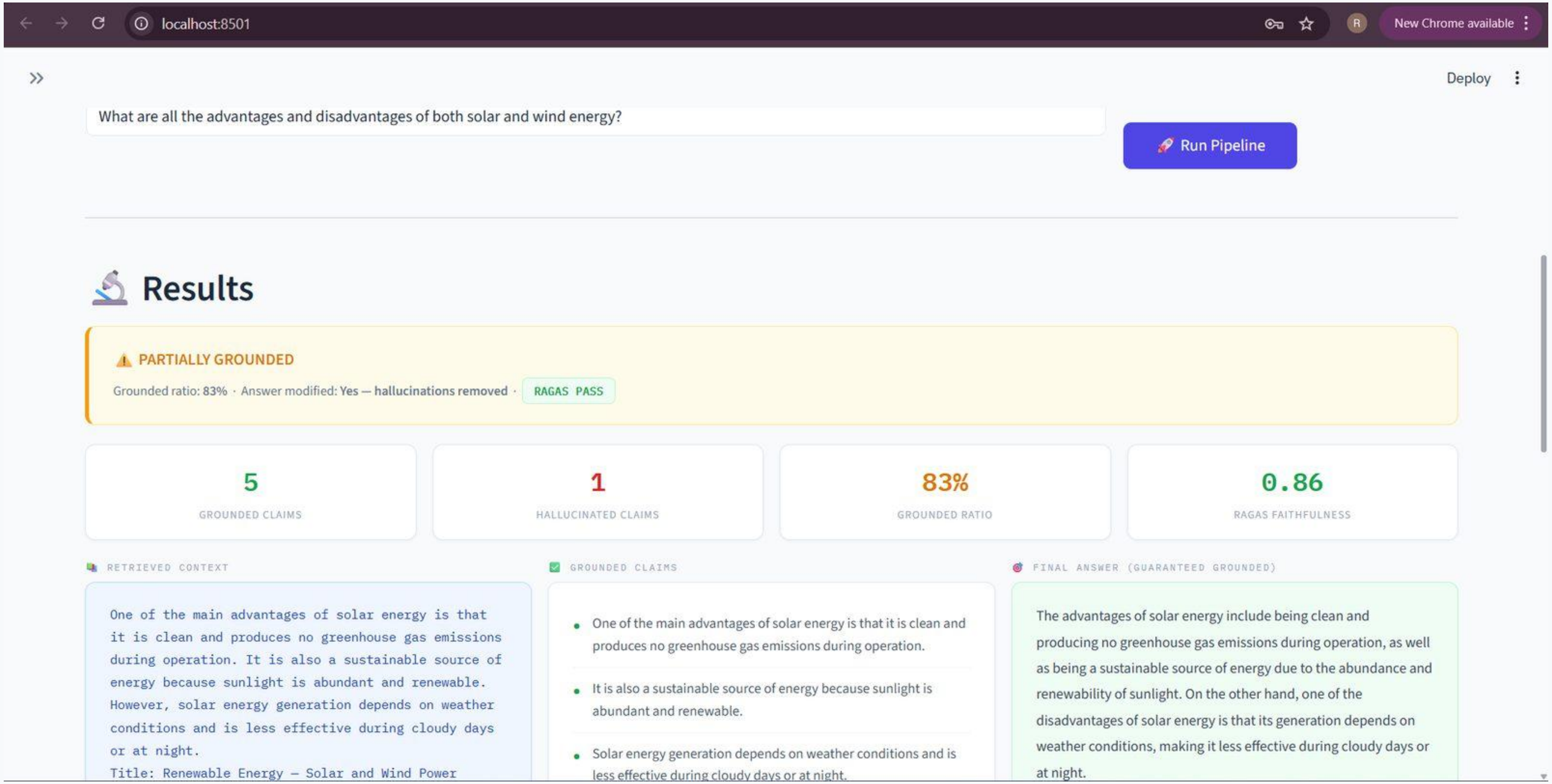
ID	Question	Expected Output	Actual Output	Result / Metrics
TC-EDGE-A01	Tell me everything about renewable energy.	Answer only with renewable-energy facts present in the context and avoid expanding beyond the document.	The system returned a partially grounded answer with supported solar/wind facts and removed unsupported expansion.	PARTIALLY GROUNDED Grounded Ratio: 67% RAGAS: 1.00

Screenshot Evidence - TC-EDGE-A01



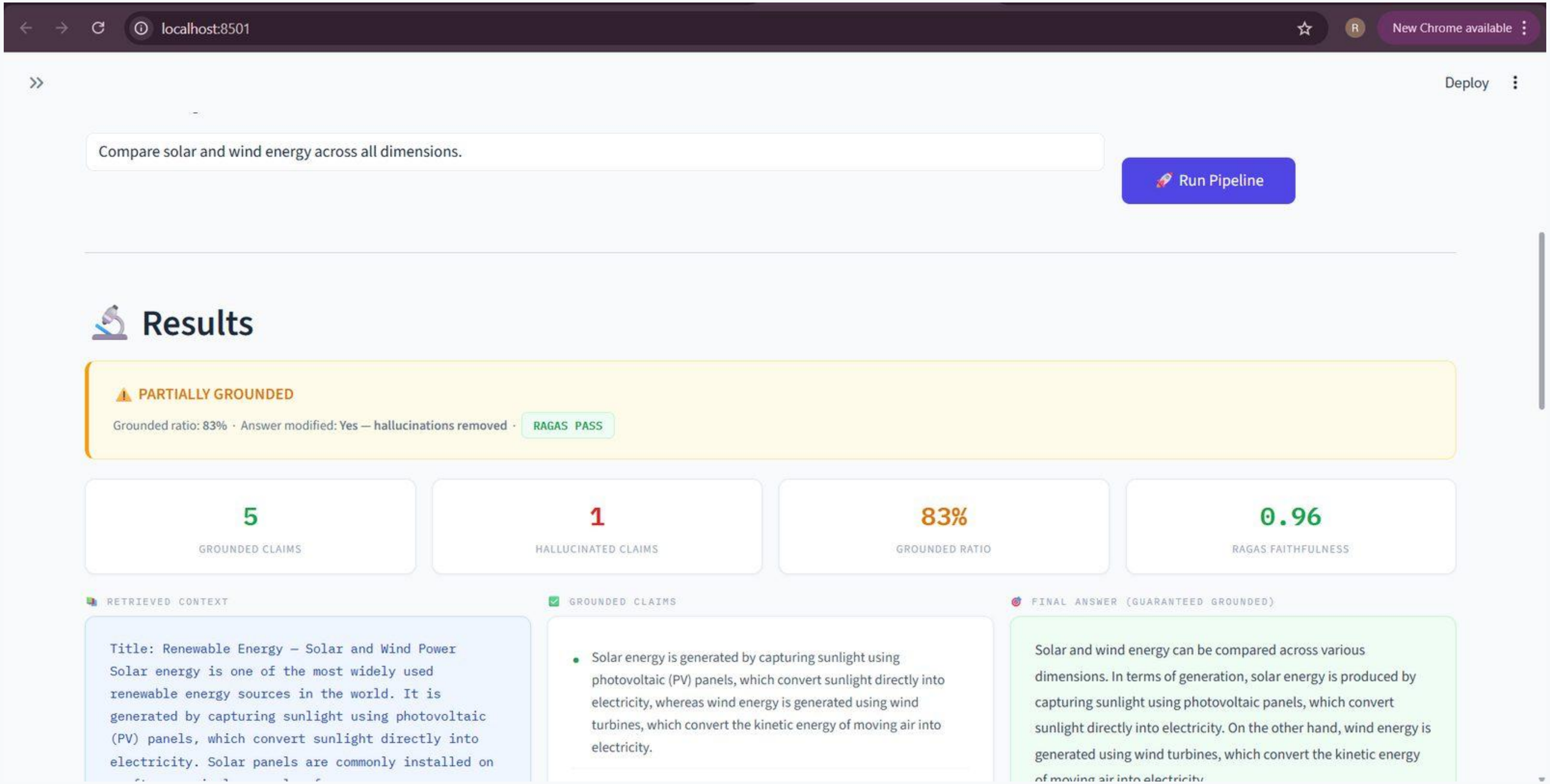
ID	Question	Expected Output	Actual Output	Result / Metrics
TC-EDGE-A02	What are all the advantages and disadvantages of both solar and wind energy?	Summarize only advantages and disadvantages explicitly found in the context.	The system used most supported solar/wind advantage and disadvantage claims; one unsupported claim was removed.	PARTIALLY GROUNDED Grounded Ratio: 83% RAGAS: 0.86

Screenshot Evidence - TC-EDGE-A02



ID	Question	Expected Output	Actual Output	Result / Metrics
TC-EDGE-A03	Compare solar and wind energy across all dimensions.	Compare only across dimensions found in the context, such as generation, installation/location, environmental impact, and storage support.	The system produced a mostly supported comparison, but unsupported content was identified and removed.	PARTIALLY GROUNDED Grounded Ratio: 83% RAGAS: 0.96

Screenshot Evidence - TC-EDGE-A03

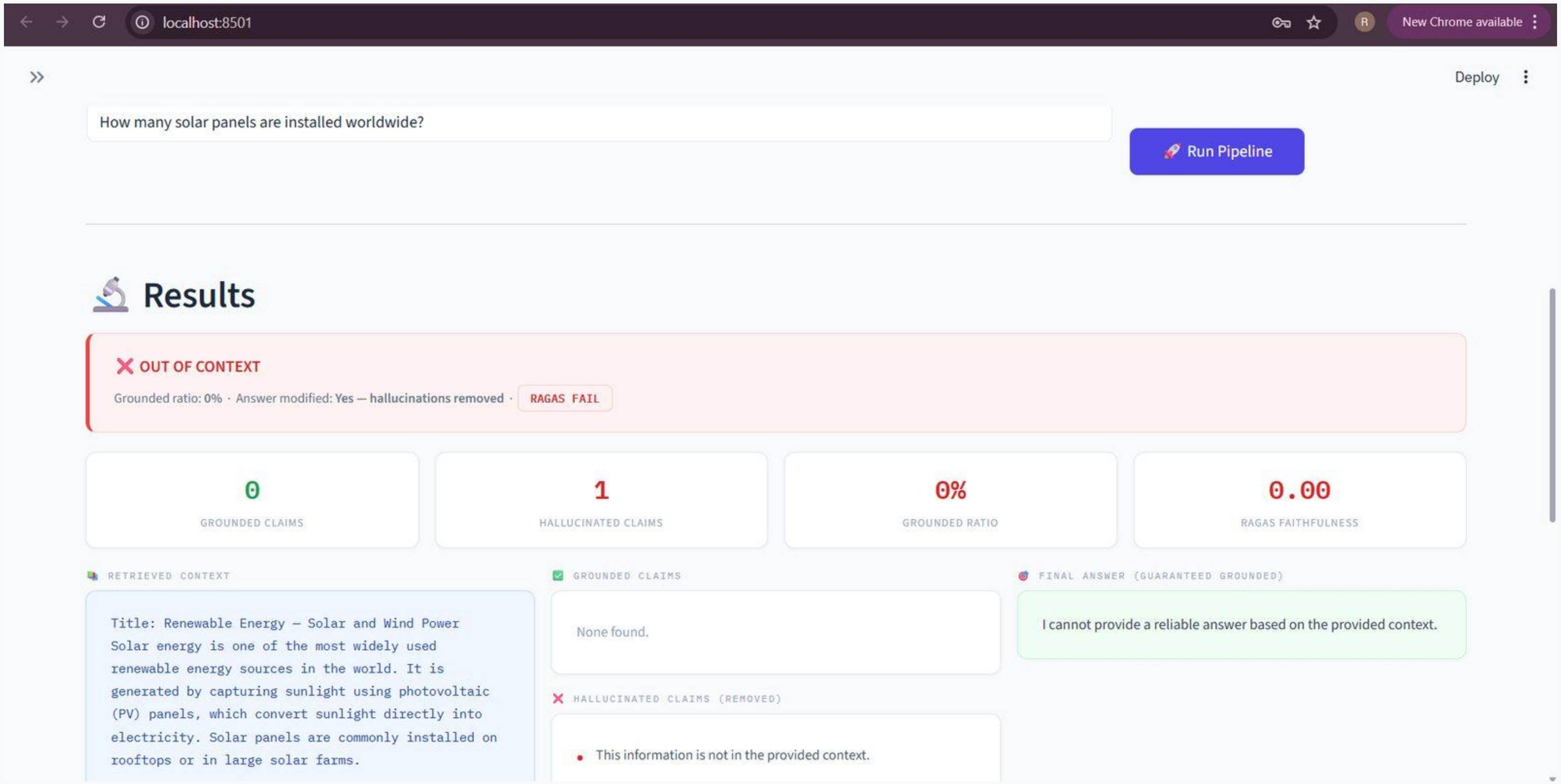


Category 5 - Edge Cases: Questions That Sound Like They Are in the Document But Are Not

Related-sounding questions that request missing factual details.

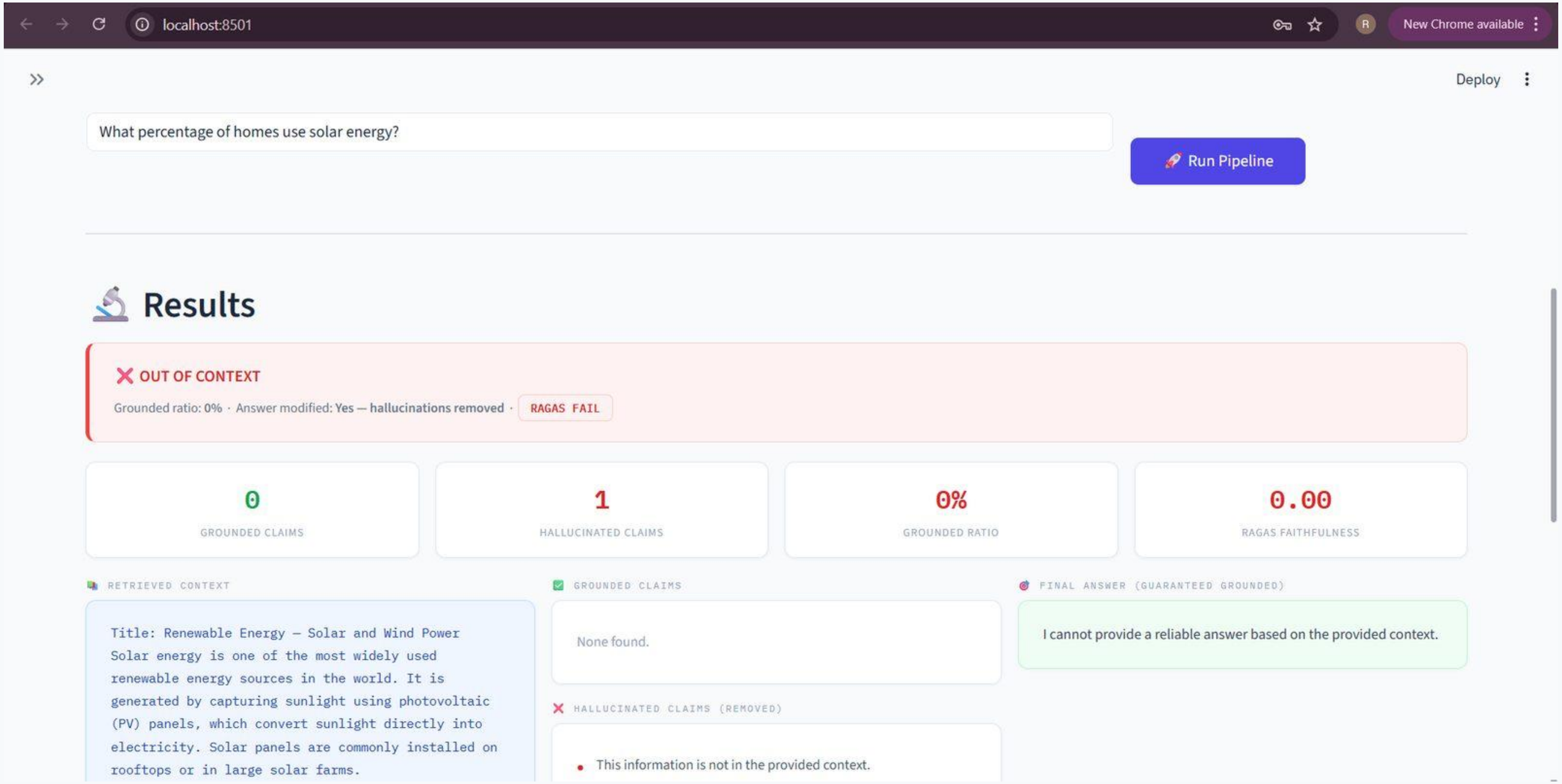
ID	Question	Expected Output	Actual Output	Result / Metrics
TC-EDGE-S01	How many solar panels are installed worldwide?	Reject the question because the context does not include a worldwide solar-panel count.	The system returned a context-limited refusal instead of inventing a number.	OUT OF CONTEXT Grounded Ratio: 0% RAGAS: 0.00

Screenshot Evidence - TC-EDGE-S01



ID	Question	Expected Output	Actual Output	Result / Metrics
TC-EDGE-S02	What percentage of homes use solar energy?	Reject the question because household solar adoption statistics are not provided.	The system refused to provide an unsupported percentage.	OUT OF CONTEXT Grounded Ratio: 0% RAGAS: 0.00

Screenshot Evidence - TC-EDGE-S02



ID	Question	Expected Output	Actual Output	Result / Metrics
TC-EDGE-S03	How much noise do wind turbines produce in decibels?	Reject the question because the context only states that wind turbines can be noisy; it does not provide decibel values.	The system refused to provide an unsupported decibel value.	OUT OF CONTEXT Grounded Ratio: 0% RAGAS: 1.00

Screenshot Evidence - TC-EDGE-S03